

Intraspecific predator interference promotes biodiversity in ecosystems

Ju Kang, Shijie Zhang, Yiyuan Niu, Fan Zhong, Xin Wang 

School of Physics, Sun Yat-sen University, Guangzhou, China • School of Mathematics, Sun Yat-sen University, Guangzhou, China • Department of Mechanical Engineering, Massachusetts Institute of Technology, Cambridge, MA, USA

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Reviewed Preprint

v2 • August 5, 2024

Revised by authors

Reviewed Preprint

v1 • January 11, 2024

Abstract

Explaining biodiversity is a fundamental issue in ecology. A long-standing puzzle lies in the paradox of the plankton: many species of plankton feeding on a limited variety of resources coexist, apparently flouting the competitive exclusion principle (CEP), which holds that the number of predator (consumer) species cannot exceed that of the resources at a steady state. Here, we present a mechanistic model and demonstrate that intraspecific interference among the consumers enables a plethora of consumer species to coexist at constant population densities with only one or a handful of resource species. This facilitated biodiversity is resistant to stochasticity, either with the stochastic simulation algorithm or individual-based modeling. Our model naturally explains the classical experiments that invalidate the CEP, quantitatively illustrates the universal S-shaped pattern of the rank-abundance curves across a wide range of ecological communities, and can be broadly used to resolve the mystery of biodiversity in many natural ecosystems.

eLife assessment

This manuscript is an **important** contribution, assessing the role of intraspecific consumer interference in maintaining diversity using a mathematical model. Consistent with long-standing ecological theory, the authors **convincingly** show that predator interference allows for the coexistence of multiple species on a single resource, beyond the competitive exclusion principle. Notably, the model matches observed rank-abundance curves in several natural ecosystems.

<https://doi.org/10.7554/eLife.93115.2.sa3>

Introduction

The most prominent feature of life on Earth is its remarkable species diversity: countless macro- and micro-species fill every corner on land and in the water (Pennisi, 2005 ; Hoorn et al., 2010 ; Vargas et al., 2015 ; Daniel, 2005 ). In tropical forests, thousands of plant and vertebrate species coexist (Hoorn et al., 2010 ). Within a gram of soil, the number of microbial species is estimated

to be 2,000–18,000 (Daniel, 2005 [\[1\]](#)). In the photic zone of the world ocean, there are roughly 150,000 eukaryotic plankton species (Vargas et al., 2015 [\[2\]](#)). Explaining this astonishing biodiversity is a major focus in ecology (Pennisi, 2005 [\[3\]](#)). A great challenge stems from the well-known competitive exclusion principle (CEP): two species competing for a single type of resources cannot coexist at constant population densities (Gause, 1934 [\[4\]](#); Hardin, 1960 [\[5\]](#)), or generically, in the framework of consumer-resource models, the number of consumer species cannot exceed that of resources at a steady state (MacArthur and Levins, 1964 [\[6\]](#); Levin, 1970 [\[7\]](#); McGehee and Armstrong, 1977 [\[8\]](#)). On the contrary, in the paradox of plankton, a limited variety of resources supports hundreds or more coexisting species of phytoplankton (Hutchinson, 1961 [\[9\]](#)). Then, how can plankton and many other organisms somehow liberate the constraint of CEP?

Ever since MacArthur and Levin proposed the classical mathematical proof for CEP in the 1960s (MacArthur and Levins, 1964 [\[6\]](#)), various mechanisms have been put forward to overcome the limits set by CEP (Chesson, 2000 [\[10\]](#)). Some suggest that the system never approaches a steady state where the CEP applies, due to temporal variations (Hutchinson, 1961 [\[9\]](#); Levins, 1979 [\[11\]](#)), spatial heterogeneity (Levin, 1974 [\[12\]](#)), or species' self-organized dynamics (Koch, 1974 [\[13\]](#); Huisman and Weissing, 1999 [\[14\]](#)). Others consider factors such as toxins (Lazaran et al., 2002 [\[15\]](#)), cross-feeding (Goyal and Maslov., 2018 [\[16\]](#); Goldford et al., 2018 [\[17\]](#); Niehaus et al., 2019 [\[18\]](#)), spatial circulation (Martín et al., 2020 [\[19\]](#); Gupta et al., 2021 [\[20\]](#)), “kill the winner” (Thingstad, 2000 [\[21\]](#)), pack hunting (Wang and Liu, 2020 [\[22\]](#)), collective behavior (Dalziel et al., 2021 [\[23\]](#)), metabolic trade-offs (Posfai et al., 2017 [\[24\]](#); Weiner et al., 2019 [\[25\]](#)), co-evolution (Xue and Goldenfeld, 2017 [\[26\]](#)), and other complex interactions among the species (Beddington, 1975 [\[27\]](#); DeAngelis et al., 1975 [\[28\]](#); Arditi and Ginzburg, 1989 [\[29\]](#); Kelsic et al., 2015 [\[30\]](#); Grilli et al., 2017 [\[31\]](#); Ratzke et al., 2020 [\[32\]](#)). However, questions remain as to what determines species diversity in nature, especially for quasi-well-mixed systems such as that of plankton (Pennisi, 2005 [\[3\]](#); Sunagawa et al., 2020 [\[33\]](#)).

Among the proposed mechanisms, predator interference, specifically the pairwise encounters among consumer individuals, emerges as a potential solution to this issue. Predator interference is commonly described by the classical Beddington-DeAngelis (B-D) phenomenological model (Beddington, 1975 [\[27\]](#); DeAngelis et al., 1975 [\[28\]](#)). Through the application of the B-D model, several studies (Cantrell et al., 2004 [\[34\]](#); Hsu et al., 2013 [\[35\]](#)) have shown that intraspecific predator interference can break CEP and facilitate species coexistence. However, from a mechanistic perspective, the functional response of the B-D model can be formally derived from a scenario solely involving chasing pairs, representing the consumption process between consumers and resources, without accounting for pairwise encounters among consumer individuals (Wang and Liu, 2020 [\[22\]](#); Huisman and Boer, 1997 [\[36\]](#)). Disturbingly, it has been established that the scenario involving only chasing pairs is subject to the constraint of CEP (Wang and Liu, 2020 [\[22\]](#)), raising doubt regarding the validity of applying the B-D model to overcome the CEP.

In this work, building upon MacArthur's consumer-resource model framework (MacArthur, 1969 [\[37\]](#), 1970 [\[7\]](#); Chesson, 1990 [\[10\]](#)), and drawing on concepts from chemical reaction kinetics (Ruxton et al., 1992 [\[38\]](#); Huisman and Boer, 1997 [\[36\]](#); Wang and Liu, 2020 [\[22\]](#)), we present a mechanistic model of predator interference that extends the B-D phenomenological model (Beddington, 1975 [\[27\]](#); DeAngelis et al., 1975 [\[28\]](#)) by providing a detailed consideration of pairwise encounters. The intraspecific interference among consumer individuals effectively constitutes a negative feedback loop, enabling a wide range of consumer species to coexist with only one or a few types of resources. The coexistence state is resistant to stochasticity and can hence be realized in practice. Our model is broadly applicable and can be used to explain biodiversity in many ecosystems. In particular, it naturally explains species coexistence in classical experiments that invalidate CEP (Ayala, 1969 [\[39\]](#); Park, 1954 [\[40\]](#)) and quantitatively illustrates the S-shaped pattern of the rank-abundance curves in an extensive spectrum of ecological communities, ranging from the communities of ocean plankton worldwide (Fuhrman et al., 2008 [\[41\]](#); Ser-Giacomi et al., 2018 [\[42\]](#)), tropical river fishes from Argentina (Cody and Smallwood, 1996 [\[43\]](#)), forest bats of Trinidad (Clarke

et al., 2005 [\[1\]](#)), rainforest trees (Hubbell, 2001 [\[2\]](#)), birds (Terborgh et al., 1990 [\[3\]](#); Martínez et al., 2023 [\[4\]](#)), butterflies (Devries et al., 1997 [\[5\]](#)) in Amazonia, to those of desert bees (Hubbell, 2001 [\[2\]](#)) in Utah and lizards from South Australia (Cody and Smallwood, 1996 [\[6\]](#)).

Results

A generic model of pairwise encounters

Here we present a mechanistic model of pairwise encounters (see **Fig. 1A** [\[1\]](#)), where S_C consumer species $\{C_1, \dots, C_{S_C}\}$ compete for S_R resource species $\{R_1, \dots, R_{S_R}\}$. The consumers are biotic, while the resources can be either biotic or abiotic. For simplicity, we assume that all species are motile and move at certain speeds, namely, v_{C_i} for consumer species C_i ($i = 1, \dots, S_C$) and v_{R_l} for resource species R_l ($l = 1, \dots, S_R$). For abiotic resources, they cannot propel themselves but may passively drift due to environmental factors. Each consumer is free to feed on one or multiple types of resources, while consumers do not directly interact with one another except through pairwise encounters.

Then, we explicitly consider the population structure of consumers and resources: some wander around freely, undergoing Brownian motions; others encounter one another, forming ephemeral entangled pairs. Specifically, when a consumer individual C_i and a resource R_l come close within a distance of $r_{il}^{(C)}$ (see **Fig. 1A** [\[1\]](#)), the consumer can chase the resource and form a chasing pair:

$C_i^{(P)} \vee R_l^{(P)}$ (see **Fig. 1B** [\[1\]](#)), where the superscript “(P)” represents “pair”. The resource can either escape at rate d_{il} or be caught and consumed by the consumer with rate k_{il} . Meanwhile, when a C_i individual encounters another consumer C_j ($j = 1, \dots, S_C$) within a distance of $r_{ij}^{(I)}$ (see **Fig. 1A** [\[1\]](#)), they can stare at, fight against, or play with each other, thus forming an interference pair: $C_i^{(P)} \vee C_j^{(P)}$ (see **Fig. 1B** [\[1\]](#)). This paired state is evanescent, with consumers separating at rate d'_{ij} . For simplicity, we assume that all $r_{il}^{(C)}$ and $r_{ij}^{(I)}$ are identical, respectively, i.e., $\forall i, j, l$, $r_{il}^{(C)} = r^{(C)}$ and $r_{ij}^{(I)} = r^{(I)}$.

In a well-mixed system of size L^2 , the encounter rates among individuals, a_{il} and a'_{ij} (see **Fig. 1B** [\[1\]](#)), can be derived using the mean-field approximation: $a_{il} = 2r^{(C)}L^{-2}\sqrt{v_{C_i}^2 + v_{R_l}^2}$ and

$a'_{ij} = 2r^{(I)}L^{-2}\sqrt{v_{C_i}^2 + v_{C_j}^2}$ (see Materials and Methods, and **Appendix-fig. 1** [\[1\]](#)). Then, we

proceed to analyze scenarios involving different types of pairwise encounters (see **Fig. 1B** [\[1\]](#)). For the scenario involving only chasing pairs, the population dynamics can be described as follows:

$$\begin{cases} \dot{x}_{il} = a_{il}C_i^{(F)}R_l^{(F)} - (d_{il} + k_{il})x_{il}, \\ \dot{C}_i = \sum_{l=1}^{S_R} w_{il}k_{il}x_{il} - D_iC_i, \quad i = 1, \dots, S_C, \\ \dot{R}_l = g_l(\{R_l\}, \{x_i\}, \{C_i\}), \quad l = 1, \dots, S_R, \end{cases} \quad (1)$$

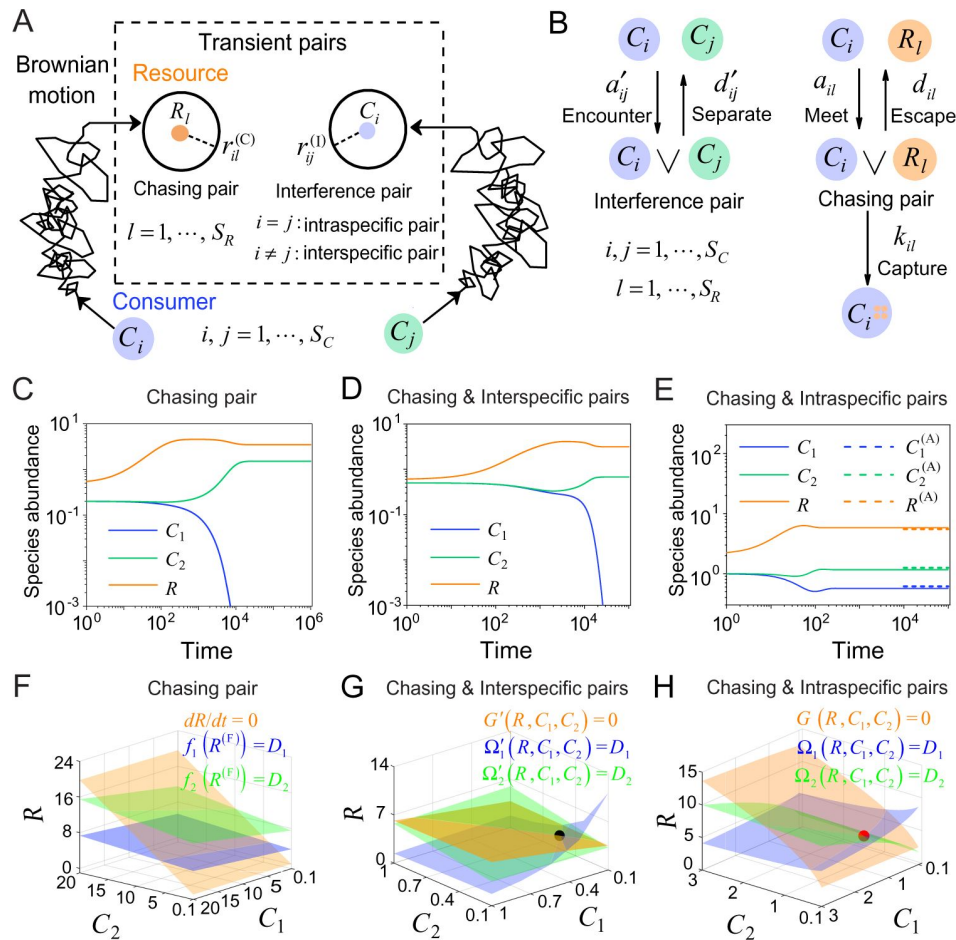


Figure 1.

A model of pairwise encounters may naturally break CEP.

(A) A generic model of pairwise encounters involving S_C consumer species and S_R resource species. (B) The well-mixed model of (A). (C-E) Time courses of two consumer species competing for one resource species. (F-H) Positive solutions to the steady-state equations (see [Eqs. S38](#) and [S65](#)): $\dot{R} = 0$ (orange surface), $\dot{C}_1 = 0$ (blue surface), $\dot{C}_2 = 0$ (green surface), i.e., the zero-growth isoclines. The black/red dot represents the unstable/stable fixed point, while the dotted lines in (E) are the analytical solutions of the steady-state abundances (marked with superscript "(A)"). See [Appendix-tables 1](#)–[2](#) for the definitions of symbols. See [Appendix VIII](#) for simulation details.

where $x_{il} \equiv C_i^{(P)} \vee R_l^{(P)}$, g_l is an unspecified function, the superscript “(F)” represents the freely wandering population, D_i denotes the mortality rate of C_i , and w_{il} is the mass conversion ratio (Wang and Liu, 2020) from resource R_l to consumer C_i . With the integration of intraspecific predator interference, we combine Eq. 1 and the following equation:

$$\dot{y}_i = a'_i [C_i^{(F)}]^2 - d'_i y_i, \quad (2)$$

where $a'_i = a'_{ii}$, $d'_i = d'_{ii}$, and $y_i \equiv C_i^{(P)} \vee C_i^{(P)}$ represents the intraspecific interference pair (see Fig. 1B). For the scenario involving chasing pairs and interspecific interference, we combine Eq. 1 with the following equation:

$$\dot{z}_{ij} = a'_{ij} C_i^{(F)} C_j^{(F)} - d'_{ij} z_{ij}, \quad i \neq j, \quad (3)$$

where $z_{ij} \equiv C_i^{(P)} \vee C_j^{(P)}$ stands for the interspecific interference pair (see Fig. 1B). In the scenario where chasing pairs and both intra- and interspecific interference are relevant, we combine Eqs. 13, and the populations of consumers and resources are given by

$$C_i = C_i^{(F)} + \sum_l x_{il} + 2y_i + \sum_{j \neq i} z_{ij} \quad \text{and} \quad R_l = R_l^{(F)} + \sum_i x_{il}, \quad \text{respectively.}$$

Generically, the consumption and interference processes are much quicker compared to the birth and death processes. Thus, in the derivation of the functional response, $F(R_l, C_i) \equiv k_{il} x_{il} / C_i$, the consumption and interference processes are supposed to be in fast equilibrium. In all scenarios involving different types of pairwise encounters, the functional response in the B-D model is a good approximation only for a special case with $d_{il} \approx 0$ and $R_l \gg \sum_{i=1}^{S_C} C_i$ (see Appendix-fig. 2 and

Appendix II for details).

To facilitate further analysis, we assume that the population dynamics of the resources follows the same construction rule as that in MacArthur’s consumer-resource model (MacArthur, 1969, 1970; Chesson, 1990). Then,

$$g_l(\{R_l\}, \{x_i\}, \{C_i\}) = \begin{cases} \eta_l R_l (1 - R_l / \kappa_l) - \sum_{i=1}^{S_C} k_{il} x_{il} & \text{(for biotic resources);} \\ \zeta_l (1 - R_l / \kappa_l) - \sum_{i=1}^{S_C} k_{il} x_{il} & \text{(for abiotic resources).} \end{cases} \quad (4)$$

In the absence of consumers, biotic resources exhibit logistic growth. Here, η_l and κ_l represent the intrinsic growth rate and the carrying capacity of species R_l . For abiotic resources, ζ_l stands for the external resource supply rate of R_l , and κ_l is the abundance of R_l at a steady state without consumers. For simplicity, we focus our analysis on abiotic resources, although all results generally apply to biotic resources as well. By applying dimensional analysis, we render all parameters dimensionless (see Appendix VI). For convenience, we retain the same notations below, with all parameters considered dimensionless unless otherwise specified.

Intraspecific predator interference facilitates species coexistence and breaks CEP

To clarify the specific mechanisms that can facilitate species coexistence, we systematically investigate scenarios involving different forms of pairwise encounters in a simple case with $S_C = 2$ and $S_R = 1$. To simplify the notations, we omit the subscript/superscript “ l ” since $S_R = 1$. For clarity, we assign each consumer species of unique competitiveness by setting that the mortality rate D_i is the only parameter that varies with the consumer species.

First, we conduct the analysis within a deterministic framework using ordinary differential equations (ODEs). In the scenario involving only chasing pairs, consumer species cannot coexist at a steady state except for special parameter settings (sets of measure zero) (Wang and Liu, 2020 [4](#)). In practice, if all species coexist, the steady-state equations of the consumer species ($\dot{C}_i = 0$, i.e., the zero-growth isolines) yield $f_i(R^{(F)}) = D_i$ ($i = 1, 2$), where $f_i(R^{(F)})$ is defined as $f_i(R^{(F)}) \equiv R^{(F)} / (R^{(F)} + K_i)$ and $K_i \equiv (d_i + k_i) / a_i$. These equations form two parallel surfaces in the (C_1, C_2, R) coordinates, making steady coexistence impossible (Wang and Liu, 2020 [4](#)) (see **Fig. 1C**, **F** [4](#) and **Appendix-fig. 3A-C** [4](#)).

Meanwhile, in the scenario involving chasing pairs and interspecific interference, if all species coexist, the zero-growth isolines of the three species (see **Eqs. S65** [4](#)) correspond to three nonparallel surfaces $\Omega'_i(R, C_1, C_2) = D_i$ ($i = 1, 2$), $G'(R, C_1, C_2) = 0$ (see **Fig. 1G** [4](#) and **Appendix-fig. 3D** [4](#); refer to **Appendix IV** [4](#) for definitions of Ω'_i and G'), which can intersect at a common point (fixed point). However, this fixed point is unstable (see **Fig. 1G** [4](#) and **Appendix-fig. 3D**, **F** [4](#)), and thus one of the consumer species is doomed to extinction (see **Fig. 1D** [4](#)).

Next, we turn to the scenario involving chasing pairs and intraspecific interference. Likewise, steady coexistence requires (see **Eqs. S38** [4](#)) that three non-parallel surfaces $\Omega_i(R, C_1, C_2) = D_i$ ($i = 1, 2$), $G(R, C_1, C_2) = 0$ cross at a common point (see **Fig. 1H** [4](#) and **Appendix-fig. 3G** [4](#); refer to **Appendix III** [4](#) for definitions of Ω_i and G). Indeed, this naturally happens, and encouragingly the fixed point can be stable. Therefore, two consumer species may stably coexist at a steady state with only one type of resources, which obviously breaks CEP (see **Fig. 1E** [4](#) and **Appendix-fig. 4A** [4](#)). In fact, the coexisting state is globally attractive (see **Appendix-fig. 4A** [4](#)), and there exists a non-zero volume of parameter space where the two consumer species stably coexist at constant population densities (see **Appendix-fig. 4B-C** [4](#)), demonstrating that the violation of CEP does not depend on special parameter settings. We further consider the scenario involving chasing pairs and both intra- and interspecific interference (see **Appendix-fig. 5** [4](#)). Much as expected, the species coexistence behavior is very similar to that without interspecific interference.

Intraspecific interference promotes biodiversity in the presence of stochasticity

Stochasticity is ubiquitous in nature. However, it is prone to jeopardize species coexistence (Xue and Goldenfeld, 2017 [4](#)). Influential mechanisms such as “kill the winner” fail when stochasticity is incorporated (Xue and Goldenfeld, 2017 [4](#)). Consistent with this, we observe that two notable cases of oscillating coexistence (Koch, 1974 [4](#); Huisman and Weissing, 1999 [4](#)) turn into species extinction when stochasticity is introduced (see **Appendix-fig. 6A-B** [4](#)), where we simulate the models with stochastic simulation algorithm (SSA) (Gillespie, 2007 [4](#)) and adopt the same parameters as those in the original references (Koch, 1974 [4](#); Huisman and Weissing, 1999 [4](#)).

Then, we proceed to investigate the impact of stochasticity on our model using SSA (Gillespie, 2007 [4](#)). In the scenario involving chasing pairs and intraspecific interference, species may coexist indefinitely in the SSA simulations (see **Fig. 2A** [4](#) and **Appendix-fig. 4D** [4](#)). In fact, the parameter region for species coexistence in this scenario is rather similar between the SSA and ODEs studies

(see **Appendix-fig. 6C-D**). Similarly, in the scenario involving chasing pairs and both inter- and intraspecific interference, all species may coexist indefinitely in company with stochasticity (see **Appendixfig. 5D**).

To further mimic a real ecosystem, we resort to individual-based modeling (IBM) (*Grimm and Railsback, 2013*; *Vetsigian, 2017*), an essentially stochastic simulation method. In the simple case of $S_C = 2$ and $S_R = 1$, we simulate the time evolution of a 2-D square system in a size of L^2 with periodic boundary conditions (see Materials and Methods for details). In the scenario involving chasing pairs and intraspecific interference, two consumer species coexist for long with only one type of resources in the IBM simulations (see **Fig. 2B-C**). Together with the SSA simulation studies, it is obvious that intraspecific interference still robustly promotes species coexistence when stochasticity is considered.

Comparison with experimental studies that reject CEP

In practice, two classical studies (*Ayala, 1969*; *Park, 1954*) reported that, in their respective laboratory systems, two species of insects coexisted for roughly years or more with only one type of resources. Evidently, these two experiments (*Ayala, 1969*; *Park, 1954*) are incompatible with CEP, while factors such as temporal variations, spatial heterogeneity, cross-feeding, etc. are clearly not involved in such systems. As intraspecific fighting is prevalent among insects (*Boomsma et al., 2005*; *Dankert et al., 2009*; *Chen et al., 2002*), we apply the model involving chasing pairs and intraspecific interference to simulate the two systems. Overall, our SSA results show good consistency with those of the experiments (see **Fig. 2D-E** and **Appendix-figs. 7**). The fluctuations in experimental time series can be mainly accounted by stochasticity.

A handful of resource species can support a wide range of consumer species regardless of stochasticity

To resolve the puzzle stated in the paradox of the plankton, we analyze the generic case where S_C consumer species compete for S_R resource species (with $S_C > S_R$) within the scenario involving chasing pairs and intraspecific interference. The population dynamics is described by equations combining **Eqs. 1**, **2**, and **4**. As with the cases above, each consumer species is assigned a unique competitiveness through a distinctive $D_i (i = 1, \dots, S_C)$.

Strikingly, a plethora of consumer species may coexist at a steady state with only one resource species ($S_C \gg S_R, S_R = 1$) in the ODEs simulations, and crucially, the facilitated biodiversity can still be maintained in the SSA simulations. The long-term coexistence behavior is exemplified in **Fig. 3** and **Appendix-fig. 8–10**, involving simulations with or without stochasticity. The number of consumer species in long-term coexistence can be up to hundreds or more (see **Fig. 3** and **Appendixfig. 8**). To mimic real ecosystems, we further analyze cases with more than one type of resources, such as systems with $S_R = 3$ ($S_C \gg S_R$). Just like the case of $S_R = 1$ ($S_C \gg S_R$), an extensive range of consumer species may coexist indefinitely regardless of stochasticity (see **Fig. 3** and **Appendixfigs. 11–14**).

We further analyze the scenario involving chasing pairs and both intra- and interspecific interference, where multiple consumer species compete for one resource species. Similar to the scenario involving chasing pairs and intraspecific interference, all species coexist indefinitely in either ODEs or SSA simulation studies (see **Appendix-fig. 5F-H** for the cases of $S_C = 6, 20$ and $S_R = 1$).

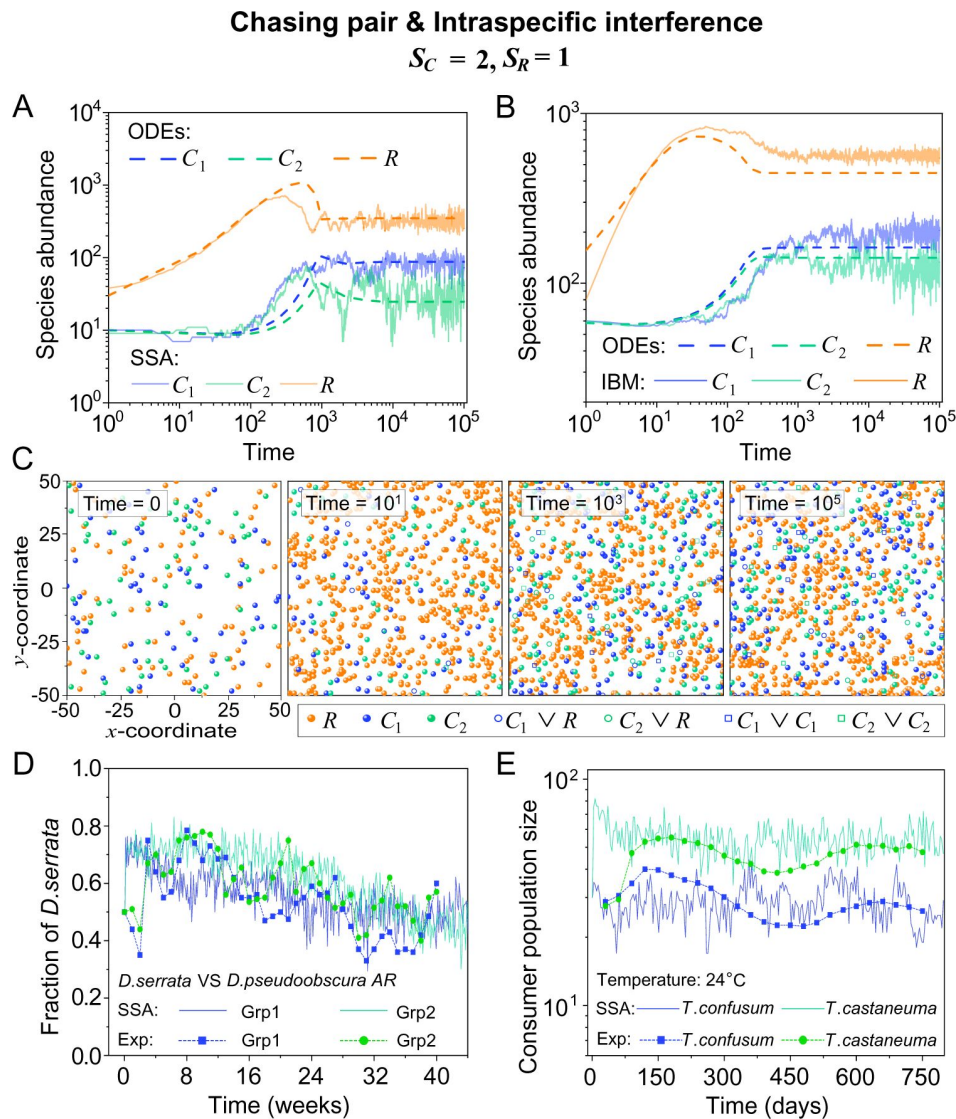


Figure 2.

Intraspecific predator interference facilitates species coexistence regardless of stochasticity.

(A-B) Time courses of the species abundances simulated with ODEs, SSA, or IBM. (C) Snapshots of the IBM simulations. (D-E) A model of intraspecific predator interference explains two classical laboratory experiments that invalidate CEP. (D) In Ayala's experiment, two *Drosophila* species coexist with one type of resources within a laboratory bottle (Ayala, 1969 [\[1\]](#)). (E) In Park's experiment, two *Tribolium* species coexist for two years with one type of food (flour) within a lab (Park, 1954 [\[2\]](#)). See **Appendix-fig. 7C-D** [\[3\]](#) for the comparison between model results and experimental data using Shannon entropies.

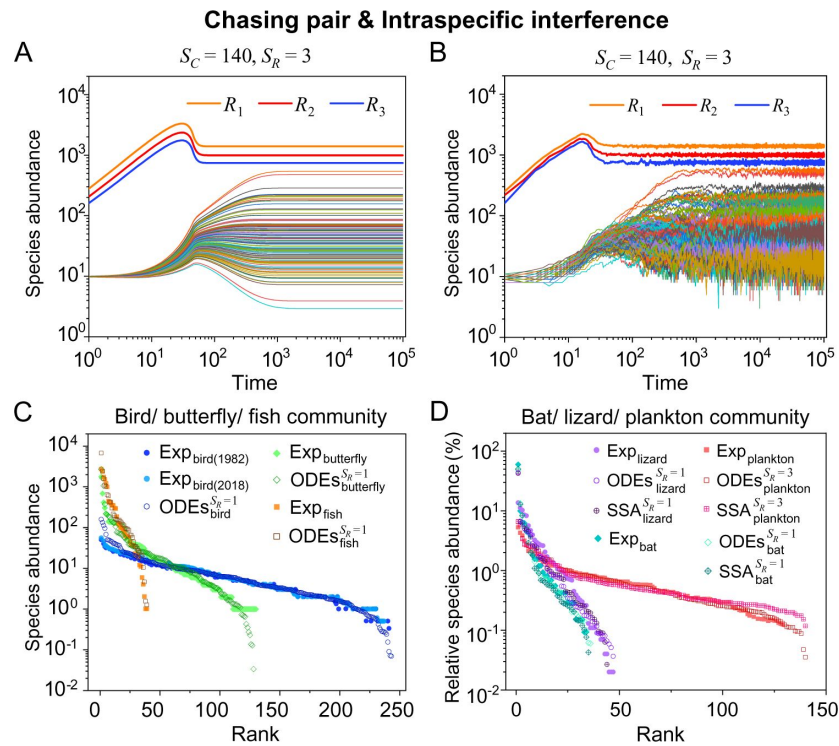


Figure 3.

Intraspecific interference enables a wide range of consumer species to coexist with only one or a handful of resource species.

(A-B) Representative time courses simulated with ODEs and SSA. (C-D) A model of intraspecific predator interference illustrates the S-shaped pattern of the species' rank-abundance curves across different ecological communities. The solid icons represent the experimental data (marked with "Exp") reported in existing studies (Fuhrman et al., 2008; Cody and Smallwood, 1996; Terborgh et al., 1990; Martínez et al., 2023; Clarke et al., 2005; Devries et al., 1997; Hubbell, 2001), where the bird community data were collected longitudinally in 1982 and 2018 (Terborgh et al., 1990; Martínez et al., 2023). The ODEs and SSA results were constructed from timestamp $t = 1.0 \times 10^5$ in the time series. In the K-S test, the probabilities (p values) that the simulation results and the corresponding experimental data come from the same distributions are: $p_{\text{ODEs}}^{\text{bird}(1982)} = 0.17$, $p_{\text{ODEs}}^{\text{bird}(2018)} = 0.26$, $p_{\text{ODEs}}^{\text{butterfly}} = 0.70$, $p_{\text{ODEs}}^{\text{fish}} = 0.88$, $p_{\text{ODEs}}^{\text{bat}} = 0.42$, $p_{\text{SSA}}^{\text{bat}} = 0.48$, $p_{\text{ODEs}}^{\text{lizard}} = 0.96$, $p_{\text{SSA}}^{\text{lizard}} = 0.54$, $p_{\text{ODEs}}^{\text{plankton}} = 0.20$, $p_{\text{SSA}}^{\text{plankton}} = 0.06$. See Appendix VIII for simulation details and the Shannon entropies.

Intuitive understanding: an underlying negative feedback loop

For the case with only one resource species ($S_R = 1$), if the total population size of the resources is much larger than that of the consumers (i.e., $R \gg \sum_{i=1}^{S_C} C_i$), the functional response $F \equiv k_i x_i / C_i$ and the steady-state population of each consumer and resource species can be obtained analytically (see [Appendix III B-C](#) for details). In fact, the functional response of a consumer species (e.g., C_i) is negatively correlated with its own population size:

$$F(R, C_i) \approx \frac{2R}{\sqrt{(R + K_i)^2 + 8\beta_i K_i^2 C_i} + R + K_i}, \quad (5)$$

where $\beta_i \equiv a'_i/d'_i$. The analytical steady-state solutions are highly consistent with the numerical results (see [Fig. 1E](#) and [Appendix-fig. 3H-I](#)) and can even quantitatively predict the coexistence region of the parameter space (see [Appendix-fig. 3I](#)).

Intuitively, the mechanisms of how intraspecific interference facilitates species coexistence can be understood from the underlying negative feedback loop. Specifically, for consumer species of higher competitiveness (e.g., C_i) in an ecological community, as the population size of C_i increases during competition, a larger portion of C_i individuals are then engaged in intraspecific interference pairs which are temporarily absent from hunting (see [Eq. S59](#) and [Appendix-fig. 15A-B](#)). Consequently, the fraction of C_i individuals within chasing pairs decreases (see [Eq. S59](#) and [Appendixfig. 15A-B](#)) and thus form a self-inhibiting negative feedback loop through the functional response (see [Eq. S59](#) and [Appendix-fig. 15C](#)). This negative feedback loop prevents further increases in C_i populations, results in an overall balance among the consumer species, and thus promotes biodiversity (see [Appendix III C](#) for details).

The S shape pattern of the rank-abundance curves in a broad range of ecological communities

As mentioned above, a prominent feature of biodiversity is that the species' rank-abundance curves follow a universal S-shaped pattern in the linear-log plot across a broad spectrum of ecological communities ([Fuhrman et al., 2008](#); [Ser-Giacomi et al., 2018](#); [Cody and Smallwood, 1996](#); [Terborgh et al., 1990](#); [Martí'nez et al., 2023](#); [Clarke et al., 2005](#); [Hubbell, 2001](#); [Devries et al., 1997](#)). Previously, this pattern was mostly explained by the neutral theory ([Hubbell, 2001](#)), which requires special parameter settings that all consumer species share identical fitness. To resolve this issue, we apply the model involving chasing pairs and intraspecific interference to simulate the ecological communities, where one or three types of resources support a large number of consumer species ($S_C \gg S_R$). In each model system, the mortality rates of consumer species follow a Gaussian distribution where the coefficient of variation was taken around 0.3 ([Menon et al., 2003](#)) (see [Appendix VIII](#) for details). For a broad array of the ecological communities, the rank-abundance curves obtained from the long-term coexisting state of both the ODEs and SSA simulation studies agree quantitatively with those of experiments (see [Fig. 3C-D](#) and [Appendix-figs. 8-14](#)), sharing roughly equal Shannon entropies and mostly being regarded as identical distributions in the Kolmogorov-Smirnov (K-S) statistical test (with a significance threshold of 0.05). Still, there is a noticeable discrepancy between the experimental data and SSA studies in terms of the species' absolute abundances (e.g., see [Appendix-fig. 8C](#)): those with experimental abundances less than 10 tend to be extinct in

the SSA simulations. This is due to the fact that the recorded individuals in an experimental sample are just a tiny portion of that in the real ecological system, whereas the species population size in a natural community is certainly much larger than 10.

Discussion

The conflict between the CEP and biodiversity, exemplified by the paradox of the plankton (Hutchinson, 1961 [\[1\]](#)), is a long-standing puzzle in ecology. Although many mechanisms have been proposed to overcome the limit set by CEP (Hutchinson, 1961 [\[1\]](#); Chesson, 2000 [\[2\]](#); Levins, 1979 [\[3\]](#); Levin, 1974 [\[4\]](#); Koch, 1974 [\[5\]](#); Huisman and Weissing, 1999 [\[6\]](#); Lcászárán et al., 2002 [\[7\]](#); Goyal and Maslov., 2018 [\[8\]](#); Goldford et al., 2018 [\[9\]](#); Martín et al., 2020 [\[10\]](#); Gupta et al., 2021 [\[11\]](#); Thingstad, 2000 [\[12\]](#); Wang and Liu, 2020 [\[13\]](#); Dalziel et al., 2021 [\[14\]](#); Posfai et al., 2017 [\[15\]](#); Weiner et al., 2019 [\[16\]](#); Xue and Goldenfeld, 2017 [\[17\]](#); Beddington, 1975 [\[18\]](#); DeAngelis et al., 1975 [\[19\]](#); Arditi and Ginzburg, 1989 [\[20\]](#); Kelsic et al., 2015 [\[21\]](#); Grilli et al., 2017 [\[22\]](#); Ratzke et al., 2020 [\[23\]](#)), it is still unclear how plankton and many other organisms can flout CEP and maintain biodiversity in quasi-well-mixed natural ecosystems. To address this issue, we investigate a mechanistic model with detailed consideration of pairwise encounters. Using numerical simulations combined with mathematical analysis, we identify that the intraspecific interference among the consumer individuals can promote a wide range of consumer species to coexist indefinitely with only one or a handful of resource species through the underlying negative feedback loop. By applying the above analysis to real ecological systems, our model naturally explains two classical experiments that reject CEP (Ayala, 1969 [\[24\]](#); Park, 1954 [\[25\]](#)), and quantitatively illustrates the universal S-shaped pattern of the rank-abundance curves for a broad range of ecological communities (Fuhrman et al., 2008 [\[26\]](#); Ser-Giacomi et al., 2018 [\[27\]](#); Cody and Smallwood, 1996 [\[28\]](#); Terborgh et al., 1990 [\[29\]](#); Martínez et al., 2023 [\[30\]](#); Clarke et al., 2005 [\[31\]](#); Hubbell, 2001 [\[32\]](#); Devries et al., 1997 [\[33\]](#)).

In fact, predator interference has been introduced long ago by the classical B-D phenomenological model (Beddington, 1975 [\[18\]](#); DeAngelis et al., 1975 [\[19\]](#)). However, the functional response of the B-D model involving intraspecific interference can be formally derived from the scenario involving only chasing pairs without consideration of pairwise encounters between consumer individuals (Wang and Liu, 2020 [\[13\]](#); Huisman and Boer, 1997 [\[34\]](#)) (see Eqs. S8 [\[35\]](#) and S24 [\[36\]](#)). Yet, it has been demonstrated that the scenario involving only chasing pairs is under the constraint of CEP (Wang and Liu, 2020 [\[13\]](#)) (see **Appendix-fig. 3A-C** [\[37\]](#)). Therefore, it is questionable regarding the validity of applying the B-D model to break CEP (Cantrell et al., 2004 [\[38\]](#); Hsu et al., 2013 [\[39\]](#)). From a mechanistic perspective, we resolve these issues and show that the B-D model corresponds to a special case of our mechanistic model yet without the escape rate (see **Appendix-fig. 2** [\[40\]](#) and **Appendix II** [\[41\]](#) for details).

Our model is broadly applicable to explain biodiversity in many ecosystems. In practice, many more factors are potentially involved, and special attention is required to disentangle confounding factors. In microbial systems, complex interactions are commonly involved (Goyal and Maslov., 2018 [\[8\]](#); Goldford et al., 2018 [\[9\]](#); Hu et al., 2022 [\[42\]](#)), and species' preference for food is shaped by the evolutionary course and environmental history (Wang et al., 2019 [\[43\]](#)). It is still highly challenging to fully explain how organisms evolve and maintain biodiversity in diverse ecosystems.

Methods and Materials

Derivation of the encounter rates with the mean-field approximation

In the model depicted in [Fig. 1A](#), consumers and resources move randomly in space, which can be regarded as Brownian motions. At moment t , a consumer individual of species $C_i (i = 1, \dots, S_C)$ moves at speed v_{C_i} with velocity $\mathbf{v}_{C_i}(t)$, while a resource individual of species $R_l (l = 1, \dots, S_R)$ moves at speed v_{R_l} with velocity $\mathbf{v}_{R_l}(t)$. Here v_{C_i} and v_{R_l} are two invariants, while the directions of $\mathbf{v}_{C_i}(t)$ and $\mathbf{v}_{R_l}(t)$ change constantly. The relative velocity between the two individuals is $\mathbf{u}_{C_i-R_l}(t) \equiv \mathbf{v}_{R_l}(t) - \mathbf{v}_{C_i}(t)$, with a relative speed of $u_{C_i-R_l}(t)$. Then, $u_{C_i-R_l}(t)^2 = v_{C_i}^2 + v_{R_l}^2 - 2v_{C_i} \cdot v_{R_l} \cdot \cos \theta_{C_i-R_l}(t)$, where $\theta_{C_i-R_l}(t)$ represents the angle between $\mathbf{v}_{C_i}(t)$ and $\mathbf{v}_{R_l}(t)$. This system is homogeneous, thus, $\overline{\cos \theta_{C_i-R_l}} = 0$, where the overline stands for the temporal average. Then, we obtain the average relative speed between the C_i and R_l individuals: $\overline{u_{C_i-R_l}} = \sqrt{v_{C_i}^2 + v_{R_l}^2}$. Likewise, the average relative speed between the C_i and C_j individuals is $\overline{u_{C_i-C_j}} = \sqrt{v_{C_i}^2 + v_{C_j}^2}$. Evidently, $\overline{u_{C_i-C_i}} = \sqrt{2}v_{C_i}$. Meanwhile, the concentrations of species C_i and R_l in a 2-D square system with a length of L are $n_{C_i} = C_i/L^2$ and $n_{R_l} = R_l/L^2$, while those of the freely wandering C_i and R_l are $n_{C_i}^{(F)} = C_i^{(F)}/L^2$ and $n_{R_l}^{(F)} = R_l^{(F)}/L^2$.

Then, we use the mean-field approximation to calculate the encounter rates a_{il} and a'_{ij} in the well-mixed system. In particular, we estimate a_{il} by tracking a randomly chosen consumer individual from species C_i and counting its encounter frequency with the freely wandering individuals from resource species R_l (see [Appendix-fig. 1](#)). At any moment, the consumer individual may form a chasing pair with a R_l individual within a radius of $r_{il}^{(C)}$ (see [Fig. 1A](#)). Over a time interval of Δt , the number of encounters between the consumer individual and R_l individuals can be estimated by the encounter area and the concentration n_{R_l} , which takes the value of $2r_{il}^{(C)} n_{R_l}^{(F)} \overline{u_{C_i-R_l}} \Delta t$ (see [Appendix-fig. 1](#)). Combined with $n_{R_l}^{(F)} = R_l^{(F)}/L^2$, for all freely wandering C_i individuals, the number of their encounters with $R_l^{(F)}$ during interval Δt is $\frac{2r_{il}^{(C)} \overline{u_{C_i-R_l}} C_i^{(F)} R_l^{(F)}}{L^2} \Delta t$. Meanwhile, in the ODEs, this corresponds to $a_{il} C_i^{(F)} R_l^{(F)} \Delta t$. Comparing both terms above, for chasing pairs, we have

$$a_{il} = 2r_{il}^{(C)} L^{-2} \overline{u_{C_i-R_l}} = 2r_{il}^{(C)} L^{-2} \sqrt{v_{C_i}^2 + v_{R_l}^2}. \text{ Likewise, for interference pairs, we obtain } a'_{ij} = 2r_{ij}^{(I)} L^{-2} \overline{u_{C_i-C_j}} = 2r_{ij}^{(I)} L^{-2} \sqrt{v_{C_i}^2 + v_{C_j}^2}. \text{ In particular, } a'_{ii} = 2\sqrt{2}v_{C_i} r_{ii}^{(I)} L^{-2}.$$

Stochastic simulations

To investigate the impact of stochasticity on species coexistence, we use the stochastic simulation algorithm (SSA) ([Gillespie, 2007](#)) and individual-based modeling (IBM) ([Vetsigian, 2017](#); [Grimm and Railsback, 2013](#)) in simulating the stochastic process. In the SSA studies, we follow the standard Gillespie algorithm and simulation procedures ([Gillespie, 2007](#)).

In the IBM studies, we consider a 2D square system with a length of L and periodic boundary conditions. In the case of $S_C = 2$ and $S_R = 1$, consumers of species $C_i (i = 1, 2)$ move at speed v_{C_i} , while the resources move at speed v_R . The unit length is $\Delta l = 1$, and all individuals move probabilistically. Specifically, when Δt is small so that $v_{C_i} \Delta t \ll 1$, C_i individuals jump a unit length with the probability $v_{C_i} \Delta t$. In practice, we simulate the temporal evolution of the model system following the procedures below.

Initialization.

We choose the initial position for each individual randomly from a uniform distribution in the square space, which rounds to the nearest integer point in the x - y coordinates.

Moving.

We choose the destination of a movement randomly from four directions (x -positive, x -negative, y -positive, y -negative) following a uniform distribution. The consumers and resources jump a unit length with probabilities $v_{C_i} \Delta t$ and $v_R \Delta t$ respectively.

Forming pairs.

When a C_i individual and a resource individual get close in space within a distance of $r^{(C)}$, they form a chasing pair. Meanwhile, when two consumer individuals C_i and C_j stand within a distance of $r^{(I)}$, they form an interference pair.

Dissociation.

We update the system with a small time step Δt so that $d_i \Delta t, k_i \Delta t, d'_{ij} \Delta t \ll 1$ ($i, j = 1, 2$). In practice, a random number ς is sampled from a uniform distribution between 0 and 1, i.e., $U(0, 1)$. If $\varsigma < d_i \Delta t$ or $\varsigma < d'_{ij} \Delta t$, then the chasing pair or interference pair dissociates into two separated individuals. One occupies the original position, while the other individual moves just out of the encounter radius in a uniformly distributed random angle. For a chasing pair, if $d_i \Delta t < \varsigma < (d_i + k_i) \Delta t$, then, the consumer catches the resource, and the biomass of the resource flows into the consumer populations (updated according to the birth procedure), while the consumer individual occupies the original position. Finally, if $\varsigma > (d_i + k_i) \Delta t$ or $\varsigma > d'_{ij} \Delta t$, the chasing pair or interference pair maintains the current status.

Birth and death.

For each species, we use two separate counters with decimal precision to record the contributions of the birth and death processes, both of which accumulate in each time step. The incremental integer part of the counter will trigger updates in this run. Specifically, a newborn would join the system following the initialization procedure in a birth action, while an unfortunate target would be randomly chosen from the living population in a death action.

Data and materials availability

All data and codes for this paper have been deposited on GitHub: <https://github.com/SchordK/Intraspecific-predator-interference-promotes-biodiversity-in-ecosystems> .

Author contributions

X.W. conceived and designed the project; J.K., S.Z., Y.N., F.Z., and X. W. performed research; J.K. and X. W. analyzed data and wrote the paper.

Competing interests

The authors declares no competing interests.

Acknowledgements

We thank Roy Kishony, Eric D. Kelsic and Yang-Yu Liu for helpful discussions. This work was supported by National Natural Science Foundation of China (Grant No.12004443), Guangzhou Municipal Innovation Fund (Grant No.202102020284) and the Hundred Talents Program of Sun Yat-sen University.

Appendix I

The classical proof of Competitive Exclusion Principle (CEP)

In the 1960s, MacArthur (MacArthur and Levins, 1964 [↗](#)) and Levin (Levin, 1970 [↗](#)) put forward the classical mathematical proof of CEP. We rephrase their idea in the simple case of $S_C = 2$ and $S_R = 1$, i.e., two consumer species C_1 and C_2 competing for one resource species R . In practice, this proof can be generalized into higher dimensions with several consumer and resource species. The population dynamics of the system can be described as follows:

$$\begin{cases} \dot{C}_i = C_i(f_i(R) - D_i), & i = 1, 2; \\ \dot{R} = g(R, C_1, C_2). \end{cases} \quad (S1)$$

Here C_i and R represent the population abundances of consumers and resources, respectively, while the functional forms of $f_i(R)$ and $g(R, C_1, C_2)$ are unspecific. D_i stands for the mortality rate of the species C_i . If all consumer species can coexist at steady state, then $f_i(R)/D_i = 1$ ($i = 1, 2$). In a 2-D representation, this requires that three lines $y = f_i(R)/D_i$ ($i = 1, 2$) and $y = 1$ share a common point, which is commonly impossible unless the model parameters satisfy special constraint (sets of Lebesgue measure zero). In a 3-D representation, the two planes corresponding to $f_i(R)/D_i = 1$ ($i = 1, 2$) are parallel, and hence do not share a common point (see Ref. (Wang and Liu, 2020 [↗](#)) for details).

Appendix II

Comparison of the functional response with Beddington-DeAngelis (B-D) model

A B-D model

In 1975, Beddington proposed a mathematical model (Beddington, 1975) to describe the influence of predator interference on the functional response with hand-waving derivations. In the same year, DeAngelis and his colleagues considered a related question and put forward a similar model (DeAngelis et al., 1975). Essentially, both models are phenomenological, and they were called B-D model in the subsequent studies. In practice, the B-D model can be extended into scenarios involving different types of pairwise encounters with Beddington's modelling method. In this section, we systematically compare the functional response in B-D model with that of our mechanistic model in all the relevant scenarios.

Recalling Beddington's analysis, the model (Beddington, 1975) consists of one consumer species C and one resource species R ($S_C = 1, S_R = 1$). In a well-mixed system, an individual consumer meets a resource with rate a , while encounters another consumer with rate a' . There are two other phenomenological parameters in this model, namely, the handling time t_h and the wasting time t_w . Both can be determined by specifying the scenario and using statistical physics modeling analysis. In fact, Beddington analyzed the searching efficiency Ξ_{B-D} rather than the functional response F_{B-D} , yet both can be reciprocally derived with $\Xi_{B-D} \equiv F_{B-D}/R$. Here R stands for the population abundance of the resources, and the specific form of Ξ_{B-D} is (Beddington, 1975):

$$\Xi_{B-D}(R, C) = \frac{a}{1 + at_h R + a't_w C'}, \quad (S2)$$

where $C' = C - 1$, and C stands for the population abundance of the consumes. Generally, $C \gg 1$, and thus $C' \approx C$.

B Scenario involving only chasing pairs

Here we consider the scenario involving only chasing pair for the simple case with one consumer species C and one resource species R ($S_C = 1, S_R = 1$). When an individual consumer is chasing a resource, they form a chasing pair:



where the superscript “(F)” stands for populations that are freely wandering, and “(+)” signifies gaining biomass (we count $C^{(F)}(+)$ as $C^{(F)}$). $C^{(P)} \vee R^{(P)}$ represents chasing pair (where “(P)” signifies pair), denoted as x . a , d and k stand for encounter rate, escape rate and capture rate, respectively. Hence, the total number of consumers and resources are $C \equiv C^{(F)} + x$ and $R \equiv R^{(F)} + x$. Then, the population dynamics of the system follows:

$$\begin{cases} \dot{x} = aC^{(F)}R^{(F)} - (k + d)x, \\ \dot{C} = wx - DC, \\ \dot{R} = g(R, x, C). \end{cases} \quad (S3)$$

Here the functional form of $g(R, x, C)$ is unspecific, while D and w represent the mortality rate of the consumer species and biomass conversion ratio (Wang and Liu, 2020), respectively. Since consumption process is generically much faster than the birth/death process, in deriving the

functional response, the consumption process is supposed to be in fast equilibrium (i.e., $\dot{x} = 0$). Then, we can solve for x with:

$$x^2 - (R + C + K)x + RC = 0, \quad (S4)$$

where $K = \frac{k+d}{a}$, and then,

$$x = \frac{2RC}{(R + C + K)} \frac{1}{(1 + \sqrt{1 - \frac{4RC}{(R+C+K)^2}})}. \quad (S5)$$

By definition, the functional response and searching efficiency are:

$$F_{CP}(R, C) = \frac{kx}{C}, \quad (S6a)$$

$$\Xi_{CP}(R, C) = \frac{kx}{RC}. \quad (S6b)$$

Hence, we obtain the functional response and searching efficiency in this chasing-pair scenario:

$$F_{CP}(R, C)_{(1)} = k \frac{(R + C + K)}{2C} \left(1 - \sqrt{1 - \frac{4RC}{(R + C + K)^2}} \right), \quad (S7a)$$

$$\Xi_{CP}(R, C)_{(1)} = k \frac{(R + C + K)}{2RC} \left(1 - \sqrt{1 - \frac{4RC}{(R + C + K)^2}} \right). \quad (S7b)$$

Since $\frac{4RC}{(R+C+K)^2} < 4\frac{C}{R} \ll 1$, using first order approximations in [Eq. S7](#), we obtain

$\sqrt{1 - \frac{4RC}{(R+C+K)^2}} \approx 1 - \frac{2RC}{(R+C+K)^2}$. Then the functional response and searching efficiency are:

$$F_{CP}(R, C)_{(2)} = k \frac{R}{R + C + K}, \quad (S8a)$$

$$\Xi_{CP}(R, C)_{(2)} = \frac{k}{R + C + K}. \quad (S8b)$$

Evidently, there is no predator interference within the chasing-pair scenario, yet the functional response form is identical to the B-D model involving intraspecific interference (see [Eq. S2](#)). Meanwhile, using first order approximations in the denominator of [Eq. S5](#), we have

$$x \approx \frac{RC}{(R+C+K) - \frac{RC}{(R+C+K)}}.$$

Hence,

$$F_{CP}(R, C)_{(3)} = k \frac{R}{(R+C+K) - \frac{RC}{(R+C+K)}}, \quad (S9a)$$

$$\Xi_{CP}(R, C)_{(3)} = \frac{k}{(R+C+K) - \frac{RC}{(R+C+K)}}. \quad (S9b)$$

In the case that $R \gg C$, then $R \gg C > x = R - R^{(F)}$. By applying $R \approx R^{(F)}$ in Eq. S3 [\[1\]](#), we obtain

$$x \approx \frac{RC}{R+K}. \text{ Then,}$$

$$F_{CP}(R, C)_{(4)} = k \frac{R}{R+K}, \quad (S10a)$$

$$\Xi_{CP}(R, C)_{(4)} = \frac{k}{R+K}. \quad (S10b)$$

To compare these functional responses with that of the B-D model, we determine the parameters t_h and t_w in the B-D model by calculating their ensemble average values in a stochastic framework.

Using the properties of waiting time distribution in the Poisson process, we obtain $\langle t_h \rangle = \frac{1}{k}$

and $\langle t_w \rangle = \frac{1}{d'}$ (in the chasing-pair scenario, $a' = 0$). By substituting these calculations into Eq. S2 [\[1\]](#), we have:

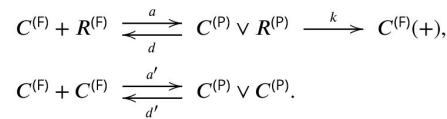
$$\Xi_{CP}^{B-D}(R, C) = \frac{a}{1 + Ra/k} = \frac{k}{k/a + R}, \quad (S11a)$$

$$F_{CP}^{B-D}(R, C) = \frac{kR}{k/a + R}. \quad (S11b)$$

In the special case with $d = 0$ and $R \gg C$, the B-D model is consistent with our mechanistic model: $\Xi_{B-D}(R, C) = \Xi_{CP}(R, C)_{(4)}$. Outside the special region, however, the discrepancy can be considerably large (see **Appendix-fig. 2A-C** [\[1\]](#) for the comparison).

C Scenario involving chasing pairs and intraspecific interference

Here we consider the scenario with additional involvement of intraspecific interference in the simple case of $S_C = 1$ and $S_R = 1$:



Here $C^{(P)} \vee C^{(P)}$ stands for the intraspecific predator interference pair, denoted as y ; a' and d' represent the encounter rate and separation rate of the interference pair, respectively. Then, the total population of consumers and resources are $C \equiv C^{(F)} + x + 2y$ and $R \equiv R^{(F)} + x$. Hence the population dynamics of the consumers and resources can be described as follows:

$$\begin{cases} \dot{x} = aC^{(F)}R^{(F)} - (k + d)x, \\ \dot{y} = a'[C^{(F)}]^2 - d'y, \\ \dot{C} = w k x - D C, \\ \dot{R} = g(R, x, C). \end{cases} \quad (S12)$$

The consumption process and interference process are supposed to be in fast equilibrium (i.e., $\dot{x} = 0, \dot{y} = 0$), then we can solve for x with:

$$x^3 + \phi_2 x^2 + \phi_1 x + \phi_0 = 0, \quad (S13)$$

where $\phi_0 = -C R^2$, $\phi_1 = 2C R + K R + R^2$, $\phi_2 = 2\beta K^2 - K - C - 2R$, with $\beta = a'/d'$. The discriminant of **Eq. S13** (denoted as Λ) is:

$$\Lambda = -4\psi^3 - 27\varphi^2, \quad (S14)$$

with $\Psi = \phi_1 - (\phi_2)^2/3$ and $\phi = \phi_0 - \phi_1\phi_2/3 + 2(\phi_2)^3/27$. When $\Lambda < 0$, there are one real solution $x_{(1)}$ and two complex solutions $x_{(2)}, x_{(3)}$, which are:

$$x_{(1)} = \theta_1 + \theta_2 - \phi_2/3, x_{(2)} = \omega\theta_1 + \omega^2\theta_2 - \phi_2/3, x_{(3)} = \omega^2\theta_1 + \omega\theta_2 - \phi_2/3, \quad (S15)$$

where $\omega = -1/2 + i\sqrt{3}/2$ (i stands for the imaginary unit), $\theta_1 = (-\varphi/2 + \sqrt{-\Lambda/108})^{1/3}$, and $\theta_2 = (-\varphi/2 - \sqrt{-\Lambda/108})^{1/3}$. On the other hand, when $\Lambda > 0$, there are three real solutions $x_{(1)}, x_{(2)}$, and $x_{(3)}$, which are:

$$x_{(1)} = \psi' \cos \varphi' - \phi_2/3, x_{(2)} = \psi' \cos(\varphi' + \frac{2\pi}{3}) - \phi_2/3, x_{(3)} = \psi' \cos(\varphi' + \frac{4\pi}{3}) - \phi_2/3, \quad (S16)$$

where $\psi^* = (-4\psi/3)^{1/2}$ and $\phi^* = \arccos(-(-\psi/3)^{3/2}\phi/2)/3$. Note that $x \in [0, \min(R, C)]$, then we obtain the exact feasible solution of x (denoted as x_{ext}), and hence the functional response and searching efficiency are:

$$\mathcal{F}_{intra}(R, C)_{(1)} = \frac{kx_{ext}}{C}, \quad (S17a)$$

$$\Xi_{intra}(R, C)_{(1)} = \frac{kx_{ext}}{RC}. \quad (S17b)$$

In the case of $R \gg C$, then $R - R^{(F)} = x < C \ll R$, and thus $R^{(F)} \approx R$. Still, the consumption process is supposed to be in fast equilibrium (i.e., $\dot{x} = 0, \dot{y} = 0$), and then we obtain:

$$x \approx \frac{RC}{\sqrt{[\frac{1}{2}(K + R)]^2 + 2C\beta K^2 + \frac{1}{2}(K + R)}}. \quad (S18)$$

Consequently,

$$\mathcal{F}_{intra}(R, C)_{(2)} = k \frac{R}{\sqrt{[\frac{1}{2}(K + R)]^2 + 2C\beta K^2 + \frac{1}{2}(K + R)}}, \quad (S19a)$$

$$\Xi_{intra}(R, C)_{(2)} = k \frac{1}{\sqrt{[\frac{1}{2}(K + R)]^2 + 2C\beta K^2 + \frac{1}{2}(K + R)}}. \quad (S19b)$$

When $\beta \ll \frac{1}{8C}$ or $8\beta C/(1+R/K)^2 \ll 1$, using first order approximations in the denominator of **Eq. S18**, we have:

$$x \approx \frac{RC}{(K + R) + \frac{2K}{(1+R/K)}\beta C}, \quad (S20)$$

and then,

$$\mathcal{F}_{\text{intra}}(R, C)_{(3)} = k \frac{R}{(K + R) + \frac{2K}{(1+R/K)}\beta C}, \quad (\text{S21a})$$

$$\Xi_{\text{intra}}(R, C)_{(3)} = k \frac{1}{(K + R) + \frac{2K}{(1+R/K)}\beta C}. \quad (\text{S21b})$$

In the case that $8\beta C/(1 + R/K)^2 \gg 1$, using first order approximations in [Eq. S18](#), we obtain:

$$x \approx \frac{RC}{K\sqrt{2C\beta} + \frac{(K+R)^2}{8K\sqrt{2C\beta}} + \frac{1}{2}(K + R)}, \quad (\text{S22})$$

and thus,

$$\mathcal{F}_{\text{intra}}(R, C)_{(4)} = k \frac{R}{K\sqrt{2C\beta} + \frac{(K+R)^2}{8K\sqrt{2C\beta}} + \frac{1}{2}(K + R)}, \quad (\text{S23a})$$

$$\Xi_{\text{intra}}(R, C)_{(4)} = k \frac{1}{K\sqrt{2C\beta} + \frac{(K+R)^2}{8K\sqrt{2C\beta}} + \frac{1}{2}(K + R)}. \quad (\text{S23b})$$

Meanwhile, the B-D model only fits to the cases with $d = 0$. By calculating the average values of t_h and t_w in the stochastic framework, we have $\langle t_h \rangle = \frac{1}{k}$, $\langle t_w \rangle = \frac{1}{d'}$. Thus, we obtain the searching efficiency and functional response in the B-D model:

$$\Xi_{\text{intra}}^{\text{B-D}}(R, C) = \frac{a}{1 + \frac{a}{k}R + \frac{a'}{d'}C} = \frac{a}{1 + R/K|_{d=0} + \beta C}, \quad (\text{S24a})$$

$$F_{\text{intra}}^{\text{B-D}}(R, C) = \frac{aR}{1 + R/K|_{d=0} + \beta C}. \quad (\text{S24b})$$

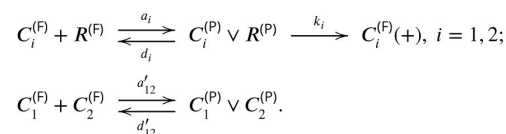
Overall, the searching efficiency (and the functional response) of the B-D model is quite different from either the rigorous form $\Xi_{\text{intra}}(R, C)_{(1)}$, the quasi rigorous form $\Xi_{\text{intra}}(R, C)_{(2)}$, or the more simplified forms $\Xi_{\text{intra}}(R, C)_{(3)}$ and $\Xi_{\text{intra}}(R, C)_{(4)}$ (**Appendix-fig. 2D-F**). Still, there is a region where the discrepancies can be small, namely $d \approx 0$ and $R \gg C$ (**Appendix-fig. 2D-F**). Intuitively, when $\beta \ll \frac{1}{8C}$ and $d = 0$, then $\Xi_{\text{intra}}(R, C)_{(3)} = \frac{a}{1 + \frac{a}{k}R + \frac{2}{(1+R/K)\beta C_i}}$. Consequently, if

$$R/K_i = x/C_i^{(F)} < 1, \text{ then } \frac{2}{(1+R/K_i)} \in [1, 2]. \text{ In this case, the difference between } \Xi_{\text{intra}}^{\text{B-D}}(R, C) \text{ and } \Xi_{\text{intra}}(R, C)_{(3)} \text{ is small.}$$

In fact, the above analysis also applies to cases with more than one types of consumer species (i.e., for cases with $S_C > 1$).

D Scenario involving chasing pairs and interspecific interference

Next, we consider the scenario involving chasing pairs and interspecific interference in the case of $S_C = 2$ and $S_R = 1$:



Here $C_1^{(P)} \vee C_2^{(P)}$ stands for the interspecific interference pair, denoted as z ; a'_{12} and d'_{12} represent the encounter rate and separation rate of the interference pair, respectively. Then, the total population of consumers and resources are $C_i \equiv C_i^{(F)} + x_i + z$ and $R \equiv R^{(F)} + x_1 + x_2$. The population dynamics of the consumers and resources follows:

$$\begin{cases} \dot{x}_i = a_i C_i^{(F)} R^{(F)} - (k_i + d_i) x_i, i = 1, 2; \\ \dot{z} = a'_{12} C_1^{(F)} C_2^{(F)} - d'_{12} z, \\ \dot{C}_i = w_i k_i x_i - D_i C_i, \\ \dot{R} = g(R, x_1, x_2, C_1, C_2). \end{cases} \quad (\text{S25})$$

where the functional form of $g(R, x_1, x_2, C_1, C_2)$ is unspecific, while D_i and w_i represents the mortality rates of the two consumers species and biomass conversion ratios. Still, the consumption/interference process is supposed to be in fast equilibrium, i.e., $\dot{x}_i = 0$, $\dot{z} = 0$. In the case that $R \gg C_1 + C_2 > x_1 + x_2$, by applying $R^{(F)} \approx R$, we obtain:

$$x_1 \approx \frac{2C_1(R/K_2 + 1)R/K_1}{\sqrt{[\gamma(C_2 - C_1) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)]^2 + 4\gamma C_1(\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1) + \gamma(C_2 - C_1) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)}}, \quad (S26a)$$

$$x_2 \approx \frac{2C_2(R/K_1 + 1)R/K_2}{\sqrt{[\gamma(C_1 - C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)]^2 + 4\gamma C_2(\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1) + \gamma(C_1 - C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)}}. \quad (S26b)$$

Then, the searching efficiencies and functional responses are:

$$\Xi_1^{\text{inter}}(RC_1, C_2)_{(0)} = \frac{2k_1(R/K_2 + 1)/K_1}{\gamma(C_2 - C_1) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1) + \sqrt{[\gamma(C_2 - C_1) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)]^2 + 4\gamma C_1(\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)}}, \quad (S27a)$$

$$\Xi_2^{\text{inter}}(RC_1, C_2)_{(0)} = \frac{2k_2(R/K_1 + 1)/K_2}{\gamma(C_1 - C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1) + \sqrt{[\gamma(C_1 - C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)]^2 + 4\gamma C_2(\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)}}, \quad (S27b)$$

$$\mathcal{F}_1^{\text{inter}}(RC_1, C_2)_{(0)} = \frac{2k_1(R/K_2 + 1)R/K_1}{\gamma(C_2 - C_1) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1) + \sqrt{[\gamma(C_2 - C_1) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)]^2 + 4\gamma C_1(\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)}}, \quad (S27c)$$

$$\mathcal{F}_2^{\text{inter}}(RC_1, C_2)_{(0)} = \frac{2k_2(R/K_1 + 1)R/K_2}{\gamma(C_1 - C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1) + \sqrt{[\gamma(C_1 - C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)]^2 + 4\gamma C_2(\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)}}. \quad (S27d)$$

Since $\frac{4\gamma^2 C_1 C_2}{[\gamma C_1 + \gamma C_2 + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)]^2} < 1$, by applying first order approximation to the denominator of Eq. S26 [↗](#), we obtain:

$$x_1 \approx \frac{C_1 R}{(R + K_1) + \frac{\gamma K_1 K_2 C_2}{(R + K_2)} - \frac{\gamma^2 K_1 K_2 C_1 C_2}{[\gamma(C_1 + C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)](R + K_2)}}, \quad (\text{S28a})$$

$$x_2 \approx \frac{C_2 R}{(R + K_2) + \frac{\gamma K_1 K_2 C_1}{(R + K_1)} - \frac{\gamma^2 K_1 K_2 C_1 C_2}{[\gamma(C_1 + C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)](R + K_1)}}, \quad (\text{S28b})$$

and the searching efficiencies and functional responses are:

$$\Xi_1^{\text{inter}}(R, C_1, C_2)_{(2)} = \frac{k_1}{(R + K_1) + \frac{\gamma K_1 K_2 C_2}{(R + K_2)} - \frac{\gamma^2 K_1 K_2 C_1 C_2}{[\gamma(C_1 + C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)](R + K_2)}}, \quad (\text{S29a})$$

$$\Xi_2^{\text{inter}}(R, C_1, C_2)_{(2)} = \frac{k_2}{(R + K_2) + \frac{\gamma K_1 K_2 C_1}{(R + K_1)} - \frac{\gamma^2 K_1 K_2 C_1 C_2}{[\gamma(C_1 + C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)](R + K_1)}}, \quad (\text{S29b})$$

$$F_1^{\text{inter}}(R, C_1, C_2)_{(2)} = \frac{k_1 R}{(R + K_1) + \frac{\gamma K_1 K_2 C_2}{(R + K_2)} - \frac{\gamma^2 K_1 K_2 C_1 C_2}{[\gamma(C_1 + C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)](R + K_2)}}, \quad (\text{S29c})$$

$$F_2^{\text{inter}}(R, C_1, C_2)_{(2)} = \frac{k_2 R}{(R + K_2) + \frac{\gamma K_1 K_2 C_1}{(R + K_1)} - \frac{\gamma^2 K_1 K_2 C_1 C_2}{[\gamma(C_1 + C_2) + (\frac{R}{K_1} + 1)(\frac{R}{K_2} + 1)](R + K_1)}}. \quad (\text{S29d})$$

Likewise, the B-D model only fits to cases with $d = 0$. By calculating the average values in a stochastic framework, we obtain $\langle t_h^i \rangle = \frac{1}{k_i}$, $\langle t_w^i \rangle = \frac{1}{d'_{12}}$ ($i = 1, 2$). Then, we obtain the searching efficiencies in the B-D model:

$$\Xi_1^{\text{B-D (inter)}}(R, C_1, C_2) = \frac{a_1}{1 + \frac{a_1}{k_1} R + \frac{a'_{12}}{d'_{12}} C_2} = \frac{a_1}{1 + R/K_1|_{d=0} + \gamma C_2}, \quad (\text{S30a})$$

$$\Xi_2^{\text{B-D (inter)}}(R, C_1, C_2) = \frac{a_2}{1 + \frac{a_2}{k_2} R + \frac{a'_{12}}{d'_{12}} C_1} = \frac{a_2}{1 + R/K_2|_{d=0} + \gamma C_1}. \quad (\text{S30b})$$

Consequently, the functional responses in the B-D model are:

$$\mathcal{F}_1^{\text{B-D (inter)}}(R, C_1, C_2) = \frac{a_1 R}{1 + R/K_1|_{d=0} + \gamma C_2}, \quad (\text{S31a})$$

$$\mathcal{F}_2^{\text{B-D (inter)}}(R, C_1, C_2) = \frac{a_2 R}{1 + R/K_2|_{d=0} + \gamma C_1}. \quad (\text{S31b})$$

Evidently, the searching efficiencies in the B-D model are overall different from either the quasi rigorous form $\Xi_i(R, C_1, C_2)_1$, or the simplified form $\Xi_i(R, C_1, C_2)_2$ (**Appendix-fig. 2G-I** [↗](#)). Still, the discrepancy can be small when $d \approx 0$ and $R \gg C$ (**Appendix-fig. 2G-I** [↗](#)). Intuitively, when $\gamma \ll \min(C_1^{-1}, C_2^{-1})$ we have:

$$\Xi_1^{\text{inter}}(R, C_1, C_2)_{(2)} \approx \frac{a_1}{(1 + \frac{a_1}{k_1} R) + \frac{\gamma C_2}{R/K_2+1}}, \quad (\text{S32a})$$

$$\Xi_2^{\text{inter}}(R, C_1, C_2)_{(2)} \approx \frac{a_2}{(1 + \frac{a_2}{k_2} R) + \frac{\gamma C_1}{R/K_1+1}}. \quad (\text{S32b})$$

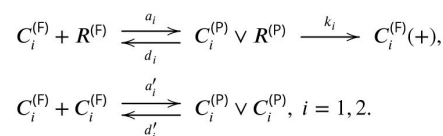
Thus, if $R/K_i = x_i/C_i^{(F)} < 1$ ($i = 1, 2$), then $\frac{1}{1+R/K_i} \in [0.5, 1]$. In this case, the difference between $\Xi_i^{\text{B-D (inter)}}(R, C_1, C_2)$ and $\Xi_i^{\text{inter}}(R, C_1, C_2)_{(2)}$ is small.

Appendix III

Scenario involving chasing pairs and intraspecific interference

A Two consumers species competing for one resource species

We consider the scenario involving chasing pairs and intraspecific interference in the simple case of $S_C = 2$ and $S_R = 1$:



Here, the variables and parameters are just extended from the case of $S_C = 1$ and $S_R = 1$ (see **Appendix II** [↗](#)). The total number of consumers and resources are $C_i \equiv C_i^{(F)} + x_i + 2y_i$ and

$R \equiv R^{(F)} + \sum_{i=1}^2 x_i$. Then, the population dynamics of the consumers and resources can be described as follows:

$$\begin{cases} \dot{x}_i = a_i C_i^{(F)} R^{(F)} - (k_i + d_i) x_i, i = 1, 2; \\ \dot{y}_i = a'_i [C_i^{(F)}]^2 - d'_i y_i, \\ \dot{C}_i = w_i k_i x_i - D_i C_i, \\ \dot{R} = g(R, x_1, x_2, C_1, C_2). \end{cases} \quad (\text{S33})$$

The functional form of $g(R, x_1, x_2, C_1, C_2)$ is unspecified. For simplicity, we limit our analysis to abiotic resources, while all results generically apply to biotic resources. Besides, we define $K_i \equiv (d_i + k_i)/a_i$, $\alpha_i \equiv D_i/(w_i k_i)$, and $\beta_i \equiv a'_i/d'_i (i = 1, 2)$. At steady state, from $\dot{x}_i = 0$, $\dot{y}_i = 0$, we have:

$$\begin{cases} x_i = C_i^{(F)} R^{(F)} / K_i, i = 1, 2; \\ y_i = \beta_i [C_i^{(F)}]^2. \end{cases} \quad (\text{S34})$$

Note that $C_i \equiv C_i^{(F)} + x_i + 2y_i$ and $R \equiv R^{(F)} + \sum_{i=1}^2 x_i$. Then,

$$R^{(F)} = R / (1 + C_1^{(F)} / K_1 + C_2^{(F)} / K_2), \quad (\text{S35a})$$

$$C_i = C_i^{(F)} + R^{(F)} C_i^{(F)} / K_i + 2\beta_i [C_i^{(F)}]^2, i = 1, 2. \quad (\text{S35b})$$

By substituting **Eq. S35a** [↗](#) into **Eq. S35b** [↗](#), we have:

$$C_2^{(F)} = \frac{K_2}{K_1} \left[\frac{R C_1^{(F)}}{C_1 - C_1^{(F)} - 2\beta_1 [C_1^{(F)}]^2} - K_1 - C_1^{(F)} \right], \quad (\text{S36a})$$

$$(C_2 - C_2^{(F)} - 2\beta_2 [C_2^{(F)}]^2) (1 + C_1^{(F)} / K_1 + C_2^{(F)} / K_2) = R C_2^{(F)} / K_2. \quad (\text{S36b})$$

Then, we can present $C_i^{(F)}$ with C_1, C_2 and $R (i = 1, 2)$. By further combining with **Eqs. S34** [↗](#), **S35a** [↗](#) and **S36a** [↗](#), we express $R^{(F)}$, x_i , and y_i using C_1, C_2 and R . In particular, for x_i , we have:

$$x_i = u_i(R, C_1, C_2), i = 1, 2. \quad (\text{S37})$$

If all species coexist, then the steady-state equations of $\dot{C}_i = 0$ ($i = 1, 2$) and $\dot{R} = 0$ are:

$$\begin{cases} \Omega_1(R, C_1, C_2) - D_1 = 0, \\ \Omega_2(R, C_1, C_2) - D_2 = 0, \\ G(R, C_1, C_2) = 0, \end{cases} \quad (\text{S38})$$

where $G(R, C_1, C_2) \approx g(R, u_1(R, C_1, C_2), u_2(R, C_1, C_2), C_1, C_2)$, and $\Omega_i(R, C_1, C_2) \equiv \frac{w_i k_i}{C_i} u_i(R, C_1, C_2)$.

In practice, **Eq. S38** corresponds to three unparallel surfaces, which share a common point (**Fig. 1H** and **Appendix-fig. 3G**). Importantly, the fixed point can be stable, and hence two consumer species may coexist at constant population densities.

1 Stability analysis of the fixed-point solution

We use linear stability analysis to study the local stability of the fixed point. Specifically, for an arbitrary fixed point $E(x_1, x_2, y_1, y_2, C_1, C_2, R)$, only when all the eigenvalues (defined as $\lambda_i, i = 1, \dots, 7$) of the Jacobian matrix at point E own negative real parts would the point be locally stable.

To investigate whether there exists a non-zero measure parameter region for species coexistence, we set D_i ($i = 1, 2$) to be the only parameter that varies with species C_1 and C_2 , and then $\Delta \equiv (D_1 - D_2)/D_2$ reflects the competitive difference between the two consumer species. As shown in **Appendix-fig. 4B**, the region below the blue surface and above the red surface corresponds to stable coexistence. Thus, there exists a non-zero measure parameter region to promote species coexistence, which breaks CEP.

2 Analytical solutions of the species abundances at steady state

At steady state, since $\dot{x}_i = \dot{y}_i = \dot{C}_i = 0$ ($i = 1, 2$), then,

$$\begin{cases} x_i = \alpha_i C_i, \\ C_i^{(F)} = K_i \alpha_i C_i / R^{(F)}, \\ y_i = \beta_i (K_i \alpha_i C_i)^2 [R^{(F)}]^{-2}. \end{cases} \quad (\text{S39})$$

Meanwhile, $C_i = C_i^{(F)} + x_i + 2y_i$, and $C_i R > 0$ ($i = 1, 2$). Then, we have:

$$C_i = \frac{(1 - \alpha_i)[R^{(F)}]^2 - K_i \alpha_i R^{(F)}}{2\beta_i (K_i \alpha_i)^2}. \quad (\text{S40})$$

If the resource species owns a much larger population abundance than the consumers (i.e., $R \gg C_1 + C_2$), then $R \gg x_1 + x_2$, and $R^{(F)} \approx R$. Thus,

$$C_i = \frac{(1 - \alpha_i)R^2 - K_i \alpha_i R}{2\beta_i (K_i \alpha_i)^2}. \quad (\text{S41})$$

By further assuming that the population dynamics of the resources follow identical construction rule as the MacArthur's consumer-resource model (MacArthur, 1970), we have:

$$g(R, x_1, x_2, C_1, C_2) = \zeta(1 - R/\kappa) - (k_1 x_1 + k_2 x_2), \quad (\text{S42})$$

Since $\dot{R} = 0$, then,

$$R = \frac{-o_1 + \sqrt{o_1^2 + 4o_2\zeta}}{2o_2}, \quad (\text{S43})$$

$$\text{where } o_1 \equiv \frac{\zeta}{\kappa} - \frac{k_1}{2\beta_1 K_1} - \frac{k_2}{2\beta_2 K_2} \text{ and } o_2 \equiv \frac{k_1(1-\alpha_1)}{2\beta_1 \alpha_1 (K_1)^2} + \frac{k_2(1-\alpha_2)}{2\beta_2 \alpha_2 (K_2)^2}.$$

Eqs. S41, S43 are the analytical solutions of species abundances at steady state when $R \gg C_1 + C_2$. As shown in Fig. 1E, the analytical solutions agree well with the numerical results (the exact solutions). To conduct a systematic comparison for different model parameters, we assign D_i ($i = 1, 2$) to be the only parameter varying with species C_1 and C_2 ($D_1 > D_2$), and define $\Delta \equiv (D_1 - D_2)/D_2$ as the competitive difference between the two consumer species. The comparison between analytical solutions and numerical results is shown in Appendix-fig. 3H. Clearly, they are close to each other, exhibiting very good consistency.

Furthermore, we test if the parameter region for species coexistence is predictable using the analytical solutions. Since D_i ($i = 1, 2$) is the only parameter that varies with the two-consumer species, the supremum of the competitive difference tolerated for species coexistence (defined as $\hat{\Delta}$) corresponds to the steady-state solutions that satisfy $R, C_2 > 0$ and $C_1 = 0^+$, where 0^+ stands for the infinitesimal positive number. To calculate the analytical solutions at the upper surface of the coexistence region, where $\Delta = \hat{\Delta}$ and $C_1 = 0^+$, we further combine Eq. S41 and then obtain (note that $R > 0$):

$$R = \frac{K_1 \alpha_1}{1 - \alpha_1}. \quad (\text{S44})$$

Meanwhile, $\alpha_1 = \alpha_2(\Delta + 1)$. Thus, for the upper surface of the coexistence region:

$$\alpha_1 = \alpha_2(\hat{\Delta} + 1). \quad (\text{S45})$$

Combining Eqs. S43–S45, we have:

$$\hat{\Delta} = \frac{1}{\alpha_2(\kappa_1 \varpi + 1)} - 1, \quad (\text{S46})$$

where $\varpi \equiv \frac{1}{2}(\frac{1}{\kappa} - \frac{k_2}{2\zeta\beta_2 K_2}) + \frac{1}{2}\sqrt{(\frac{1}{\kappa} - \frac{k_2}{2\zeta\beta_2 K_2})^2 + 2\frac{k_2(1-\alpha_2)}{\zeta\beta_2\alpha_2(K_2)^2}}$ When $R \gg C_1 + C_2$, the comparison of $\hat{\Delta}$ obtained from analytical solutions with that from numerical results (the exact solutions) are shown in **Appendix-fig. 31**, which overall exhibits good consistency.

B S_C consumers species competing for S_R resources species

Here we consider the scenario involving chasing pairs and intraspecific interference for the generic case with S_C types of consumers and S_R types of resources. Then, the population dynamics of the system can be described as follows:

$$\begin{cases} \dot{x}_{il} = a_{il}C_i^{(F)}R_l^{(F)} - (k_{il} + d_{il})x_{il}, \\ \dot{y}_i = a'_{ii}[C_i^{(F)}]^2 - d'_{ii}y_i, \\ \dot{C}_i = \sum_{l=1}^{S_R} w_{il}k_{il}x_{il} - D_iC_i, \\ \dot{R}_l = g_l(\{R_l\}, \{x_i\}, \{C_i\}), i = 1, \dots, S_C, l = 1, \dots, S_R. \end{cases} \quad (S47)$$

Note that **Eq. S47** is identical with **Eqs. 1–2**, and we use the same variables and parameters as that in the main text. Then, the populations of the consumers and resources are

$$C_i = C_i^{(F)} + \sum_{l=1}^{S_R} x_{il} + 2y_i \quad \text{and} \quad R_l = R_l^{(F)} + \sum_{i=1}^{S_C} x_{il}. \quad \text{For convenience, we define } K_{il} \equiv (d_{il} + k_{il})/a_{il} \equiv D_{il}/(k_{il}w_{il}) \text{ and } \beta_i \equiv a'_{ii}/d'_{ii} \quad (i=1, \dots, S_C, l=1, \dots, S_R).$$

1 Analytical solutions of species abundances at steady state

At steady state, from $\dot{x}_{il} = 0$, $\dot{y}_i = 0$, and $\dot{C}_i = 0$, we have:

$$\begin{cases} x_{il} = C_i^{(F)}R_l^{(F)}/K_{il}, \\ y_i = \beta_i[C_i^{(F)}]^2, \\ C_i = \sum_{l=1}^{S_R} x_{il}/\alpha_{il} = \sum_{l=1}^{S_R} C_i^{(F)}R_l^{(F)}/(K_{il}\alpha_{il}). \end{cases} \quad (S48)$$

Meanwhile $C_i = C_i^{(F)} + \sum_{l=1}^{S_R} x_{il} + 2y_i$, and note that $C_i > 0$, thus,

$$C_i^{(F)} = \frac{1}{2\beta_i}[-1 + \sum_{l=1}^{S_R} (\frac{1}{\alpha_{il}} - 1) \frac{R_l^{(F)}}{K_{il}}]. \quad (S49)$$

Combined with **Eq. S49**, and then,

$$C_i = \sum_{l=1}^{S_R} \frac{R_l^{(F)}}{2\beta_i\alpha_{il}K_{il}}[-1 + \sum_{l'=1}^{S_R} (\frac{1}{\alpha_{il'}} - 1) \frac{R_{l'}^{(F)}}{K_{il'}}]. \quad (S50)$$

We further assume that the specific function of $g_i(\{R_l\}, \{x_i\}, \{C_i\})$ satisfies **Eq. 4**, i.e.,

$$g_l(\{R_l\}, \{x_i\}, \{C_i\}) = \zeta_l(1 - R_l/\kappa_l) - \sum_{i=1}^{S_C} k_{il}x_{il}. \quad (\text{S51})$$

By combining **Eqs. S48**, **S49** and **S51**, we have:

$$\zeta_l(1 - \frac{R_l}{\kappa_l}) = \sum_{i=1}^{S_C} \frac{k_{il}}{2\beta_i K_{il}} [-1 + \sum_{l'=1}^{S_R} (\frac{1}{\alpha_{il'}} - 1) \frac{R_{l'}^{(F)}}{K_{il'}}] R_l^{(F)}. \quad (\text{S52})$$

If the population abundance of each resource species is much more than the total population of all consumers (i.e., $R_l \gg \sum_{i=1}^{S_C} C_i (l = 1, \dots, S_R)$), then $R_l \gg \sum_{i=1}^{S_C} x_{il}$ and $R_l^{(F)} \approx R_l$. Thus,

$$(\frac{\zeta_l}{\kappa_l} - \sum_{i=1}^{S_C} \frac{k_{il}}{2\beta_i K_{il}} + \sum_{l'=1}^{S_R} \sum_{i=1}^{S_C} \frac{k_{il}}{2\beta_i K_{il}} (\frac{1}{\alpha_{il'}} - 1) \frac{R_{l'}}{K_{il'}}) R_l = \zeta_l, \quad (\text{S53})$$

with $l = 1, \dots, S_R$. **Eq. S53** is a set of second-order algebraic differential equations, which is clearly solvable.

In fact, when $S_R = 1$, $S_C \geq 1$, and $R_l \gg \sum_{i=1}^{S_C} C_i (l = 1)$, we can explicitly present the analytical

solution of the steady-state species abundances. To simplify the notations, we omit the “ l ” in the sub-/super-scripts since $S_R = 1$. Then, we have:

$$\begin{cases} R = \frac{-t_1 + \sqrt{t_1^2 + 4t_2\zeta}}{2t_2}, \\ C_i = \frac{1}{2\beta_i \alpha_i K_i} [(\frac{1}{\alpha_i} - 1) \frac{R}{K_i} - 1] R, \quad i = 1, \dots, S_C. \end{cases} \quad (\text{S54})$$

Here $t_1 \equiv \frac{\zeta}{\kappa} - \sum_{i=1}^{S_C} \frac{k_i}{2\beta_i K_i}$ and $t_2 \equiv \sum_{i=1}^{S_C} \frac{k_i(1-\alpha_i)}{2\beta_i \alpha_i (K_i)^2}$.

C Intuitive understanding: an underlying negative feedback loop

Intuitively, how can intraspecific predator interference promote biodiversity? Here we solve this question by considering the case that S_C types of consumers compete for one resource species. The population dynamics of the system are described in **Eqs. S47** and **S51** with $S_R = 1$. To simplify

the notations, we omit the “ l ” in the subscript since $S_R = 1$. The consumption process and interference process are supposed to be in fast equilibrium (i.e., $\dot{x}_i = 0, \dot{y}_i = 0$). Then, we have a set of equations to solve for x_i and y_i given the population size of each species:

$$\begin{cases} x_i = C_i^{(F)} R^{(F)} / K_i, \\ y_i = \beta_i [C_i^{(F)}]^2, \\ C_i = C_i^{(F)} + x_i + 2y_i, \\ R = R^{(F)} + \sum_{i=1}^{S_C} x_i. \end{cases} \quad (\text{S55})$$

In the first three sub-equations of **Eq. S55**, by getting rids of $C_i^{(F)}$ we have,

$$\begin{cases} 2\beta_i [\frac{x_i}{R^{(F)}}]^2 + x_i + \frac{K_i x_i}{R^{(F)}} - C_i = 0, \\ y_i = \frac{1}{2} [C_i - x_i - \frac{x_i}{R^{(F)}}]. \end{cases} \quad (\text{S56})$$

Then, by regarding $R^{(F)}$ as a temporary parameter, we solve for x_i and y_i :

$$\begin{cases} x_i = \frac{2R^{(F)}C_i}{\sqrt{(R^{(F)}+K_i)^2 + 8\beta_i K_i^2 C_i + R^{(F)} + K_i}}, \\ y_i = \frac{1}{2} [C_i - (1 + \frac{K_i}{R^{(F)}})x_i]. \end{cases} \quad (\text{S57})$$

If the total population size of the resources is much larger than that of consumers (i.e., $R \gg \sum_{i=1}^{S_C} C_i$), then $R \gg \sum_{i=1}^{S_C} x_i$ and $R^{(F)} \approx R$, and thus we get the analytical expressions of x_i and y_i :

$$\begin{cases} x_i \approx \frac{2RC_i}{\sqrt{(R+K_i)^2 + 8\beta_i K_i^2 C_i + R + K_i}}, \\ y_i \approx \frac{C_i}{2} [1 - \frac{2(R+K_i)}{\sqrt{(R+K_i)^2 + 8\beta_i K_i^2 C_i + R + K_i}}]. \end{cases} \quad (\text{S58})$$

Note that the fraction of C_i individuals engaged in chasing pairs is x_i/C_i , while that for individuals trapped in intraspecific interference pairs is y_i/C_i . With **Eq. S58**, it is straightforward to obtain these fractions:

$$\begin{cases} x_i/C_i \approx \frac{2R}{\sqrt{(R+K_i)^2 + 8\beta_i K_i^2 C_i + R + K_i}}, \\ y_i/C_i \approx \frac{1}{2} (1 - \frac{2(R+K_i)}{\sqrt{(R+K_i)^2 + 8\beta_i K_i^2 C_i + R + K_i}}). \end{cases} \quad (\text{S59})$$

where both x_i/C_i and y_i/C_i are bivariate functions of R and C_i . From [Eq. S59](#), it is clear that for a given population size of the resource species, y_i/C_i is a monotonously increasing function of C_i , while x_i/C_i is a monotonously decreasing function of C_i . In [Appendix-fig. 15A-B](#), we see that the analytical results are highly consistence with the exact numerical solutions. By definition, the functional response of C_i species is $F \equiv k_i x_i/C_i$, and thus,

$$F(R, C_i) \approx \frac{2R}{\sqrt{(R + K_i)^2 + 8\beta_i K_i^2 C_i + R + K_i}}, \quad (\text{S60})$$

Evidently, the function response of C_i species is negatively correlated with the population size of itself, which effectively constitutes a self-inhibiting negative feedback loop ([Appendix-fig. 15C](#)).

Then, we have a simple intuitive understanding of species coexistence through the mechanism of intraspecific interference. In an ecological community, consumer species that of higher/lower competitiveness tend to increase/decrease their population size in the competition process. Without intraspecific interference, the increasing/decreasing trend would continue until the system obeys CEP. In the scenario involving intraspecific interference, however, for species of higher competitiveness (e.g., C_i), with the increase of C_i 's population size, a larger portion of C_i individuals are then engaged in intraspecific interference pair which are temporarily absent from hunting ([Appendix-fig. 15A-B](#)). Consequently, the functional response of C_i drops, which prevents further increase of C_i 's population size, results in an overall balance among the consumer species, and thus promotes species coexistence.

Appendix IV

Scenario involving chasing pairs and interspecific interference

Here we consider the scenario involving chasing pairs and interspecific interference in the case of $S_C = 2$ and $S_R = 1$, with all settings follow that depicted in [Appendix II](#). Then,

$C_i \equiv C_i^{(F)} + x_i + z$, $R \equiv R^{(F)} + x_1 + x_2$, and the population dynamics follows (identical with [Eq. S25](#)):

$$\begin{cases} \dot{x}_i = a_i C_i^{(F)} R^{(F)} - (k_i + d_i) x_i, i = 1, 2; \\ \dot{z} = a'_{12} C_1^{(F)} C_2^{(F)} - d'_{12} z, \\ \dot{C}_i = w_i k_i x_i - D_i C_i, \\ \dot{R} = g(R, x_1, x_2, C_1, C_2). \end{cases} \quad (\text{S61})$$

Here the functional form of $g(R, x_1, x_2, C_1, C_2)$ is unspecified. For convenience, we define $K_i \equiv (d_i + k_i)/a_i$, $\alpha_i \equiv D_i/(w_i k_i)$ ($i = 1, 2$), and $\gamma \equiv a'_{12}/d'_{12}$. At steady state, from $\dot{x}_i = 0$ ($i = 1, 2$) and $\dot{z} = 0$, we have:

$$\begin{cases} x_i = C_i^{(F)} R^{(F)} / K_i, i = 1, 2; \\ z = \gamma C_1^{(F)} C_2^{(F)}. \end{cases} \quad (\text{S62})$$

Note that $C_i \equiv C_i^{(F)} + x_i + z$ and $R \equiv R^{(F)} + x_1 + x_2$, then,

$$\begin{cases} C_1 = C_1^{(F)} + R^{(F)}C_1^{(F)}/K_1 + \gamma C_1^{(F)}C_2^{(F)}, \\ C_2 = C_2^{(F)} + R^{(F)}C_2^{(F)}/K_2 + \gamma C_1^{(F)}C_2^{(F)}, \\ R = R^{(F)}(1 + C_1^{(F)}/K_1 + C_2^{(F)}/K_2). \end{cases} \quad (S63)$$

Then, we can express $C_1^{(F)}, C_2^{(F)}$ and $R^{(F)}$ with C_1, C_2 and R . Combined with [Eq. S62](#), x_i and z can also be expressed using C_1, C_2 and R . In particular, for x_i , we have:

$$x_i = u'_i(R, C_1, C_2), i = 1, 2. \quad (S64)$$

If all species coexist, by defining $\Omega'_i(R, C_1, C_2) \equiv \frac{w_i k_i}{C_i} u'_i(R, C_1, C_2)$, then, the steady-state equations of $\dot{C}_i = 0 (i = 1, 2)$ and $\dot{R} = 0$ are:

$$\begin{cases} \Omega'_1(R, C_1, C_2) - D_1 = 0, \\ \Omega'_2(R, C_1, C_2) - D_2 = 0, \\ G'(R, C_1, C_2) = 0, \end{cases} \quad (S65)$$

where $G'(R, C_1, C_2) \equiv g(R, u'_1(R, C_1, C_2), u'_2(R, C_1, C_2), C_1, C_2)$.

Here, [Eq. S65](#) corresponds to three unparallel surfaces and share a common point ([Fig. 1G](#) and [Appendix-fig 3A](#)). However, all the fixed points are unstable ([Appendix-fig. 3F](#)), and hence the consumer species cannot stably coexist at steady state ([Fig. 1D](#)).

A Analytical results of the fixed-point solution

We proceed to investigate the unstable fixed point where $R, C_1, C_2 > 0$. From $\dot{x}_i = 0 (i = 1, 2)$, $\dot{z} = 0$, $\dot{C}_i = 0$, and note that $C_i \equiv C_i^{(F)} + x_i + z$, we have:

$$\begin{cases} C_i = K_i \alpha_i C_i (R^{(F)})^{-1} + \alpha_i C_i + z, i = 1, 2; \\ z = \gamma K_1 \alpha_1 K_2 \alpha_2 (R^{(F)})^{-2} C_1 C_2. \end{cases} \quad (S66)$$

Since $C_i > 0$, then:

$$\begin{cases} C_1 = \frac{(1-\alpha_2)(R^{(F)})^2 - K_2 \alpha_2 R^{(F)}}{\gamma K_1 \alpha_1 K_2 \alpha_2}, \\ C_2 = \frac{(1-\alpha_1)(R^{(F)})^2 - K_1 \alpha_1 R^{(F)}}{\gamma K_1 \alpha_1 K_2 \alpha_2}. \end{cases} \quad (S67)$$

If $R \gg C_1 + C_2$, then $R \gg x_1 + x_2$ and $R^{(F)} \approx R$, we have:

$$\begin{cases} C_1 = \frac{(1-\alpha_2)R^2 - K_2\alpha_2 R}{\gamma K_1\alpha_1 K_2\alpha_2}, \\ C_2 = \frac{(1-\alpha_1)R^2 - K_1\alpha_1 R}{\gamma K_1\alpha_1 K_2\alpha_2}. \end{cases} \quad (S68)$$

Still, we assume that the population dynamics of the resource species follows [Eq. S42](#). At the fixed point, $\dot{R} = 0$. We have:

$$\zeta(1 - \frac{R}{K}) = k_1\alpha_1 C_1 + k_2\alpha_2 C_2. \quad (S69)$$

Combined with [Eq. S68](#), we can solve for R :

$$R = \frac{-\varrho_1 + \sqrt{\varrho_1^2 + 4\varrho_2\zeta}}{2\varrho_2}. \quad (S70)$$

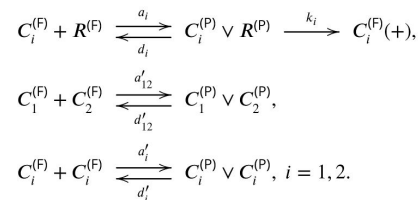
where $\varrho_1 \equiv \frac{\zeta}{K} - \frac{k_1}{\gamma K_1} - \frac{k_2}{\gamma K_2}$ and $\varrho_2 \equiv \frac{k_1(1-\alpha_2)}{\gamma K_1 K_2 \alpha_2} + \frac{k_2(1-\alpha_1)}{\gamma K_1 K_2 \alpha_1}$.

[Eqs. S68](#), [S70](#) are the analytical solutions of the fixed point when $R \gg C_1 + C_2$. As shown in [Appendix-fig. 3E](#), the analytical predictions agree well with the numerical results (the exact solutions).

Appendix V

Scenario involving chasing pairs and both intra- and interspecific interference

Here we consider the scenario involving chasing pairs and both intra- and inter-specific interference in the simple case of $S_C = 2$ and $S_R = 1$:



We adopt the same notations as that depicted in [Appendix III](#). and. Then,

$C_i \equiv C_i^{(F)} + x_i + 2y_i + z$ and $R \equiv R^{(F)} + x_1 + x_2$, and the population dynamics of the system can be described as follows:

$$\begin{cases} \dot{x}_i = a_i C_i^{(F)} R^{(F)} - (k_i + d_i) x_i, \\ \dot{z} = a'_{12} C_1^{(F)} C_2^{(F)} - d'_{12} z, \\ \dot{y}_i = a'_i [C_i^{(F)}]^2 - d'_i y_i, \\ \dot{C}_i = w_i k_i x_i - D_i C_i, \\ \dot{R} = g(R, x_1, x_2, C_1, C_2), i = 1, 2. \end{cases} \quad (S71)$$

Here, the functional form of $g(R, x_1, x_2, C_1, C_2)$ follows [Eq. S42](#). For convenience, we define $K_i \equiv (d_i + k_i)/a_i \alpha_i \equiv D_i/(w_i k_i)$, $\beta_i \equiv a'_i/d'_i$, and $\gamma \equiv a'_{12}/d'_{12}$, ($i = 1, 2$). At steady state, from $\dot{x}_i = 0$, $\dot{y}_i = 0$, $\dot{z} = 0$, and $\dot{C}_i = 0$, ($i = 1, 2$), we have:

$$\begin{cases} x_i = \alpha_i C_i, \\ C_i^{(F)} = K_i \alpha_i C_i (R^{(F)})^{-1}, \\ y_i = \beta_i (K_i \alpha_i C_i)^2 [R^{(F)}]^{-2}, \\ z = \gamma K_1 \alpha_1 K_2 \alpha_2 [R^{(F)}]^{-2} C_1 C_2. \end{cases} \quad (S72)$$

Combined with $C_i \equiv C_i^{(F)} + x_i + 2y_i + z$, and since $C_i > 0$ ($i = 1, 2$), then,

$$\begin{cases} (1 - \alpha_1)(R^{(F)})^2 - K_1 \alpha_1 R^{(F)} = 2\beta_1 (K_1 \alpha_1)^2 C_1 + \gamma K_1 \alpha_1 K_2 \alpha_2 C_2, \\ (1 - \alpha_2)(R^{(F)})^2 - K_2 \alpha_2 R^{(F)} = 2\beta_2 (K_2 \alpha_2)^2 C_2 + \gamma K_1 \alpha_1 K_2 \alpha_2 C_1. \end{cases} \quad (S73)$$

A Analytical solutions of species abundances at steady state

If $R \gg C_1 + C_2$, then $R \gg x_1 + x_2$ and thus $R^{(F)} \approx R$. Combined with [Eq. S73](#), we obtain:

$$\begin{cases} C_1 = R \frac{(2\beta_2 K_2 \alpha_2 (1 - \alpha_1) - \gamma K_1 \alpha_1 (1 - \alpha_2)) R + (\gamma - 2\beta_2) K_1 \alpha_1 K_2 \alpha_2}{K_1^2 \alpha_1^2 K_2 \alpha_2 (4\beta_1 \beta_2 - \gamma^2)}, \\ C_2 = R \frac{(2\beta_1 K_1 \alpha_1 (1 - \alpha_2) - \gamma K_2 \alpha_2 (1 - \alpha_1)) R + (\gamma - 2\beta_1) K_1 \alpha_1 K_2 \alpha_2}{K_1 \alpha_1 K_2^2 \alpha_2^2 (4\beta_1 \beta_2 - \gamma^2)}. \end{cases} \quad (S74)$$

Using $\dot{R} = 0$ and $R > 0$, we have:

$$R = \frac{-\chi_2 + \sqrt{(\chi_2)^2 + 4\chi_1 \zeta}}{2\chi_1}, \quad (S75)$$

where $\chi_1 \equiv \frac{k_2 \gamma (\alpha_1 - 1)}{K_1 K_2 \alpha_1 (4\beta_1 \beta_2 - \gamma^2)} + \frac{k_1 \gamma (\alpha_2 - 1)}{K_1 K_2 \alpha_2 (4\beta_1 \beta_2 - \gamma^2)} - \frac{k_1 2\beta_2 (\alpha_1 - 1)}{K_1^2 \alpha_1 (4\beta_1 \beta_2 - \gamma^2)} - \frac{k_2 2\beta_1 (\alpha_2 - 1)}{K_2^2 \alpha_2 (4\beta_1 \beta_2 - \gamma^2)}$, and

$$\chi_2 \equiv \frac{k_1 (\gamma - 2\beta_2)}{K_1 (4\beta_1 \beta_2 - \gamma^2)} + \frac{k_2 (\gamma - 2\beta_1)}{K_2 (4\beta_1 \beta_2 - \gamma^2)} + \frac{\zeta}{\kappa}$$

[Eqs. S74](#)–[S75](#) are the analytical solutions of the species abundances at steady state when $R \gg C_1 + C_2$. As shown in [Appendix-fig. 5E](#), the analytical calculations agree well with the numerical results (the exact solutions).

B Stability analysis of the coexisting state

In the scenario involving chasing pairs and both intra- and inter-specific interference, the behavior of species coexistence is similar to that without interspecific interference. Evidently, the influence of interspecific interference would be negligible if d'_{12} is extremely large, and vice versa for intraspecific interference if both d'^1 and d'^2 are tremendous. In the deterministic framework, the two-consumer species may coexist at constant population densities (**Appendix-fig. 5B**), and the fixed points are globally attracting (**Appendix-fig. 5C**). Furthermore, there is a non-zero measure of parameter set where both consumer species can coexist at steady state with only one type of resources (**Appendix-fig. 5A**). In the stochastic framework, just as the scenario involving chasing pairs and intraspecific interference, the coexistence state can be maintained along with stochasticity (**Appendix-fig. 5D**).

Appendix VI

Dimensional analysis for the scenario involving chasing pairs and both intra- and inter-specific interference

The population dynamics of the system involving chasing pairs and both intra- and inter-specific interference are shown in **Eqs. 1–4**:

$$\begin{cases} \dot{x}_{il} = a_{il} C_i^{(F)} R_l^{(F)} - (d_{il} + k_{il}) x_{il}, \\ \dot{y}_i = a'_i [C_i^{(F)}]^2 - d'_i y_i, \\ \dot{z}_{ij} = a'_{ij} C_i^{(F)} C_j^{(F)} - d'_{ij} z_{ij} \\ \dot{C}_i = \sum_{l=1}^{S_R} w_{il} k_{il} x_{il} - D_i C_i, \\ \dot{R}_l = \zeta_l (1 - R_l / \kappa_l) - \sum_{i=1}^{S_C} k_{il} x_{il}, \end{cases} \quad (S76)$$

with $l = 1, \dots, S_R$; $i, j = 1, \dots, S_C$, and $i \neq j$. Here $C_i = C_i^{(F)} + \sum_l x_{il} + 2y_i + \sum_{i \neq j} z_{ij}$ and

$R_l = R_l^{(F)} + \sum_i x_{il}$ represent the population abundances of the consumers and resources in the system. In fact, there are already several dimensionless variables and parameter in **Eq. S76**, namely $x_{il}, y_i, z_{ij}, C_i^{(F)}, R_l^{(F)}, C_i, R_l, w_{il}, k_{il}$. To make all terms dimensionless, we define $\tilde{t} = t/\tau$, where $\tau = \tilde{D}_1/D_1$ and $\tilde{D}_1$ is a reducible dimensionless parameter which is freely to take any

positive values. Besides, we define dimensionless parameters $\tilde{a}_{il} = a_{il}\tau$, $\tilde{d}_{il} = d_{il}\tau$,

$\tilde{k}_{il} = k_{il}\tau$, $\tilde{a}'_i = a'_i\tau$, $\tilde{d}'_i = d'_i\tau$, $\tilde{a}'_{ij} = a'_{ij}\tau$, $\tilde{d}'_{ij} = d'_{ij}\tau$, $\tilde{D}_i = D_i\tau$ and $\tilde{\zeta}_l = \zeta_l\tau$.

By substituting all the dimensionless terms into [Eq. S76](#), we have:

$$\begin{cases} \dot{x}_{il} = \tilde{a}_{il} C_i^{(F)} R_l^{(F)} - (\tilde{d}_{il} + \tilde{k}_{il}) x_{il}, \\ \dot{y}_i = \tilde{a}'_i [C_i^{(F)}]^2 - \tilde{d}'_i y_i, \\ \dot{z}_{ij} = \tilde{a}'_{ij} C_i^{(F)} C_j^{(F)} - \tilde{d}'_{ij} z_{ij} \\ \dot{C}_i = \sum_{l=1}^{S_R} w_{il} \tilde{k}_{il} x_{il} - \tilde{D}_i C_i, \\ \dot{R}_l = \tilde{\zeta}_l (1 - R_l/\kappa_l) - \sum_{i=1}^{S_C} \tilde{k}_{il} x_{il}. \end{cases} \quad (S77)$$

For convenience, we omit the notation “~” and use dimensionless variables and parameters in the simulation studies unless otherwise specified.

Appendix VII

Approximations applied in the pairwise encounter model

For consumers within a paired state, either in a chasing pair or an interference pair, the consumer may die following the mortality rate. Thus, in the scenario involving chasing pairs and both intra- and inter-specific interference, the population dynamics of the system should be described as follows:

$$\begin{cases} \dot{x}_{il} = a_{il} C_i^{(F)} R_l^{(F)} - (d_{il} + k_{il} + D_i) x_{il}, \\ \dot{y}_i = a'_i [C_i^{(F)}]^2 - (d'_i + D_i) y_i, \\ \dot{z}_{ij} = a'_{ij} C_i^{(F)} C_j^{(F)} - (d'_{ij} + D_i + D_j) z_{ij} \\ \dot{C}_i = \sum_{l=1}^{S_R} w_{il} k_{il} x_{il} - D_i C_i, \quad i = 1, \dots, S_C, \\ \dot{R}_l = \zeta_l (1 - R_l/\kappa_l) - \sum_{i=1}^{S_C} k_{il} x_{il}, \quad l = 1, \dots, S_R. \end{cases} \quad (S78)$$

However, since predation or interference processes are generally much faster than birth and death processes, i.e., $D_i \ll k_{il}, d_{il}, d'_{ij}, d'_{ij}$, the influence of mortality rate in a paired state is negligible. Therefore, we have used the following approximations throughout our manuscript: $(k_{il} + d_{il} + D_i) \approx (k_{il} + d_{il})$, $(d'_{ij} + D_i) \approx d'_{ij}$, $(d'_{ij} + D_i + D_j) \approx d'_{ij}$. Hence, the approximated population dynamics is described as follows:

$$\begin{cases} \dot{x}_{il} = a_{il} C_i^{(F)} R_l^{(F)} - (d_{il} + k_{il}) x_{il}, \\ \dot{y}_i = a'_i [C_i^{(F)}]^2 - d'_i y_i, \\ \dot{z}_{ij} = a'_{ij} C_i^{(F)} C_j^{(F)} - d'_{ij} z_{ij} \\ \dot{C}_i = \sum_{l=1}^{S_R} w_{il} k_{il} x_{il} - D_i C_i, \quad i = 1, \dots, S_C, \\ \dot{R}_l = \zeta_l (1 - R_l / \kappa_l) - \sum_{i=1}^{S_C} k_{il} x_{il}, \quad l = 1, \dots, S_R, \end{cases} \quad (S79)$$

which is identical to those shown in the main text.

Appendix VIII

Simulation details of the main text figures

In **Fig. 1C**, F: $a_i = 0.1$, $d_i = 0.5$, $w_i = 0.1$, $k_i = 0.1$ ($i = 1, 2$); $D_1 = 0.002$, $D_2 = 0.001$, $\kappa = 5$, $\zeta = 0.05$. In **Fig. 1D**, G: $a_i = 0.02$, $a'_i = 0.021$, $d_i = 0.5$, $d'_{ij} = 0.01$, $w_i = 0.08$, $k_i = 0.03$, $i, j = 1, 2$, $i \neq j$, $D_1 = 0.0011$, $\kappa = 20$, $\zeta = 0.01$. In **Fig. 1E**, H: $a_i = 0.5$, $a'_i = 0.625$, $d_i = 0.5$, $d'_i = 0.02$, $w_i = 0.2$, $k_i = 0.4$ ($i = 1, 2$), $D_1 = 0.0286$, $D_2 = 0.022$, $\kappa = 10$, $\zeta = 0.5$. **Fig. 1C**, F were calculated or simulated from **Eqs. 1**, **4**. **Fig. 1D**, G were calculated or simulated from **Eqs. 1**, **3**, **4**. **Fig. 1E**, H were calculated or simulated from **Eqs. 1**, **2**, **4**. The analytical solutions in **Fig. 1E** were calculated from **Eqs. S41** and **S43**.

In **Fig. 2A**: $a_i = 0.02$, $a'_i = 0.025$, $d_i = 0.7$, $d'_i = 0.7$, $w_i = 0.4$, $k_i = 0.05$ ($i = 1, 2$); $D_1 = 0.0160$, $D_2 = 0.0171$, $\kappa = 2000$, $\zeta = 5.5$. In **Fig. 2B-C**: $L = 100$, $r^{(C)} = 5$, $r^{(I)} = 5$, $v_{C_i} = v_R = 0.1$, $a_i = 0.2010$, $a'_i = 0.2828$, $d'_i = 0.8$, $d_i = 0.7$, $w_i = 0.33$, $k_i = 0.2$ ($i = 1, 2$); $D_1 = 0.0605$, $D_2 = 0.0600$, $\kappa = 1000$, $\zeta = 100$. In **Fig. 2D**: $a_i = 0.3$, $a'_i = 0.33$, $w_i = 0.018$, $k_i = 4.8$, $d'_i = 5$, $d_i = 5.5$ ($i = 1, 2$); $D_2 = 0.010$, $\zeta = 35$, $\kappa = 10000$, $D_1 = 0.011$. In **Fig. 2E**: $w_i = 0.02$, $k_i = 4.5$, $d'_i = 4$, $d_i = 4.5$ ($i = 1, 2$); $D_2 = 0.010$, $\zeta = 35$, $\kappa = 10000$, $a_i = 0.2$, $a'_i = 0.24$ ($i = 1, 2$); $D_1 = 0.0120$. In **Fig. 2D-E**: We set $\tau = 0.4$ Day (see). This results in an expected lifespan of *Drosophila serrata* in the model settings of $\tau/D_2 = 40$ days and that of *Drosophila pseudoobscura* $\tau/D_1 = 36.4$ days, which roughly agrees with experimental data showing that the average lifespan of *D. serrata* is 34 days for males and 54 days for females (Narayan et al., 2022), and the average lifespan of *D. pseudoobscura* is around 40 days for females (Gowaty et al., 2010). The time averages ($\bar{c}_i$) and standard deviations (δc_i) of the species' relative/absolute abundances

for the experimental data or SSA results are as follows: $\overline{C_{D.serrata_AR_Grp1}^{Exp(SSA)}}^{(R)} = 0.53 (0.55)$,

$$\delta^{(R)} C_{D.serrata_AR_Grp1}^{Exp(SSA)} = 0.12 (0.09), \quad \overline{C_{D.serrata_AR_Grp2}^{Exp(SSA)}}^{(R)} = 0.59 (0.61),$$

$$\delta^{(R)} C_{D.serrata_AR_Grp2}^{Exp(SSA)} = 0.10 (0.12), \quad \overline{C_{T.confusum_24^\circ C}^{Exp(SSA)}} = 29.1 (28.6),$$

$$\delta C_{T.confusum_24^\circ C}^{Exp(SSA)} = 5.4 (5.2), \quad \overline{C_{T.castaneuma_24^\circ C}^{Exp(SSA)}} = 45.9 (54.5),$$

$$\delta C_{T.castaneuma_24^\circ C}^{Exp(SSA)} = 7.2 (8.6) \text{ where the superscript “(R)” represents relative abundances. A}$$

comparison of Shannon entropies in the time series between experimental data and SSA results is presented in **Appendix-fig. 7C-D**. **Fig. 2A-E** were simulated from **Eqs. 1, 2, 4**. See **Appendix-fig. 7E, G** for the long-term time series of all species in **Fig. 2D-E**, respectively.

Model settings in **Fig. 3A-B**, D (plankton): $a_{il} = 0.1$, $a_i' = 0.125$, $d_{il} = 0.5$, $d_i' = 0.2$, $w_{il} = 0.3$, $k_{il} = 0.2$, $\kappa_1 = 8 \times 10^4$, $\kappa_2 = 5 \times 10^4$, $\kappa_3 = 3 \times 10^4$, $\zeta_1 = 280$, $\zeta_2 = 200$, $\zeta_3 = 150$, $D_i = 0.03 \times N(1, 0.25)$ ($i = 1, \dots, S_C$), $S_C = 140$ and $S_R = 3$. Model settings in **Fig. 3C** (bird): $a_i = 0.1$, $a_i' = 0.125$, $d_i = 0.5$, $d_i' = 0.5$, $w_i = 0.3$, $k_i = 0.2$, $D_i = 0.02 \times N(1, 0.28)$ ($i = 1, \dots, S_C$); $\zeta = 110$, $\kappa = 10^5$, $S_C = 250$ and $S_R = 1$. Model settings in **Fig. 3C** (fish): $a_i = 0.1$, $a_i' = 0.14$, $d_i = 0.5$, $d_i' = 0.5$, $w_i = 0.2$, $k_i = 0.1$, $D_i = 0.015 \times N(1, 0.32)$ ($i = 1, \dots, 45$); $\zeta = 550$, $\kappa = 10^6$, $S_C = 45$ and $S_R = 1$. Model settings in **Fig. 3C** (butterfly): $a_i = 0.1$, $a_i' = 0.125$, $d_i = 0.5$, $d_i' = 0.3$, $w_i = 0.3$, $k_i = 0.2$, $D_i = 0.034 \times N(1, 0.35)$ ($i = 1, \dots, S_C$); $\zeta = 300$, $\kappa = 10^5$, $S_C = 150$ and $S_R = 1$. Model settings in **Fig. 3D** (bat): $a_i = 0.1$, $a_i' = 0.125$, $d_i = 0.5$, $d_i' = 0.5$, $w_i = 0.2$, $k_i = 0.1$, $D_i = 0.013 \times N(1, 0.34)$ ($i = 1, \dots, S_C$); $\zeta = 250$, $\kappa = 10^6$, $S_C = 40$ and $S_R = 1$. Model settings in **Fig. 3D** (lizard): $a_i = 0.1$, $a_i' = 0.125$, $d_i = 0.5$, $d_i' = 0.5$, $w_i = 0.2$, $k_i = 0.1$, $D_i = 0.014 \times N(1, 0.34)$ ($i = 1, \dots, S_C$); $\zeta = 250$, $\kappa = 10^6$, $S_C = 55$ and $S_R = 1$. In **Fig. 3A-D**, the mortality rate D_i ($i = 1, \dots, S_C$) is the only parameter that varies with the consumer species, which was randomly sampled from a Gaussian distribution $N(\mu, \sigma)$, where μ and σ are the mean and standard deviation of the distribution. The coefficient of variation of the mortality rates (i.e., σ/μ) was chosen to be around 0.3, or more precisely, the best-fit in the range of 0.15–0.43. This range was estimated from experimental results (Menon et al., 2003) using the two-sigma rule. These settings for the mortality rates also apply to those in **Appendix-figs. 8–14**. **Fig. 3A-D** were simulated from **Eqs. 1, 2, 4**. See **Appendix-figs. 14C, 10K, C, D, H, I, J**, **Fig. 3A**, **Fig. 3B** for the time series of **Fig. 3C**

3C (ODEs ^{$S_R=1$} _{bird}), 3C (ODEs ^{$S_R=1$} _{butterfly}), 3C (ODEs ^{$S_R=1$} _{fish}), 3D (ODEs ^{$S_R=1$} _{bat}), 3D (SSA ^{$S_R=1$} _{bat}),

3D (ODEs ^{$S_R=1$} _{lizard}), 3D (SSA ^{$S_R=1$} _{lizard}), 3D (ODEs ^{$S_R=3$} _{plankton}) and 3D (SSA ^{$S_R=3$} _{plankton}), respectively. The

Shannon entropies of the experimental data and simulation results for each ecological community

are: $H_{\text{Exp(ODEs)}}^{\text{bird(1982)}} = 5.67(6.79)$, $H_{\text{Exp(ODEs)}}^{\text{bird(2018)}} = 6.63(6.79)$, $H_{\text{Exp(ODEs)}}^{\text{butterfly}} = 4.78(4.12)$,

$H_{\text{Exp(ODEs)}}^{\text{fish}} = 3.78(3.40)$; $H_{\text{Exp(ODEs,SSA)}}^{\text{bat}} = 3.00(2.95, 2.84)$,

$H_{\text{Exp(ODEs,SSA)}}^{\text{lizard}} = 4.05(3.57, 3.50)$, $H_{\text{Exp(ODEs,SSA)}}^{\text{plankton}} = 4.68(6.43, 6.48)$. Here the Shannon

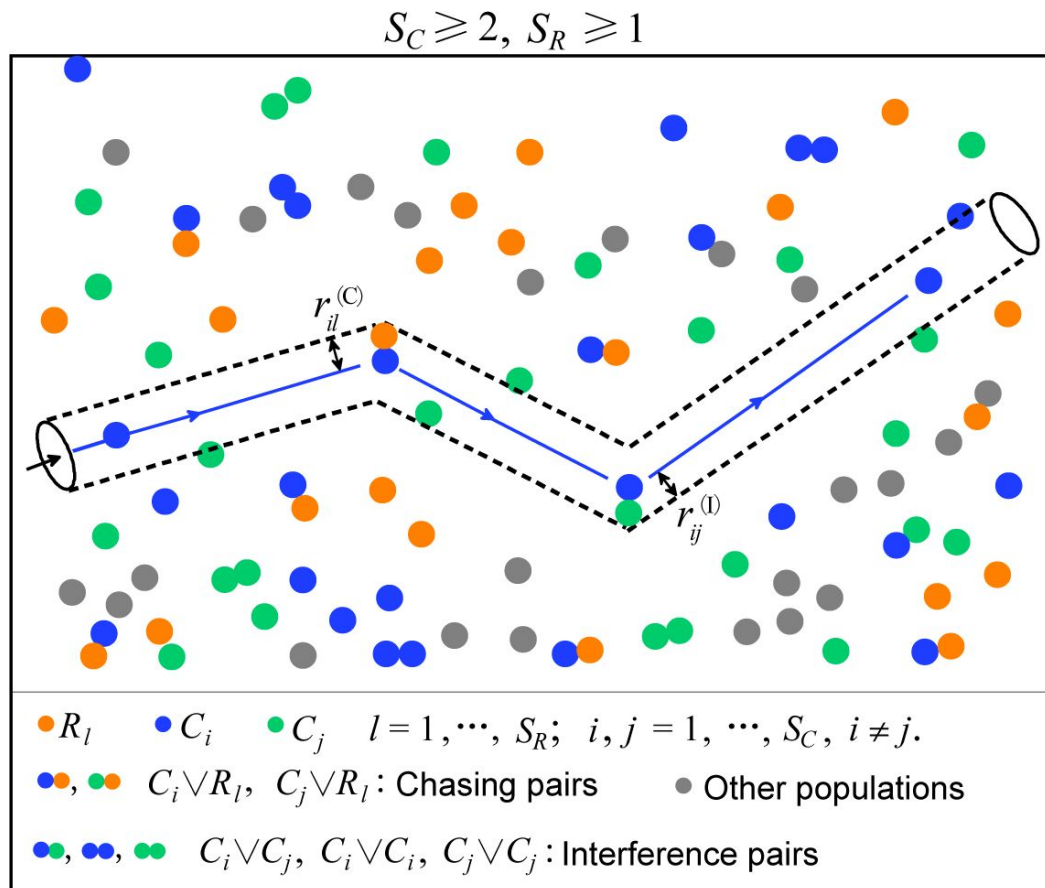
entropy $H = - \sum_{i=1}^{S_C} P_i \log_2(P_i)$, where P_i is the probability that a consumer individual belongs to species C_i .

Appendix Tables

Appendix-table 1.

Illustrations of symbols in our generic model of pairwise encounters

Appendix Figures



Appendix-figure 1.

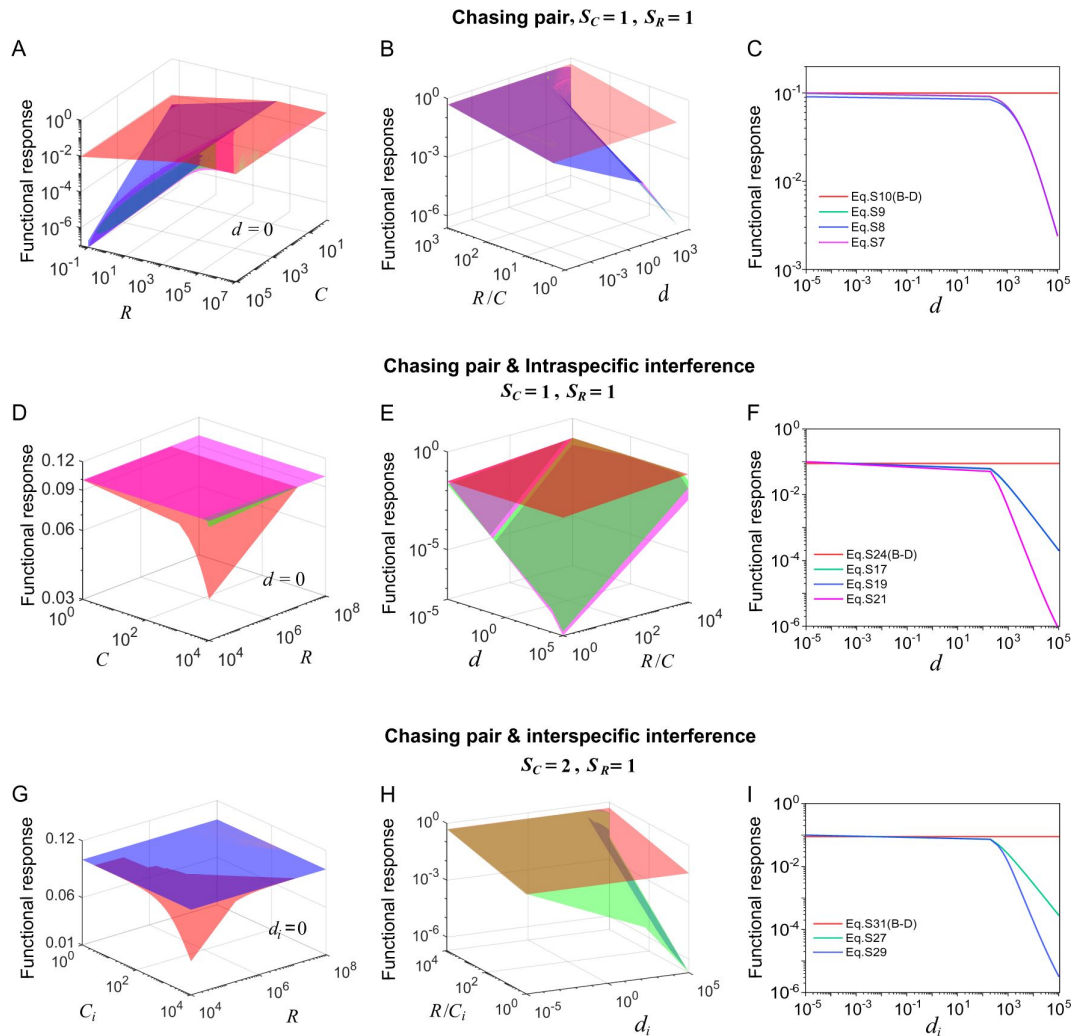
Estimation of the encounter rates with the mean-field approximations.

To calculate a_{ij} in the chasing pair, we suppose that all individuals of species R_l stand still while a C_i individual moves at the speed of $\overline{u_{C_i-R_l}}$ (the relative speed). Over a time interval of Δt , the length of zigzag trajectory of the C_i individual is approximately $\overline{u_{C_i-R_l}} \Delta t$, while the encounter area (marked with dashed lines) is estimated to be $2r_{il}^{(C)} \overline{u_{C_i-R_l}} \Delta t$. Then, we can estimate the encounter rate a_{ij} using the encounter area and concentrations of the species (see Materials and Methods for details). Similarly, we can estimate a'_{ij} in the interference pair.

Symbols	Illustrations/Definitions
$\mathcal{F}$	The functional response.
Ξ	The searching efficiency.
ς	A random number sampled from a uniform distribution.
$\mathcal{U}$	The uniform distribution.
$\mathcal{N}(\mu, \sigma)$	A Gaussian distribution with a mean of μ and a standard deviation of σ .
$\equiv$	An equal sign for equations defining the symbol on the left-hand side.
Δ	The competitive difference between two consumer species, defined as $\Delta \equiv (D_1 - D_2)/D_2$.
$\hat{\Delta}$	The supremum of the competitive difference tolerated for species coexistence.
Δt	A short time interval.
P_i	The probability that a consumer individual of the ecological community belongs to species C_i .
$p_{\text{ODEs}}, p_{\text{SSA}}$	The p-value assessing the similarity of simulation results and experimental data.
H	The Shannon entropy: $H = -\sum_{i=1}^{S_C} P_i \log_2(P_i)$.
$\overline{X}$	The time average of an arbitrary quantity X .
δX	The standard deviation of an arbitrary quantity X .
$\langle X \rangle$	The expectation of a random variable X .
τ	The parameter that sets the time scale of a system.
K_{il}	$K_{il} \equiv \frac{k_{il} + d_{il}}{a_{il}}$.
α_{il}	$\alpha_{il} \equiv \frac{D_i}{w_{il} k_{il}}$.
β_i	$\beta_i \equiv \frac{d_i'}{d_i''}$.
γ	$\gamma \equiv \frac{d_{12}'}{d_{12}''}$.
$f_i(\mathbf{R}^{(\text{F})})$	$f_i(\mathbf{R}^{(\text{F})}) \equiv \mathbf{R}^{(\text{F})} / (\mathbf{R}^{(\text{F})} + K_i)$.
ϕ_0, ϕ_1, ϕ_2	$\phi_0 \equiv -CR^2, \phi_1 \equiv 2CR + KR + R^2, \phi_2 \equiv 2\beta K^2 - K - C - 2R$.
Λ	The discriminant of Eq. S13.
ψ, φ	$\psi \equiv \phi_1 - (\phi_2)^2/3, \varphi \equiv \phi_0 - \phi_1\phi_2/3 + 2(\phi_2)^3/27$.
$\omega, \theta_1, \theta_2$	$\omega \equiv -1/2 + i\sqrt{3}/2, \theta_1 \equiv (-\varphi/2 + \sqrt{-\Lambda/108})^{1/3}, \theta_2 \equiv (-\varphi/2 - \sqrt{-\Lambda/108})^{1/3}$.
ψ', φ'	$\psi' \equiv (-4\psi/3)^{1/2}, \varphi' \equiv \arccos(-(-\psi/3)^{-3/2}\varphi/2)/3$.
t_h	The handling time in the B-D model.
t_w	The wasting time in the B-D model.
$u_i(R, C_1, C_2)$	Expression of x_i using C_1, C_2 , and R in Eq. S33 involving intraspecific interference, see Eq. S37.
$\Omega_i(R, C_1, C_2)$	$\Omega_i(R, C_1, C_2) \equiv \frac{w_i k_i}{C_i} u_i(R, C_1, C_2)$.
$G(R, C_1, C_2)$	$G(R, C_1, C_2) \equiv g(R, u_1(R, C_1, C_2), u_2(R, C_1, C_2), C_1, C_2)$, see Eqs. S33 and S38.
o_1, o_2	$o_1 \equiv \frac{\zeta}{\kappa} - \frac{k_1}{2\beta_1 K_1} - \frac{k_2}{2\beta_2 K_2}, o_2 \equiv \frac{k_1(1-\alpha_1)}{2\beta_1 \alpha_1 (K_1)^2} + \frac{k_2(1-\alpha_2)}{2\beta_2 \alpha_2 (K_2)^2}$.
ϖ	$\varpi \equiv \frac{1}{2}(\frac{1}{\kappa} - \frac{k_2}{2\zeta\beta_2 K_2}) + \frac{1}{2}\sqrt{(\frac{1}{\kappa} - \frac{k_2}{2\zeta\beta_2 K_2})^2 + 2\frac{k_2(1-\alpha_2)}{\zeta\beta_2 \alpha_2 (K_2)^2}}$.
l_1, l_2	$l_1 \equiv \frac{\zeta}{\kappa} - \sum_{i=1}^{S_C} \frac{k_i}{2\beta_i K_i}, l_2 \equiv \sum_{i=1}^{S_C} \frac{k_i(1-\alpha_i)}{2\beta_i \alpha_i (K_i)^2}$.
$u'_i(R, C_1, C_2)$	Expression of x_i using C_1, C_2 , and R in Eq. S61 involving interspecific interference, see Eq. S64.
$\Omega'_i(R, C_1, C_2)$	$\Omega'_i(R, C_1, C_2) \equiv \frac{w_i k_i}{C_i} u'_i(R, C_1, C_2)$.
$G'(R, C_1, C_2)$	$G'(R, C_1, C_2) \equiv g(R, u'_1(R, C_1, C_2), u'_2(R, C_1, C_2), C_1, C_2)$, see Eqs. S61 and S65.
ϱ_1, ϱ_2	$\varrho_1 \equiv \frac{\zeta}{\kappa} - \frac{k_1}{\gamma K_1} - \frac{k_2}{\gamma K_2}, \varrho_2 \equiv \frac{k_1(1-\alpha_2)}{\gamma K_1 K_2 \alpha_2} + \frac{k_2(1-\alpha_1)}{\gamma K_1 K_2 \alpha_1}$.
χ_1	$\chi_1 \equiv \frac{k_2 \gamma (\alpha_1 - 1)}{K_1 K_2 \alpha_1 (4\beta_1 \beta_2 - \gamma^2)} + \frac{k_1 \gamma (\alpha_2 - 1)}{K_1 K_2 \alpha_2 (4\beta_1 \beta_2 - \gamma^2)} - \frac{k_1 2\beta_2 (\alpha_1 - 1)}{K_1^2 \alpha_1 (4\beta_1 \beta_2 - \gamma^2)} - \frac{k_2 2\beta_1 (\alpha_2 - 1)}{K_2^2 \alpha_2 (4\beta_1 \beta_2 - \gamma^2)}$.
χ_2	$\chi_2 \equiv \frac{k_1 (\gamma - 2\beta_2)}{K_1 (4\beta_1 \beta_2 - \gamma^2)} + \frac{k_2 (\gamma - 2\beta_1)}{K_2 (4\beta_1 \beta_2 - \gamma^2)} + \frac{\zeta}{\kappa}$.

Appendix-table 2.

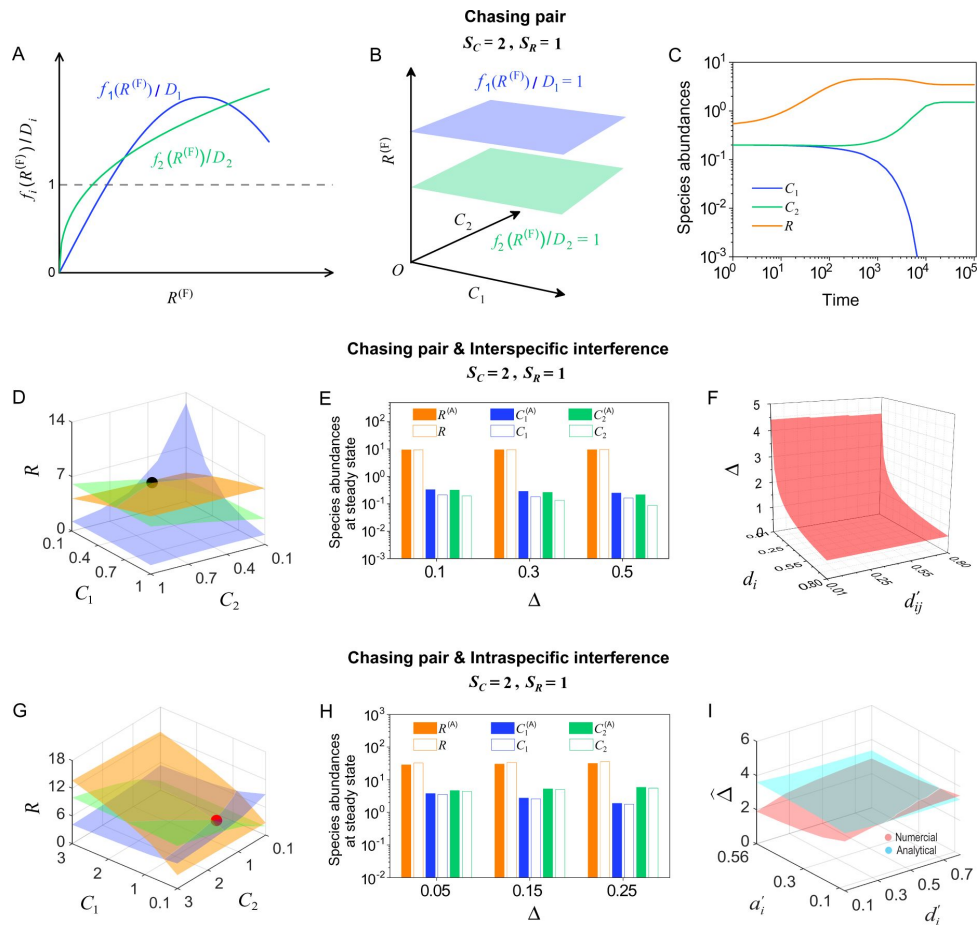
Illustrations of other symbols used in our manuscript



Appendix-figure 2.

Functional response in scenarios involving different types of pairwise encounter

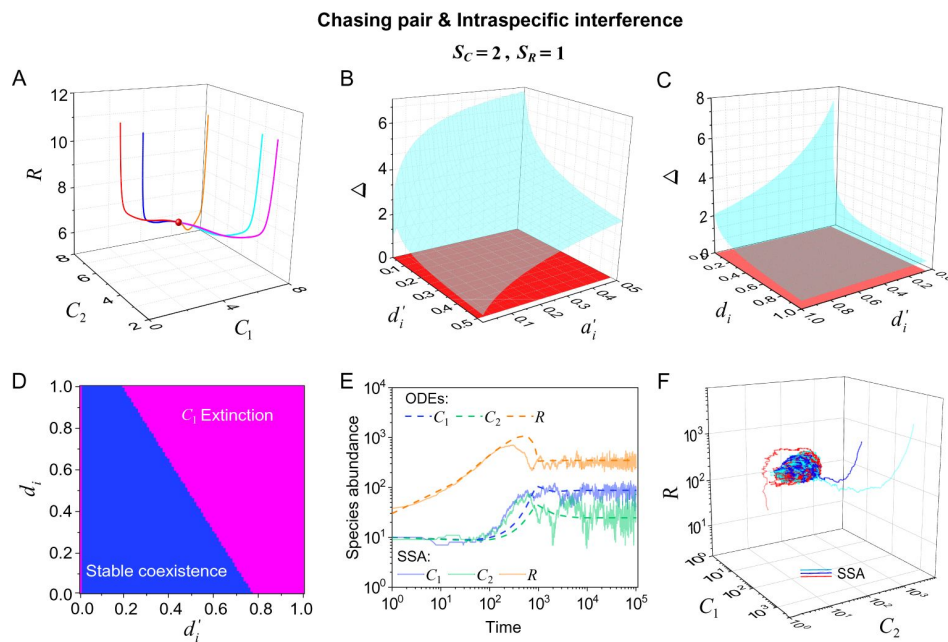
(A-C) In the scenario involving only chasing pair, the red surface/line corresponds to the B-D model (calculated from Eq.S10), while the green surface/line represents the exact solutions to our mechanistic model (calculated from Eq. S7). The magenta (calculated from Eq. S8) and blue (calculated from Eq. S9) surfaces/lines represent the approximate solutions to our model (see Appendix II). (D-F) In the scenario involving chasing pairs and intraspecific interference, the red surface/line corresponds to the B-D model (calculated from Eq.S24), while the green/line surface represents the exact solutions to our mechanistic model (calculated from Eq. S17). The blue surface/line (calculated from Eq. S19) and the magenta surface/line (calculated from Eq. S21) represent the quasi-rigorous and the approximate solutions to our model, respectively (see Appendix II). (G-I) In the scenario involving chasing pairs and interspecific interference, the red surface/line corresponds to the B-D model (calculated from Eq. S31), while the green surface/line represents the quasi-rigorous solutions to our mechanistic model (calculated from Eq. S27). The blue surface/line (calculated from Eq. S29) represents the approximate solutions to our model (see Appendix II). In (A-C): $k = 0.1$, $a = 0.25$. In (A): $d = 0$. In (C): $R = 10^4$, $C = 10^3$. In (D-F): $a = 0.1$, $k = 0.1$, $d' = 0.1$, $a' = 0.12$. In (D): $d = 0$. In (F): $R = 10^6$, $C = 10^5$. In (G-I): $a_1 = a_2 = 0.1$, $k_1 = k_2 = 0.1$, $d'_{12} = 0.1$, $a'_{12} = 0.12$. In (G): $d_i = 0$ ($i = 1, 2$). In (I): $R = 10^6$, $C_i = 10^5$.



Appendix-figure 3.

Numerical solutions in scenarios involving chasing pairs and different types of predator interference

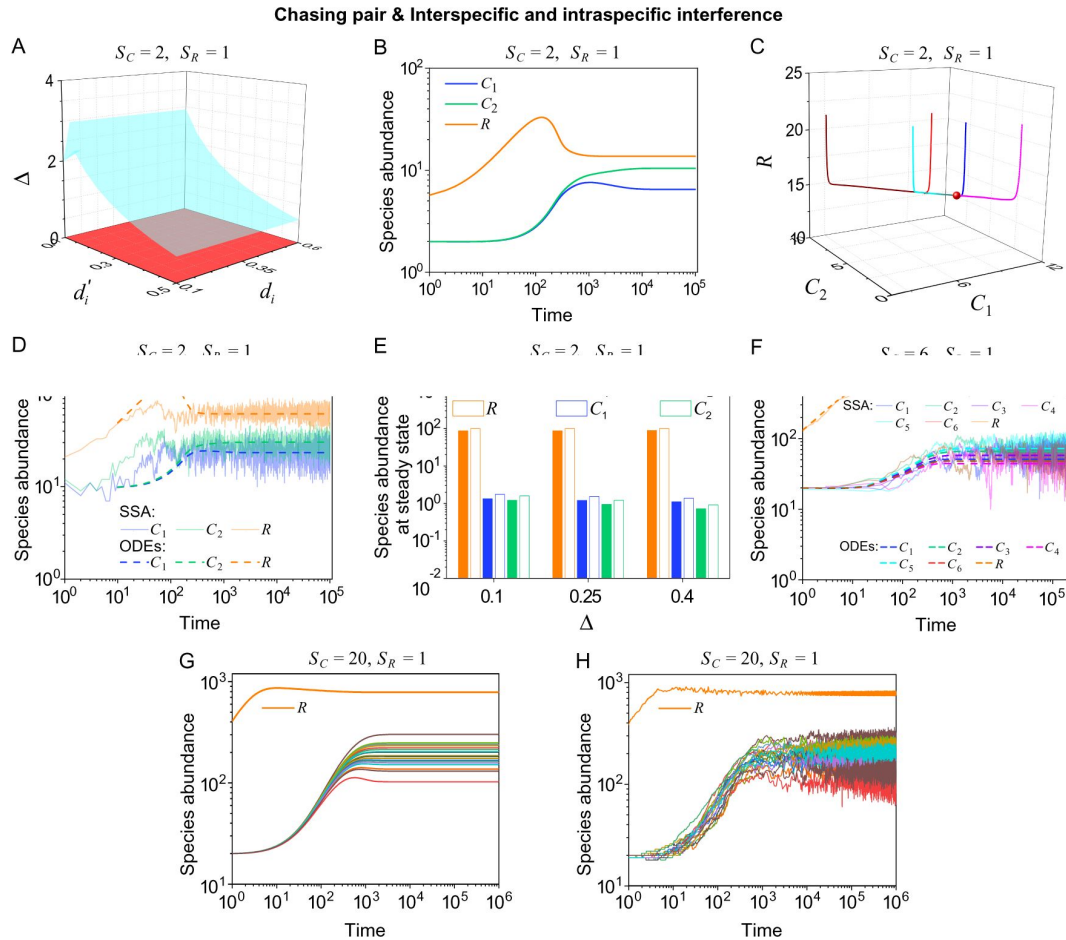
Here, $S_C = 2$ and $S_R = 1$. D_i ($i = 1, 2$) is the only parameter varying with the consumer species (with $D_1 > D_2$), and $\Delta \equiv (D_1 - D_2)/D_2$ represents the competitive difference between the two species. **(A-C)** Scenario involving only chasing pairs. **(A)** If all consumer species coexist at steady state, then $f_i(R^{(F)})/D_i = 1$ ($i = 1, 2$), where $f_i(R^{(F)}) = R^{(F)}/(R^{(F)} + K_i)$ and $K_i = (d_i + k_i)/a_i$. This requires that the three lines $y = f_i(R)/D_i$ ($i = 1, 2$) and $y = 1$ share a common point, which is generally impossible. **(B)** The blue plane is parallel to the green one, and hence they do not have a common point. **(C)** Time courses of the species abundances in the scenario involving only chasing pairs. The two consumer species cannot coexist at steady state. **(D-F)** Scenario involving chasing pairs and interspecific interference. **(G-I)** Scenario involving chasing pairs and intraspecific interference. **(D, G)** Positive solutions to the steady-state equations: $\dot{R} = 0$ (orange surface), $\dot{C}_1 = 0$ (blue surface), $\dot{C}_2 = 0$ (green surface). The intersection point marked by black/red dots is an unstable/stable fixed point. **(E, H)** Comparisons between the numerical results and analytical solutions of the species abundances at fixed points. Color bars are analytical solutions while hollow bars are numerical results. The analytical solutions in **(E)** and **(H)** (marked with superscript "(A)") were calculated from [Eqs. S68](#), [S70](#) and [Eqs. S41](#), [S43](#), respectively. **(F)** In this scenario, there is no parameter region for stable coexistence. The region below the red surface and above $\Delta = 0$ represents unstable fixed points. **(I)** Comparisons between the numerical results and analytical solutions of the coexistence region. Here $\hat{\Delta}$ represents the maximum competitive difference tolerated for species coexistence. The red and cyan surfaces represent the analytical solutions (calculated from [Eq. S46](#)) and numerical results, respectively. The numerical results in **(C)**, **(D-F)** and **(G-I)** were calculated from [Eqs. 1](#), [4](#), [Eqs. S42](#), [S61](#) and [Eqs. S33](#), [S42](#), respectively. In **(C)**: $a_i = 0.1$, $k_i = 0.1$, $w_i = 0.1$, $d_i = 0.5$ ($i = 1, 2$), $D_1 = 0.002$, $D_2 = 0.001$, $\kappa = 5$, $\zeta = 0.05$. In **(D)**: $a_i = 0.05$, $d_i = 0.05$, $k_i = 0.02$, $w_i = 0.08$ ($i = 1, 2$), $\kappa = 20$, $a'_{12} = 0.06$, $d'_{12} = 0.01$, $D_1 = 0.0011$, $D_2 = 0.001$, $\zeta = 0.01$. In **(E)**: $a_i = 0.04$, $d_i = 0.2$, $k_i = 0.1$, $w_i = 0.3$ ($i = 1, 2$), $\kappa = 10$, $a'_{12} = 0.048$, $d'_{12} = 0.001$, $D_2 = 0.0008$, $\zeta = 0.2$. In **(F)**: $a_i = 0.1$, $k_i = 0.1$, $w_i = 0.1$ ($i = 1, 2$), $D_2 = 0.001$, $a'_{12} = 0.12$, $\zeta = 0.05$, $\kappa = 100$. In **(G)**: $a_i = 0.5$, $a'_i = 0.625$, $d_i = 0.5$, $d'_i = 0.5$, $k_i = 0.4$, $w_i = 0.5$ ($i = 1, 2$), $D_2 = 0.02$, $\kappa = 10$, $D_1 = 1.2D_2$, $\zeta = 0.1$. In **(H)**: $a_i = 0.1$, $a' = 0.12$, $k_i = 0.12$, $w_i = 0.3$, $d_i = 0.5$, $d'_i = 0.05$ ($i = 1, 2$), $D_2 = 0.02$, $\kappa = 100$, $\zeta = 0.8$. In **(I)**: $a_i = 0.5$, $k_i = 0.2$, $d_i = 0.8$, $w_i = 0.2$ ($i = 1, 2$), $D_2 = 0.008$, $\kappa = 60$, $\zeta = 0.8$.



Appendix-figure 4.

Intraspecific predator interference facilitates species coexistence regardless of stochasticity

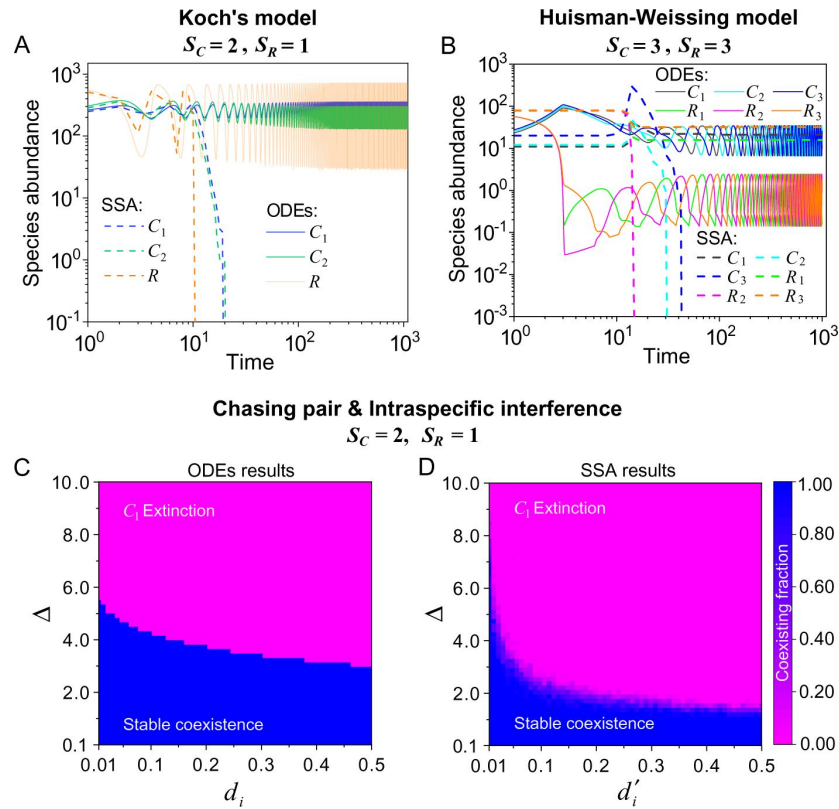
Here we consider the case of $S_C = 2, S_R = 1$. **(A)** A representative trajectory of species coexistence in the phase space simulated with ODEs. The fixed point (shown in red) is stable and globally attractive. **(B-C)** 3D phase diagrams in the ODEs studies. Here, D_i ($i = 1, 2$) is the only parameter that varies with the two consumer species, and $\Delta = (D_1 - D_2)/D_2$ measures the competitive difference between the two species. The parameter region below the blue surface yet above the red surface represents stable coexistence. The region below the red surface and above $\Delta = 0$ represents unstable fixed points (an empty set). **(D)** An exemplified transection corresponding to the $\Delta = 0.3$ plane in **(C)**. **(E)** Time courses of the species abundances simulated with ODEs or SSA. **(F)** Representative trajectories of species coexistence in the phase space simulated with SSA. The coexistence state is stable and globally attractive (see **(E)** for time courses, SSA results). **(A-F)** were calculated or simulated from [Eqs. 1](#), [2](#), [4](#). In **(A)**: $a_i = 0.1, a'_i = 0.125, d_i = 0.1, d'_i = 0.05, w_i = 0.1, k_i = 0.1$ ($i = 1, 2$); $D_1 = 0.0035, D_2 = 0.0038, \kappa = 100, \zeta = 0.3$. In **(B)**: $a_i = 0.1, d_i = 0.1, w_i = 0.1, k_i = 0.1$ ($i = 1, 2$); $D_2 = 0.001, \Delta \equiv (D_1 - D_2)/D_2, \kappa = 100, \zeta = 0.1$. In **(C-D)**: $a_i = 0.05, a'_i = 0.065, w_i = 0.1, k_i = 0.1$ ($i = 1, 2$); $D_2 = 0.002, \Delta \equiv (D_1 - D_2)/D_2, \kappa = 10, s = 0.1$. In **(D)**: $\Delta = 0.3$. In **(E-F)**: $a_i = 0.02, a'_i = 0.025, d_i = 0.7, d'_i = 0.7, w_i = 0.4, k_i = 0.05$ ($i = 1, 2$); $D_1 = 0.0160, D_2 = 0.0171, \kappa = 2000, \zeta = 5.5$.



Appendix-figure 5.

Outcomes of multiple consumers species competing for one resource species involving chasing pairs and intra- and inter-specific interference

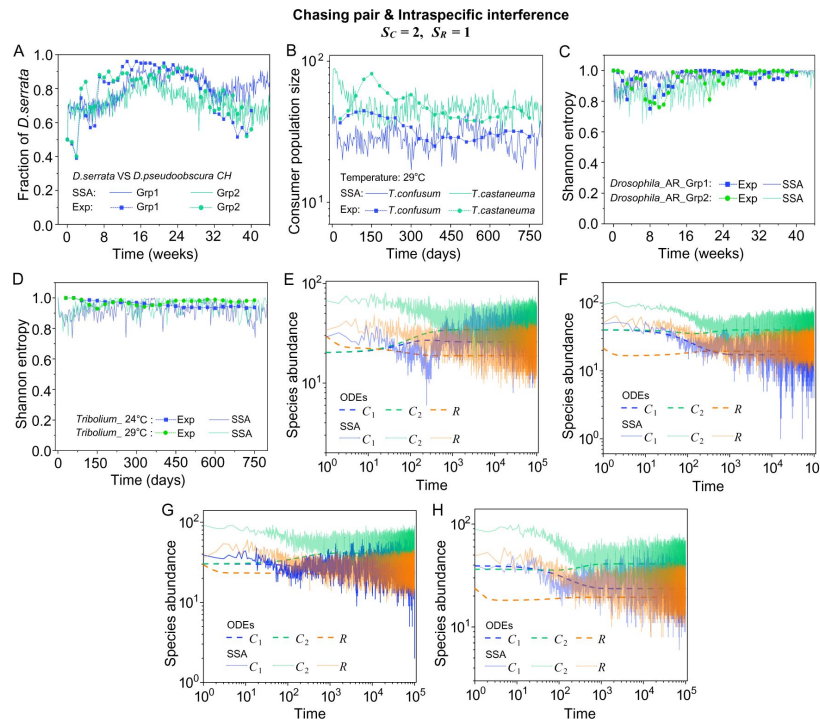
(A-E) The case involving two consumer species and one resource species ($S_C = 2, S_R = 1$). Here, D_i ($i = 1, 2$) is the only parameter that varies with the consumer species (with $D_1 > D_2$), and $\Delta = (D_1 - D_2)/D_2$ measures the competitive difference between the two species. (A) A 3D phase diagram. The parameter region below the blue surface yet above the red surface represents stable coexistence, while that below the red surface and above $\Delta = 0$ represents unstable fixed points (an empty set). (B) Time courses of the species abundances. Two consumer species may coexist with one type of resources at steady state. (C) Representative trajectories of species coexistence in the phase space. The fixed point (shown in red) is stable and globally attractive. (D) Consumer species may coexist indefinitely with the resources regardless of stochasticity. (E) Comparisons between numerical results and analytical solutions of the species abundances at fixed points. Color bars are analytical solutions while hollow bars are numerical results. The analytical solutions (marked with superscript “A”) were calculated from Eqs. S74–S75. (F-H) Time courses of species abundances in cases involving 6 or 20 consumer species and one type of resources ($S_C = 6$ or $20, S_R = 1$). All consumer species may coexist with one type of resource at a steady state, and this coexisting state is robust to stochasticity. (A-C, E, G) ODEs results. (D, F) ODEs and SSA results. (H) SSA results. The numerical results in (A-H) were calculated or simulated from Eqs. 1–4. In (A-C): $a_1 = 0.1, a'_1 = 0.12, k_1 = 0.1, w_1 = 0.1, (i = 1, 2); D_2 = 0.004, \kappa = 100, \zeta = 0.8, a'_{12} = 0.12, d'_2 = 0.5$. In (B-C): $d_i = 0.3, d'_i = 0.5, (i = 1, 2)$. In (D): $a_i = 0.1, a'_i = 0.14, k_i = 0.12, w_i = 0.15, d_i = 0.3, d'_i = 0.5, (i = 1, 2); D_1 = 0.0125, D_2 = 0.012, \kappa = 300, \zeta = 5.5, a'_{12} = 0.14, d'_2 = 5$. In (E): $a_i = 0.05, a'_i = 0.06, k_i = 0.1, w_i = 0.2, d_i = 0.5, d'_i = 0.15, (i = 1, 2); D_2 = 0.008, \kappa = 100, \zeta = 0.8, a'_{12} = 0.06, d'_{12} = 0.0005$. In (F): $a_i = 0.1, a'_i = 0.14, k_i = 0.15, w_i = 0.18, d_i = 2.8, d'_i = 0.02, a'_{ij} = 0.14, d'_{ij} = 0.15 (i, j = 1, \dots, 6; i \neq j); \kappa = 600, s = 100, D_1 = 0.0091, D_2 = 0.0084, D_3 = 0.0088, D_4 = 0.0096, D_5 = 0.0082, D_6 = 0.0093$. In (G-H): $a_i = 0.1, a'_i = 0.14, k_i = 0.2w_i = 0.18, d_i = 2.8, d'_i = 0.02, a'_{ij} = 0.14, d'_{ij} = 0.8, D_i = N(1, 0.1) \times 0.005 (i, j = 1, \dots, 20; i \neq j); \kappa = 1000, s = 500$.



Appendix-figure 6.

The influence of stochasticity on species coexistence

(A-B) Stochasticity jeopardizes species coexistence. Koch's model (Koch, 1974) and Huisman-Weissing model (Huisman and Weissing, 1999) were simulated with SSA using the same parameter settings as their deterministic model. Nevertheless, both cases of oscillating coexistence are vulnerable to stochasticity. See Ref. (Koch, 1974) and (Huisman and Weissing, 1999) for the parameters. (C-D) Phase diagrams in the scenario involving chasing pairs and intraspecific interference. Here, $S_C = 2$ and $S_R = 1$. d_i ($i = 1, 2$) is the only parameter varying with the consumer species (with $D_1 > D_2$), and $\Delta = (D_1 - D_2)/D_2$ represents the competitive difference between the two species. (C) The ODEs results. (D) The SSA results (with the same parameter region as (C)). The species' coexisting fraction in each pixel was calculated from 16 random repeats. (C) and (D) were calculated from Eqs. 1, 2, 4. In (C-D): $a_i = 0.1$, $a'_i = 0.125$, $d_i = 0.5$, $w_i = 0.1$, $k_i = 0.1$ ($i = 1, 2$); $\kappa = 100$, $\zeta = 5$, $D_2 = 0.0014$.



Appendix-figure 7.

A model of intraspecific predator interference explains two classical experiments that invalidate CEP. **(A)** In Ayala's experiment (Ayala, 1969), two *Drosophila* species (consumers) coexisted for 40 weeks with the same type of abiotic resources within a laboratory bottle. The time averages ($\bar{C}_i$) and standard deviation (δC_i) of the species' relative abundances for the experimental data or SSA results are:

$$\overline{C_{D.serrata_CH_Grp1}^{Exp(SSA)}} = 0.77 (0.80), \delta^{(R)} C_{D.serrata_CH_Grp1}^{Exp(SSA)} = 0.17 (0.08), \overline{C_{D.serrata_CH_Grp2}^{Exp(SSA)}} = 0.78 (0.73),$$

$$\delta^{(R)} C_{D.serrata_CH_Grp2}^{Exp(SSA)} = 0.14 (0.07),$$

where the superscript "(R)" represents relative abundances. **(B)** In Park's

experiment (Park, 1954), two *Tribolium* species coexisted for 750 days with the same food (flour). The time averages ($\bar{C}_i$)

and standard deviations (δC_i) of the species' abundances are: $\overline{C_{T.confusum_29^\circ C}^{Exp(SSA)}} = 33.4 (28.8),$

$$\delta C_{T.confusum_29^\circ C}^{Exp(SSA)} = 6.0 (5.4), \overline{C_{T.castaneuma_29^\circ C}^{Exp(SSA)}} = 48.8 (47.7), \delta C_{T.castaneuma_29^\circ C}^{Exp(SSA)} = 11.9 (9.9).$$

(A-B) The solid icons represent the experimental data, which are connected by the dotted lines for the sake of visibility. The solid lines stand for the SSA simulation results. **(C-D)** The Shannon entropies of each time point for the experimental or model-simulated

communities shown in **(B)** and **Fig. 2D-E**. Here, we calculated the Shannon entropies with $H(t) = -\sum_{i=1}^{S_C} P_i(t) \log_2(P_i(t))$,

where $P_i(t)$ is the probability that a consumer individual belongs to species C_i at the time stamp of t . The time averages ($\bar{H}$)

and standard deviations (δH) of the Shannon entropies are: $\bar{H}_{Exp(SSA)}^{Drosophila_AR_Grp1} = 0.95 (0.97),$

$$\delta H_{Exp(SSA)}^{Drosophila_AR_Grp1} = 0.06 (0.04), \bar{H}_{Exp(SSA)}^{Drosophila_AR_Grp2} = 0.94 (0.92), \delta H_{Exp(SSA)}^{Drosophila_AR_Grp2} = 0.07 (0.07),$$

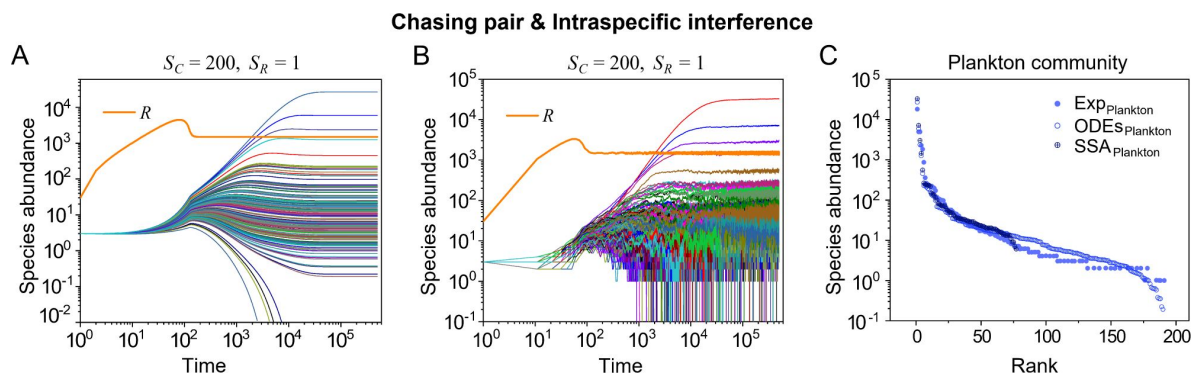
$$\bar{H}_{Exp(SSA)}^{Tribolium_24^\circ C} = 0.96 (0.92), \delta H_{Exp(SSA)}^{Tribolium_24^\circ C} = 0.02 (0.05), \bar{H}_{Exp(SSA)}^{Tribolium_29^\circ C} = 0.97 (0.94),$$

$$\delta H_{Exp(SSA)}^{Tribolium_29^\circ C} = 0.02 (0.05).$$

(E-H) Time courses of the species abundances in the scenario involving chasing pairs

and intraspecific interference. The time series in **(E-H)** correspond to the long-term version of that shown in **Fig. 2D**,

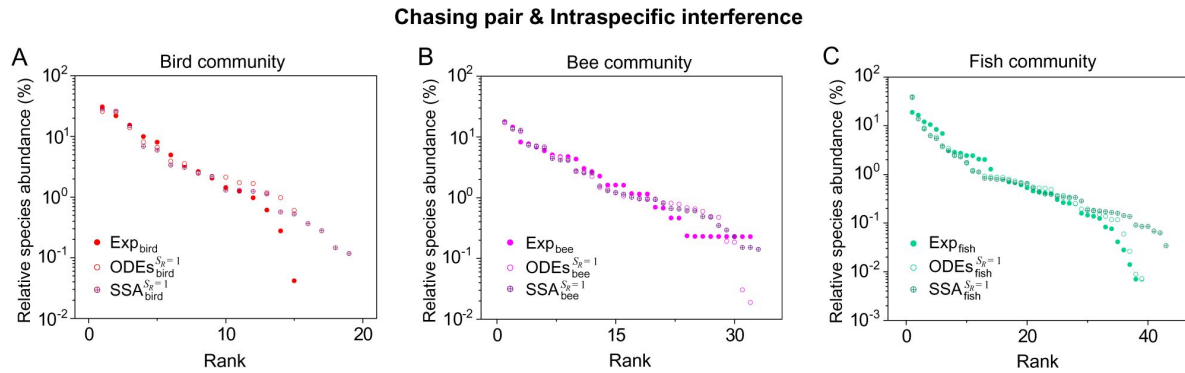
Appendix-fig. 7A, **Fig. 2E**, **Appendix-fig. 7B**, respectively. **(A-B, F-I)** were simulated from **Eqs. 1**, **2**, **4**. In **(A)**: $a_i = 0.3, a'_i = 0.33, w_i = 0.018, k_i = 4.8, d'_i = 5, d_i = 5.5 (i = 1, 2); D_2 = 0.010, \zeta = 35, \kappa = 10000, D_1 = 0.0132$. In **(B)**: $w_i = 0.02, k_i = 4.5, d'_i = 4, d_i = 4.5 (i = 1, 2); D_2 = 0.010, \zeta = 35, \kappa = 10000, a_i = 0.3, a'_i = 0.36 (i = 1, 2); D_1 = 0.0122$. In **(A-B)**: $\tau = 0.4$ Day (see).



Appendix-figure 8.

A model of intraspecific interference semi-quantitatively illustrates the rank-abundance curve of a plankton community ($S_C \gg S_R$).

(A-B) Intraspecific interference enables a wide range of consumers species to coexist with one type of resources. (A) Time courses of the species abundances simulated with ODEs. (B) Time series of the species abundances simulated with SSA (with the same parameter settings as (A)). (C) The rank-abundance curve of a plankton community. The solid dots represent the experimental data (marked with "Exp") reported in a recent study ([Ser-Giacomi et al., 2018](#)) (TARA_139.SUR.180.2000.DNA), while the hollow dots and those with "+" center are the ODEs and SSA results constructed from timestamp $t = 5.0 \times 10^5$ in the time series (see (A) and (B)), respectively. The Shannon entropies of the experimental data and simulation results for the plankton community are: $H_{\text{Exp(ODEs,SSA)}^{\text{Plankton}}} = 2.85(2.18, 2.00)$. In the model settings, $S_C = 200$ and $S_R = 1$. D_i ($i = 1, \dots, S_C$) is the only parameter that varies with the consumer species, which was randomly drawn from a Gaussian distribution $N(\mu, \sigma)$. Here, μ and σ are the mean and standard deviation of the distribution. The numerical results in (A-C) were simulated from Eqs. 1, 2, 4. In (A-C): $a_i = 0.1$, $a'_i = 0.125$, $d'_i = 0.5$, $d_i = 0.2$, $w_i = 0.2$, $k_i = 0.1$, $D_i = N(1, 0.38) \times 0.008$ ($i = 1, \dots, 200$), $\kappa = 10^5$, $\zeta = 150$.



Appendix-figure 9.

A model of intraspecific interference illustrates the rank-abundance curves across different ecological communities ($S_C \gg S_R$).

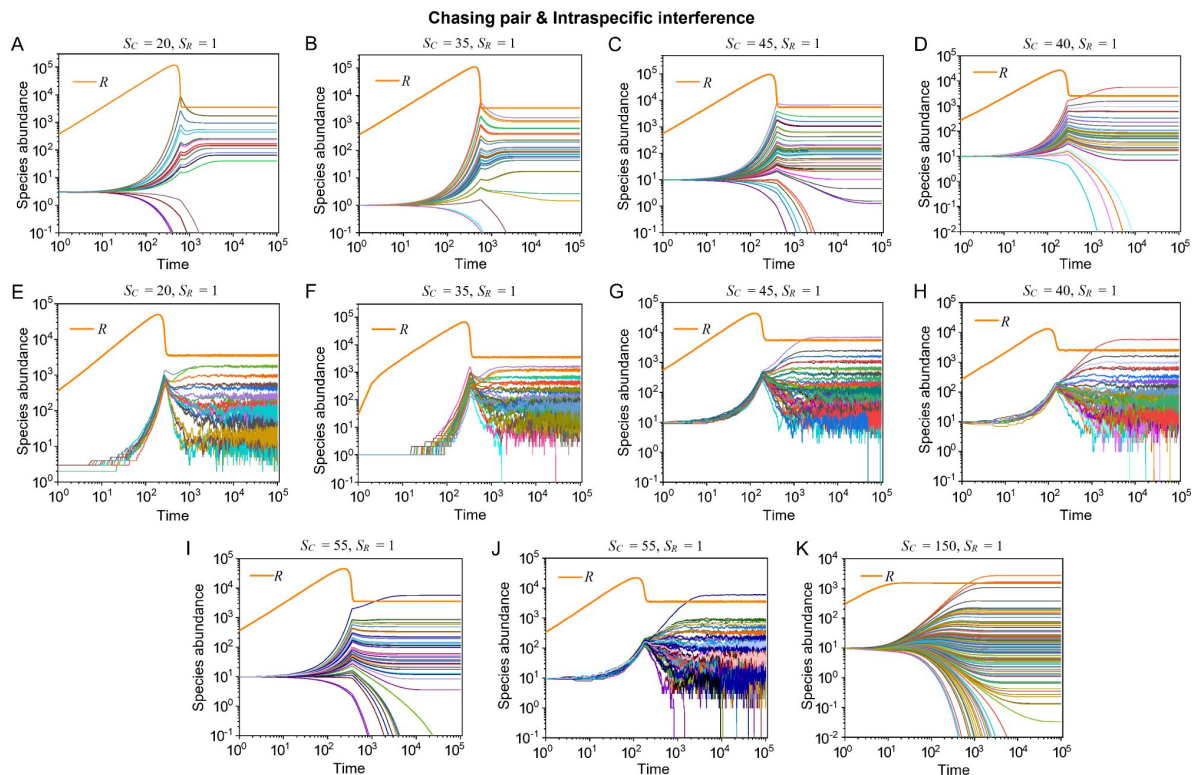
The solid dots represent the experimental data (marked with “Exp”) reported in existing studies ([Hubbell, 2001](#)), ([Holmes et al., 1986](#)), ([Cody et al., 1996](#)), while the hollow dots and those with “+” center are the ODEs and SSA results constructed from timestamp $t = 1.0 \times 10^5$ in the time series (see [Appendix-fig. 10A-G](#)), respectively. In the model settings, $S_R = 1$, $S_C = 20$ (in **(A)**), 35 (in **(B)**) or 45 (in **(C)**). D_i ($i = 1, \dots, S_C$) is the only parameter varying with the consumer species, which was randomly drawn from a Gaussian distribution. The Shannon entropies of the experimental data and simulation results for each ecological community are: $H_{\text{Exp(ODEs,SSA)}}^{\text{bird}} = 2.98(3.06, 2.98)$, $H_{\text{Exp(ODEs,SSA)}}^{\text{bee}} = 4.04(4.02, 4.02)$,

$H_{\text{Exp(ODEs,SSA)}}^{\text{fish}} = 3.78(3.40, 3.41)$. In the Kolmogorov-Smirnov (K-S) test, the probabilities (p values) that the

simulation results and the corresponding experimental data come from identical distributions are: $p_{\text{ODEs}}^{\text{bird}} = 0.89$,

$p_{\text{SSA}}^{\text{bird}} = 0.88$, $p_{\text{ODEs}}^{\text{bee}} = 0.47$, $p_{\text{SSA}}^{\text{bee}} = 0.75$, $p_{\text{ODEs}}^{\text{fish}} = 0.88$, $p_{\text{SSA}}^{\text{fish}} = 0.77$. With a significance threshold of 0.05,

none of the p values suggest there exists a statistically significant difference. The numerical results in **(A-C)** were simulated from [Eqs. 1](#), [2](#), [4](#). In **(A)**: $a_i = 0.1$, $a'_i = 0.125$, $d_i = 0.5$, $d'_i = 0.6$, $w_i = 0.22$, $k_i = 0.1$, $D_i = 0.016 \times N(1, 0.35)$ ($i = 1, \dots, 20$); $\zeta = 350$, $\kappa = 10^6$. In **(B)**: $a_i = 0.1$, $a'_i = 0.125$, $d_i = 0.5$, $d'_i = 0.6$, $w_i = 0.22$, $k_i = 0.1$, $D_i = 0.012 \times N(1, 0.35)$ ($i = 1, \dots, 35$); $\zeta = 350$, $\kappa = 10^6$. In **(C)**: $a_i = 0.1$, $a'_i = 0.14$, $d_i = 0.5$, $d'_i = 0.5$, $w_i = 0.2$, $k_i = 0.1$, $D_i = 0.015 \times N(1, 0.32)$ ($i = 1, \dots, 45$); $\zeta = 550$, $\kappa = 10^6$.

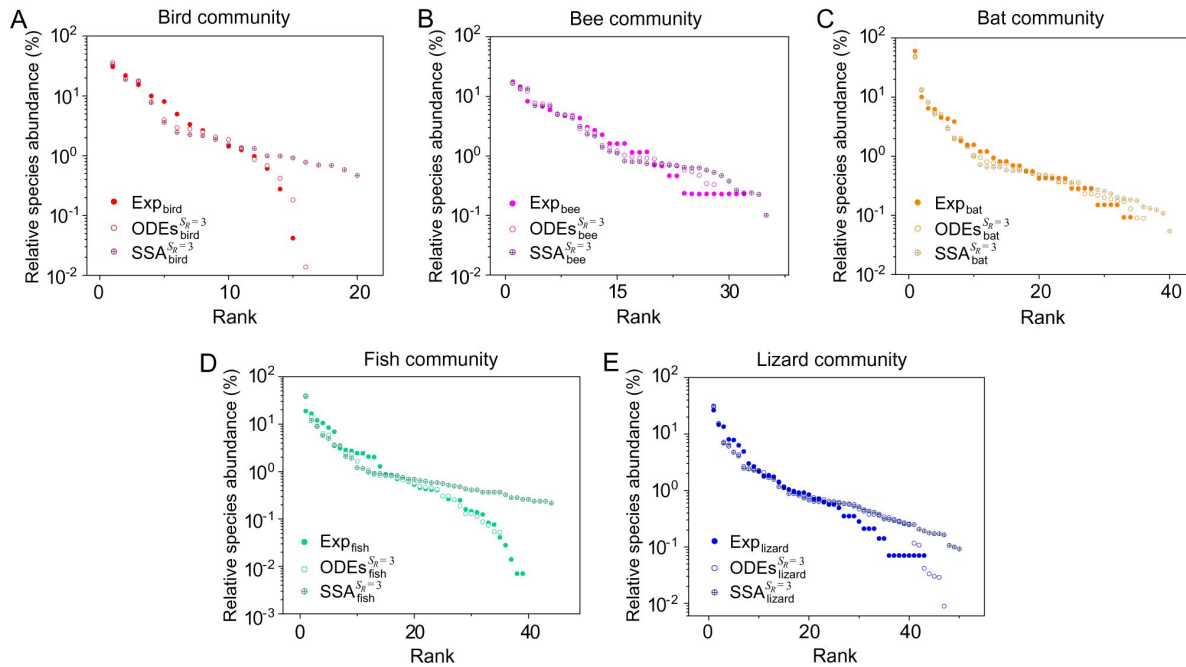


Appendix-figure 10.

Time courses of the species abundances in the scenario involving chasing pairs and intraspecific interference.

The time series in (A, E), (B, F), (C, G), (D, H), (I, J) and (K) correspond to that shown in [Appendix-fig. 9A-C](#), [Fig. 3D](#) (bat), [Fig. 3D](#) (lizard) and [Fig. 3C](#) (butterfly), respectively.

Chasing pair & Intraspecific interference



Appendix-figure 11.

A model of intraspecific interference illustrates the rank-abundance curves across different ecological communities ($S_C \gg S_R$).

The solid dots represent the experimental data (marked with "Exp") reported in existing studies ([Hubbell, 2001](#)), ([Holmes et al., 1986](#)), ([Cody et al., 1996](#)), ([Clarke et al., 2005](#)), while the hollow dots and those with "+" center are the ODEs and SSA results constructed from timestamp $t = 1.0 \times 10^9$ in the time series (see [Appendix-fig. 13](#)), respectively. In the model settings, $S_R = 3$, $S_C = 20$ (in (A)), 35 (in (B)), 40 (in (C)), 45 (in (D)) or 50 (in (E)). D_i ($i = 1, \dots, S_C$) is the only parameter varying with the consumer species, which was randomly drawn from a Gaussian distribution. The Shannon entropies of the experimental data and simulation results for each ecological community are: $H_{\text{Exp(ODEs,SSA)}}^{\text{Bird}} = 2.98(2.98, 3.26)$,

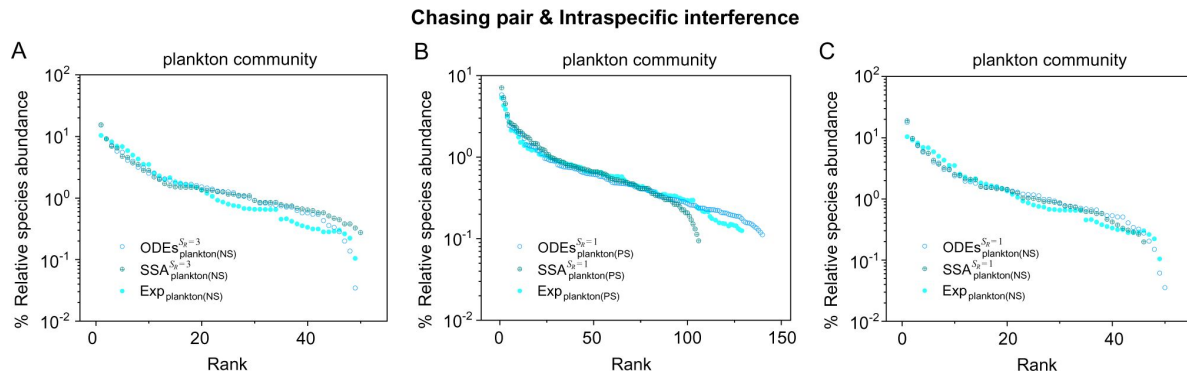
$$H_{\text{Exp(ODEs,SSA)}}^{\text{Bee}} = 4.04(4.34, 4.35), \quad H_{\text{Exp(ODEs,SSA)}}^{\text{Bat}} = 3.00(3.00, 3.00), \quad H_{\text{Exp(ODEs,SSA)}}^{\text{Fish}} =$$

$$3.78(3.28, 3.64), \quad H_{\text{Exp(ODEs,SSA)}}^{\text{lizard}} = 4.05(3.94, 3.94). \text{ In the K-S test, the } p \text{ values that the simulation results and the}$$

corresponding experimental data come from identical distributions are: $p_{\text{ODEs}}^{\text{Bird}} = 0.59$, $p_{\text{SSA}}^{\text{Bird}} = 0.43$, $p_{\text{ODEs}}^{\text{Bee}} = 0.47$,

$$p_{\text{SSA}}^{\text{Bee}} = 0.33, \quad p_{\text{ODEs}}^{\text{Bat}} = 0.42, \quad p_{\text{SSA}}^{\text{Bat}} = 0.27, \quad p_{\text{ODEs}}^{\text{Fish}} = 0.22, \quad p_{\text{SSA}}^{\text{Fish}} = 0.06, \quad p_{\text{ODEs}}^{\text{lizard}} = 0.56, \quad p_{\text{SSA}}^{\text{lizard}} = 0.36.$$

With a significance threshold of 0.05, none of the p values suggest there exists a statistically significant difference. The numerical results in (A-E) were simulated from [Eqs. 1](#), [2](#), [3](#), [4](#). In (A-E): $a_{ij} = 0.1$, $a'_i = 0.125$, $d_{ij} = 0.5$. In (A): $d'_i = 0.3$, $w_{ij} = 0.2$, $k_{ij} = 0.12$, $\kappa_1 = 8 \times 10^4$, $\kappa_2 = 5 \times 10^4$, $\kappa_3 = 3 \times 10^4$, $D_i = 0.021 \times N(1, 0.28)$ ($i = 1, \dots, 20$, $l = 1, 2, 3$); $\zeta_1 = 180$, $\zeta_2 = 160$, $\zeta_3 = 140$. In (B): $d'_i = 0.6$, $w_{ij} = 0.2$, $k_{ij} = 0.12$, $\kappa_1 = 8 \times 10^4$, $\kappa_2 = 5 \times 10^4$, $\kappa_3 = 3 \times 10^4$, $D_i = 0.017 \times N(1, 0.3)$ ($i = 1, \dots, 35$, $l = 1, 2, 3$); $\zeta_1 = 180$, $\zeta_2 = 160$, $\zeta_3 = 110$. In (C): $d'_i = 0.4$, $w_{ij} = 0.3$, $k_{ij} = 0.12$, $\kappa_1 = 10^5$, $\kappa_2 = 5 \times 10^4$, $\kappa_3 = 3 \times 10^4$, $D_i = 0.023 \times N(1, 0.34)$ ($i = 1, \dots, 40$, $l = 1, 2, 3$); $\zeta_1 = 180$, $\zeta_2 = 120$, $\zeta_3 = 40$. In (D): $d'_i = 0.3$, $w_{ij} = 0.3$, $k_{ij} = 0.12$, $\kappa_1 = 8 \times 10^4$, $\kappa_2 = 5 \times 10^4$, $\kappa_3 = 3 \times 10^4$, $D_i = 0.027 \times N(1, 0.32)$ ($i = 1, \dots, 45$, $l = 1, 2, 3$); $\zeta_1 = 80$, $\zeta_2 = 60$, $\zeta_3 = 40$. In (E): $d'_i = 0.3$, $w_{ij} = 0.3$, $k_{ij} = 0.2$, $\kappa_1 = 3 \times 10^5$, $\kappa_2 = 10^5$, $\kappa_3 = 3 \times 10^4$, $D_i = 0.034 \times N(1, 0.34)$ ($i = 1, \dots, 50$, $l = 1, 2, 3$); $\zeta_1 = 380$, $\zeta_2 = 260$, $\zeta_3 = 140$.



Appendix-figure 12.

A model of intraspecific interference illustrates the rank-abundance curves across different plankton communities

($S_C \gg S_R$). The solid dots represent the experimental data (marked with “Exp”) reported in a recent study *Fuhrman et al.* (2008), while the hollow dots and those with “+” center are the ODEs and SSA results constructed from timestamp $t = 1.0 \times 10^5$ in the time series (see **Appendix-fig. 13**), respectively. The plankton community data were obtained separately from the Norwegian Sea (NS) and Pacific Station (PS). In the model settings, $S_R = 1$ (in **(B-C)**), 3 (in **(A)**); $S_C = 50$ (in **(A, C)**), 150 (in **(B)**). D_i ($i = 1, \dots, S_C$) is the only parameter varying with the consumer species, which was randomly drawn from a Gaussian distribution. The Shannon entropies of the experimental data and simulation results for each plankton community are:

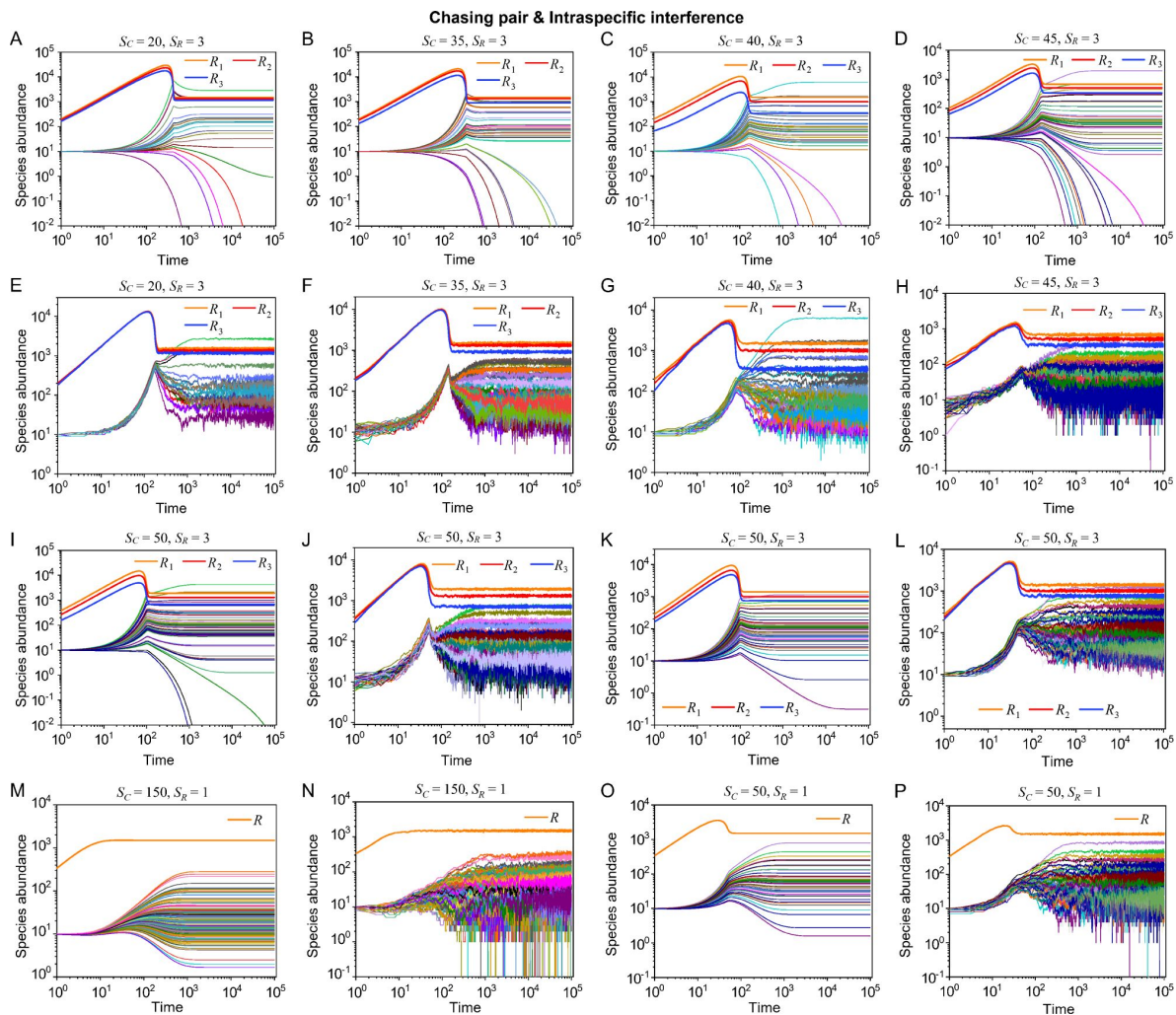
$$H_{\text{Exp(ODEs,SSA)}}^{\text{plankton(NS)}} = 4.67(4.85, 4.90) \text{ for } S_R = 3; H_{\text{Exp(ODEs,SSA)}}^{\text{plankton(PS)}} = 4.68(6.53, 6.16),$$

$$H_{\text{Exp(ODEs,SSA)}}^{\text{plankton(NS)}} = 4.67(4.74, 4.64) \text{ for } S_R = 1. \text{ In the K-S test, the } p \text{ values that the simulation results and the}$$

corresponding experimental data come from identical distributions are: $p_{\text{ODEs}}^{\text{plankton(NS)}} = 0.31$, $p_{\text{SSA}}^{\text{plankton(NS)}} = 0.14$

for $S_R = 3$; $p_{\text{ODEs}}^{\text{plankton(PS)}} = 0.08$; $p_{\text{SSA}}^{\text{plankton(PS)}} = 0.28$, $p_{\text{ODEs}}^{\text{plankton(NS)}} = 0.46$, $p_{\text{SSA}}^{\text{plankton(NS)}} = 0.37$ for $S_R = 1$. With a

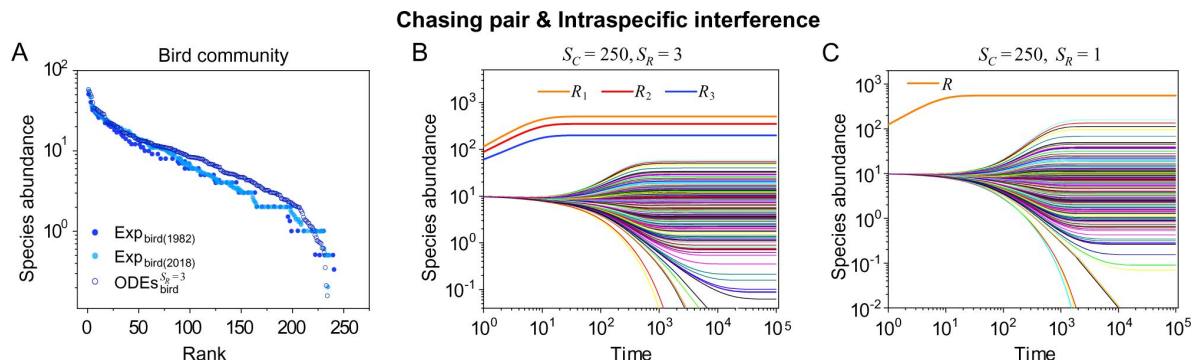
significance threshold of 0.05, none of the p values suggest there exists a statistically significant difference. The numerical results in **(A-C)** were simulated from **Eqs. 1**, **2**, **4**. In **(A)**: $a_{ij} = 0.1$, $a'_i = 0.125$, $d_{ij} = 0.5$, $d'_i = 0.2$, $w_{ij} = 0.3$, $k_{ij} = 0.2$, $\kappa = 8 \times 10^4$, $\kappa_2 = 5 \times 10^4$, $\kappa_3 = 3 \times 10^4$, $\zeta_1 = 280$, $\zeta_2 = 200$, $\zeta_3 = 150$, $D_i = 0.035 \times N(1, 0.25)$ ($i = 1, \dots, 50$, $l = 1, 2, 3$). In **(B)**: $a_i = 0.1$, $a'_i = 0.125$, $d_i = 0.3$, $d'_i = 0.3$, $w_i = 0.3$, $k_i = 0.2$, $D_i = 0.025 \times N(1, 0.25)$ ($i = 1, \dots, 150$, $l = 1, 2, 3$); $\zeta = 350$, $\kappa = 10^4$. In **(C)**: $a_i = 0.1$, $a'_i = 0.125$, $d_i = 0.3$, $d'_i = 0.3$, $w_i = 0.3$, $k_i = 0.2$, $\zeta = 350$, $\kappa = 10^4$, $D_i = 0.03 \times N(1, 0.27)$ ($i = 1, \dots, S_C$).



Appendix-figure 13.

Time courses of the species abundances in the scenario involving chasing pairs and intraspecific interference.

The time series in (A, E), (B, F), (C, G), (D, H), (I, J), (K, L), (M, N) and (O, P) correspond to that shown in **Appendix-figs. 11A-E** and **12A-C**, respectively.



Appendix-figure 14.

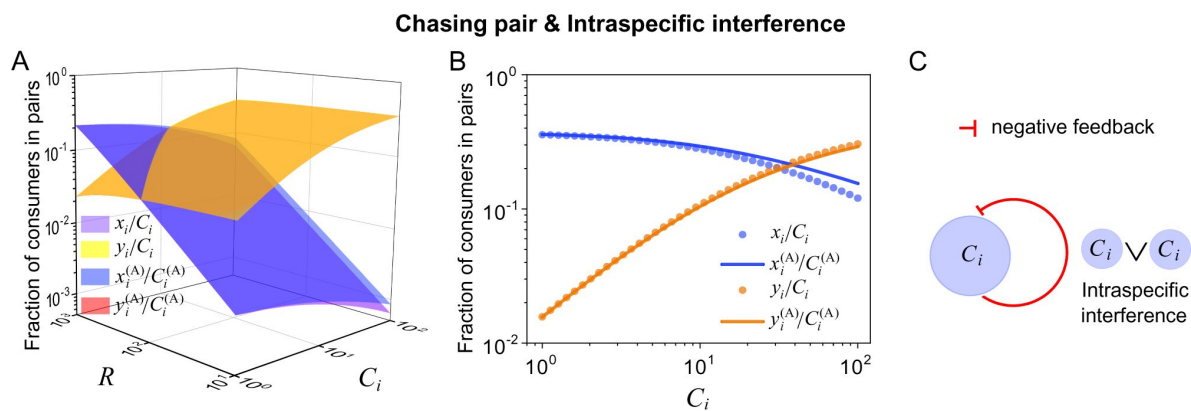
A model of intraspecific interference illustrates the rank-abundance curve of a bird community ($S_C \gg S_R$).

(A) The rank-abundance curve. The solid dots represent the bird community data (marked with “Exp”) collected longitudinally within the same Amazonian region in 1982 (blue) and 2018 (cyan) (Terborgh et al., 1990, (Martinez et al., 2023)). The hollow dots are the ODEs results constructed from timestamp $t = 1.0 \times 10^5$ in the time series (see (B)). (B) Time courses of the species abundances simulated with ODEs. In the model settings, $S_R = 3$ and $S_C = 250$. D_i ($i = 1, \dots, S_C$) is the only parameter varying with the consumer species, which was randomly drawn from a Gaussian distribution. The Shannon entropies of the experimental data and simulation results for the bird community are

$$H_{\text{Exp(ODEs)}}^{\text{bird(1982/2018)}} = 5.67/6.63(7.31). \text{ In the K-S test, the } p \text{ values that the simulation results and the corresponding}$$

experimental data come from identical distributions are: $p_{\text{ODEs}}^{\text{bird(1982)}} = 0.28$, $p_{\text{ODEs}}^{\text{bird(2018)}} = 0.46$. With a significance

threshold of 0.05, none of the p values suggest there exists a statistically significant difference. (C) Time courses of the species abundances simulated with ODEs corresponding to Fig. 3C (bird), and the simulation parameters are the same as Fig. 3C (bird). The numerical results in (A-C) were simulated from Eqs. 1, 2, 4. In (A-B): $a_{ij} = 0.1$, $a'_i = 0.125$, $d_{ij} = 0.5$, $d'_i = 0.6$, $w_{ij} = 0.3$, $k_{ij} = 0.2$, $D_i = 0.032 \times N(1, 0.17)$ ($i = 1, \dots, 250$, $l = 1, 2, 3$); $\kappa_1 = 5 \times 10^4$, $\kappa_2 = 3 \times 10^4$, $\kappa_3 = 10^4$, $\zeta_1 = 100$, $\zeta_2 = 70$, $\zeta_3 = 40$.



Appendix-figure 15.

Intraspecific interference results in an underlying negative feedback loop and thus promotes biodiversity.

(A-B) The fraction of consumer individuals engaged in pairwise encounter. Here, $S_C = 40$ and $S_R = 1$. x_i represents $C_i^{(P)} \vee R^{(P)}$, and y_i represents $C_i^{(P)} \vee C_i^{(P)}$. x_i/C_i and y_i/C_i stand for the fractions of consumer individuals within a chasing pair and an interference pair, respectively. The numerical results were calculated from [Eq. S55](#), while the analytical solutions (marked with superscript "(A)") were calculated from [Eq. S59](#). The orange surface in **(A)** is an overlap of the red and yellow surfaces. **(C)** The formation of intraspecific interference results in a self-inhibiting negative feedback loop. In **(A-B)**: $a_i = 0.0015$, $d'_i = 0.0021$, $d_i = 0.1$, $d'_i = 0.05$, $k_i = 5$. In **(B)**: $R = 2000$.

References

- Arditi R, Ginzburg LR (1989) **Coupling in predator-prey dynamics: ratio-dependence** *Journal of Theoretical Biology* **139**:311–326 [https://doi.org/10.1016/S0022-5193\(89\)80211-5](https://doi.org/10.1016/S0022-5193(89)80211-5)
- Ayala FJ (1969) **Experimental invalidation of the principle of competitive exclusion** *Nature* **224**:1076–1079 <https://doi.org/10.1038/2241076a0>
- Beddington JR (1975) **Mutual interference between parasites or predators and its effect on searching efficiency** *Journal of Animal Ecology* **44**:331–340 <https://doi.org/10.2307/3866>
- Boomsma JJ, Baer B, Heinze J (2005) **The evolution of male traits in social insects** *Annual Review of Entomology* **50**:395–420 <https://doi.org/10.1146/annurev.ento.50.071803.130416>
- Cantrell RS, Cosner C, Ruan S (2004) **Intraspecific interference and consumer-resource dynamics** *Discrete and Continuous Dynamical Systems - B* **4**:527–546 <https://doi.org/10.3934/dcdsb.2004.4.527>
- Chen S, Lee AY, Bowens NM, Huber R, Kravitz EA (2002) **Fighting fruit flies: a model system for the study of aggression** *PNAS* **99**:5664–5668 <https://doi.org/10.1073/pnas.082102599>
- Chesson P (1990) **Macarthur's consumer-resource model** *Journal of Theoretical Biology* **37**:26–38 [https://doi.org/10.1016/0040-5809\(90\)90025-Q](https://doi.org/10.1016/0040-5809(90)90025-Q)
- Chesson P (2000) **Mechanisms of maintenance of species diversity** *Annual Review of Ecology and Systematics* **31**:343–336
- Clarke FM, Rostant LV, Racey PA (2005) **Life after logging: post-logging recovery of a neotropical bat community** *Journal of Applied Ecology* **42**:409–420 <https://doi.org/10.1111/j.1365-2664.2005.01024.x>
- Cody ML, Smallwood JA (1996) **Long-Term Studies of Vertebrate Communities**
- Dalziel BD, Novak M, Watson JR, Ellner SP (2021) **Collective behaviour can stabilize ecosystems** *Nature Ecology & Evolution* **5**:1435–1440 <https://doi.org/10.1038/s41559-021-01517-w>
- Daniel R (2005) **The metagenomics of soil** *Nature Reviews Microbiology* **3**:470–478 <https://doi.org/10.1038/nrmicro1160>
- Dankert H, Wang L, Hoopfer ED, Anderson DJ, Perona P (2009) **Automated monitoring and analysis of social behavior in drosophila** *Nature Methods* **6**:297–303 <https://doi.org/10.1038/nmeth.1310>
- DeAngelis DL, Goldstein RA, O'Neill RV (1975) **A model for tropic interaction** *Ecology* **56**:881–892 <https://doi.org/10.2307/1936298>
- Devries PJ, Murray D, Lande R (1997) **Species diversity in vertical, horizontal, and temporal dimensions of a fruitfeeding butterfly community in an ecuadorian rainforest** *Biological Journal of the Linnean Society* **62**:343–364 <https://doi.org/10.1006/bjil.1997.0155>

- Fuhrman JA, Steele JA, Hewson I, Schwalbach MS, Brown MV, Green JL, Brown JH (2008) **A latitudinal diversity gradient in planktonic marine bacteria** *PNAS* **105**:7774–7778 <https://doi.org/10.1073/pnas.0803070105>
- Gause GF (1934) **The Struggle for Existence**
- Gillespie DT (2007) **Stochastic simulation of chemical kinetics** *Annual Review of Physical Chemistry* **58**:35–55 <https://doi.org/10.1146/annurev.physchem.58.032806.104637>
- Goyal A, Lu N, Bajic D, Estrela S, Tikhonov M, Sanchez-Gorostiaga A, Segre D, Mehtal P, Sanchezvz A (2018) **Emergent simplicity in microbial community assembly** *Science* **361**:469–474 <https://doi.org/10.1126/science.aat1168>
- Goyal A, Maslov S (2018) **Diversity, stability, and reproducibility in stochastically assembled microbial ecosystems** *Physical Review Letters* **120** <https://doi.org/10.1103/PhysRevLett.120.158102>
- Grilli J, Barabás G, Michalska-Smith MJ, Allesina S (2017) **Higher-order interactions stabilize dynamics in competitive network models** *Nature* **548**:210–213 <https://doi.org/10.1038/nature23273>
- Grimm V, Railsback SF (2013) **Individual-based Modeling and Ecology**
- Gupta D, Garlaschi S, Suweis S, Azaele S, Maritan A (2021) **Effective resource competition model for species coexistence** *Physical Review Letters* **127** <https://doi.org/10.1103/PhysRevLett.127.208101>
- Hardin G (1960) **The competitive exclusion principle** *Science* **131**:1292–1297 <https://doi.org/10.1126/science.131.3409.1292>
- Hsu SB, Ruan S, Yang TH (2013) **On the dynamics of two-consumers-one-resource competing systems with Beddington-DeAngelis functional response** *Discrete and Continuous Dynamical Systems-B* **18**:2331–2353 <https://doi.org/10.3934/dcdsb.2013.18.2331>
- Hoorn C *et al.* (2010) **Amazonia through time: Andean uplift, climate change, landscape evolution, and biodiversity** *Science* **330**:927–931 <https://doi.org/10.1126/science.1194585>
- Hu J, Amor DR, Barbier M, Bunin G, Gore J (2022) **Emergent phases of ecological diversity and dynamics mapped in microcosms** *Science* **378**:85–89 <https://doi.org/10.1126/science.abm7841>
- Hubbell SP (2001) **The Unified Neutral Theory of Biodiversity and Biogeography**
- Huisman G, Boer RJD (1997) **A formal derivation of the “beddington” functional response** *Journal of Theoretical Biology* **185**:389–400 <https://doi.org/10.1006/jtbi.1996.0318>
- Huisman J, Weissing FJ (1999) **Biodiversity of plankton by species oscillations and chaos** *Nature* **402**:407–410 <https://doi.org/10.1038/46540>
- Hutchinson GE (1961) **The paradox of the plankton** *The American Naturalist* **95**:137–145 <https://doi.org/10.1086/282171>

- Kelsic ED, Zhao J, Vetsigian K, Kishony R (2015) **Counteraction of antibiotic production and degradation stabilizes microbial communities** *Nature* **521**:516–519 <https://doi.org/10.1038/nature14485>
- Koch AL (1974) **Competitive coexistence of two predators utilizing the same prey under constant environmental conditions** *Journal of Theoretical Biology* **44**:387–395 [https://doi.org/10.1016/0022-5193\(74\)90169-6](https://doi.org/10.1016/0022-5193(74)90169-6)
- Lczárán T, Hoekstra RF, Pagie L (2002) **Chemical warfare between microbes promotes biodiversity** *PNAS* **99**:786–790 <https://doi.org/10.1073/pnas.012399899>
- Levin SA (1970) **Community equilibria and stability, and an extension of the competitive exclusion principle** *The American Naturalist* **104**:413–423
- Levin SA (1974) **Dispersion and population interactions** *The American Naturalist* **108**:207–228 <https://doi.org/10.1086/282900>
- Levins R (1979) **Coexistence in a variable environment** *The American Naturalist* **114**:765–783 <https://doi.org/10.1086/283527>
- MacArthur R (1969) **Species packing, and what competition minimizes** *PNAS* **64**:1369–1371 <https://doi.org/10.1073/pnas.64.4.1369>
- MacArthur R (1970) **Species packing and competitive equilibrium for many species** *Journal of Theoretical Biology* **1**:1–11 [https://doi.org/10.1016/0040-5809\(70\)90039-0](https://doi.org/10.1016/0040-5809(70)90039-0)
- MacArthur R, Levins R (1964) **Competition, habitat selection, and character displacement in a patchy environment** *PNAS* **51**:1207–1210 <https://doi.org/10.1073/pnas.51.6.1207>
- Martín PV, Bucek A, Bourguignon T, Pigolotti S (2020) **Ocean currents promote rare species diversity in protists** *Science Advances* **6** <https://doi.org/10.1126/sciadv.aaz9037>
- Martmez AE *et al.* (2023) **The structure and organisation of an Amazonian bird community remains little changed after nearly four decades in Manu National Park** *Ecology Letters* **26**:335–34 <https://doi.org/10.1111/ele.14159>
- McGehee R, Armstrong RA (1977) **Some mathematical problems concerning the ecological principle of competitive exclusion** *Journal of Differential Equations* **23**:30–52 [https://doi.org/10.1016/0022-0396\(77\)90135-8](https://doi.org/10.1016/0022-0396(77)90135-8)
- Menon P, Billen G, Servais P (2003) **Mortality rates of autochthonous and fecal bacteria in natural aquatic ecosystems** *Water Research* **37**:4151–4158 [https://doi.org/10.1016/S0043-1354\(03\)00349-X](https://doi.org/10.1016/S0043-1354(03)00349-X)
- Niehaus L *et al.* (2019) **Microbial coexistence through chemical-mediated interactions** *Nature Communications* **10**:2052–474 <https://doi.org/10.1038/s41467-019-10062-x>
- Park T (1954) **Experimental studies of interspecies competition ii temperature, humidity, and competition in two species of tribolium** *Physiological Zoology* **27**:177–238 <https://doi.org/10.1086/physzool.27.3.30152164>
- Pennisi E (2005) **What determines species diversity?** *Science* **309**:90–90 <https://doi.org/10.1126/science.309.5731.9>

- Posfai A, Taillefumier T, Wingreen NS (2017) **Metabolic trade-offs promote diversity in a model ecosystem** *Physical Review Letters* **118** <https://doi.org/10.1103/PhysRevLett.118.028103>
- Ratzke C, Barrere J, Gore J (2020) **Strength of species interactions determines biodiversity and stability in microbial communities** *Nature Ecology & Evolution* **4**:376–383 <https://doi.org/10.1038/s41559-020-1099-4>
- Ruxton GD, Gurney WSC, de Roos AM (1992) **Interference and generation cycles** *Theoretical Population Biology* **42**:235–253 [https://doi.org/10.1016/0040-5809\(92\)90014-K](https://doi.org/10.1016/0040-5809(92)90014-K)
- Ser-Giacomi E, Zinger L, Malviya S, Vargas CD, Karsenti E, Bowler C, Monte SD (2018) **Ubiquitous abundance distribution of non-dominant plankton across the global ocean** *Nature Ecology & Evolution* **2**:1243–1249 <https://doi.org/10.1038/s41559-018-0587-2>
- Sunagawa S *et al.* (2020) **Tara oceans: towards global ocean ecosystems biology** *Nature Reviews Microbiology* **18**:428–445 <https://doi.org/10.1038/s41579-020-0364-5>
- Terborgh J, Robinson SK, Parker TA, Munn CA, Pierpont N (1990) **Structure and organization of an amazonian forest bird community** *Ecological Monographs* **60**:213–238 <https://doi.org/10.2307/1943045>
- Thingstad TF (2000) **Elements of a theory for the mechanisms controlling abundance, diversity, and biogeochemical role of lytic bacterial viruses in aquatic systems** *Limnology and Oceanography* **45**:1320–1328 <https://doi.org/10.4319/lo.2000.45.6.1320>
- Vargas CD *et al.* (2015) **Eukaryotic plankton diversity in the sunlit ocean** *Science* **348** <https://doi.org/10.1126/science.1261605>
- Vetsigian K (2017) **Diverse modes of eco-evolutionary dynamics in communities of antibiotic producing microorganisms** *Nature Ecology & Evolution* **1** <https://doi.org/10.1038/s41559-017-0189>
- Wang X, Liu YY (2020) **Overcome competitive exclusion in ecosystems** *iScience* **23** <https://doi.org/10.1016/j.isci.2020.101009>
- Wang X, Xia K, Yang X, Tang C (2019) **Growth strategy of microbes on mixed carbon sources** *Nature Communications* **10** <https://doi.org/10.1038/s41467-019-09261-3>
- Weiner BG, Posfai A, Wingreen NS (2019) **Spatial ecology of territorial populations** *PNAS* **116**:17874–17879 <https://doi.org/10.1073/pnas.1911570116>
- Xue C, Goldenfeld N (2017) **Coevolution maintains diversity in the stochastic “kill the winner” model** *Physical Review Letters* **119** <https://doi.org/10.1103/PhysRevLett.119.268101>
- Ayala FJ (1969) **Experimental invalidation of the principle of competitive exclusion** *Nature* **224**:1076–1079
- Beddington J R (1975) **Mutual interference between parasites or predators and its effect on searching efficiency** *J. Anim. Ecol* **44**:331–340
- Clarke FM, Racey L (2005) **Life after logging: post-logging recovery of a neotropical bat community** *J. Appl. Ecol* **42**:409–420
- Cody ML, Smallwood JA (1996) **Long-Term Studies of Vertebrate Communities**

- DeAngelis DL, Goldstein RA, O'Neill RV (1975) **A model for tropic interaction** *Ecology* **56**:881–892
- Fuhrman JA, et al. (2008) **A latitudinal diversity gradient in planktonic marine bacteria** *Proc. Natl. Acad. Sci. U.S.A* **105**:7774–7778
- Gowaty PA, et al. (2010) **Polyandry increases offspring viability and mother productivity but does not decrease mother survival in *Drosophila pseudoobscura*** *The Proceedings of the National Academy of Sciences* **107**:13771–13776
- Holmes RT, Sherry TW, Sturges FW (1986) **Bird Community Dynamics in a Temperate Deciduous Forest: Long-Term Trends at Hubbard Brook** *Ecol. Monogr* **56**:201–220
- Hubbell SP (2001) **The Unified Neutral Theory of Biodiversity and Biogeography**
- Huisman J, Weissing FJ (1999) **Biodiversity of plankton by species oscillations and chaos** *Nature* **402**:407–410
- Koch AL (1974) **Competitive coexistence of two predators utilizing the same prey under constant environmental conditions** *J. Theor. Biol* **44**:387–395
- Levin SA (1970) **Community equilibria and stability, and an extension of the competitive exclusion principle** *Am. Nat* **104**:413–423
- MacArthur R (1970) **Species packing and competitive equilibrium for many species** *Theor. Popul. Biol* **1**:1–11
- MacArthur R, Levins R (1964) **Competition, habitat selection, and character displacement in a patchy environment** *Proc. Natl. Acad. Sci. U.S.A* **51**:1207–1210
- Martínez AE, et al. (2023) **The structure and organisation of an Amazonian bird community remains little changed after nearly four decades in Manu National Park** *Ecol. Lett* **26**:335–346
- Menon P, Billen G, Servais P (2003) **Mortality rates of autochthonous and fecal bacteria in natural aquatic ecosystems** *Water Res* **37**:4151–4158
- Narayan VP, Wilson AJ, Chenoweth SF (2022) **Genetic and social contributions to sex differences in lifespan in *Drosophila serrata*** *Journal of Evolutionary Biology* **35**:657–663
- Park T (1954) **Experimental studies of interspecies competition ii. temperature, humidity, and competition in two species of tribolium** *Physiol. Zool* **27**:177–238
- Ser-Giacomi E, et al. (2018) **Ubiquitous abundance distribution of non-dominant plankton across the global ocean** *Nat. Ecol. Evol* **2**:1243–1249
- Terborgh J, Robinson SK, Parker TA, Munn CA, Pierpont N (1990) **Structure and organization of an amazonian forest bird community** *Ecol. Mono* **60**:213–238
- Wang X, Liu YY (2020) **Overcome competitive exclusion in ecosystems** *iScience* **23**

Editors

Reviewing Editor

Babak Momeni

Boston College, Chestnut Hill, United States of America

Senior Editor

Aleksandra Walczak

École Normale Supérieure - PSL, Paris, France

Reviewer #1 (Public Review):

Summary:

The manuscript considers a mechanistic extension of MacArthur's consumer-resource model to include chasing down of food and potential encounters between the chasers (consumers) that lead to less efficient feeding in the form of negative feedback. After developing the model, a deterministic solution and two forms of stochastic solutions are presented, in agreement with each other. Finally, the model is applied to explain observed coexistence and rank-abundance data.

Strengths:

- The application of the theory to natural rank-abundance curves is impressive.
- The comparison with the experiments that reject the competitive exclusion principle is promising. It would be fascinating to see if in, e.g. insects, the specific interference dynamics could be observed and quantified and whether they would agree with the model.
- The results are clearly presented; the methods adequately described; the supplement is rich with details.
- There is much scope to build upon this expansion of the theory of consumer-resource models. This work can open up new avenues of research.

Weaknesses:

- Though more and better data could be used to constrain and validate the modeling, given this is a theory-driven manuscript, their results are sufficient.

<https://doi.org/10.7554/eLife.93115.2.sa2>

Reviewer #3 (Public Review):

Summary:

In this manuscript, the authors extend previous work on the role of predator interference in species coexistence. Previous theoretical work (for example, using the Beddington-DeAngelis model) has shown that predator interference allows for multiple predators to coexist on the same prey. While the Beddington-DeAngelis has been influential in theoretical ecology, it has also been criticized at times for several unusual assumptions, most critically, that predators interfere with each other regardless of whether they are already engaged in another interaction. There has been considerable work since then which has sought either to find sets of assumptions that lead to the B-D equation or to derive alternative equations from a more realistic set of assumptions (Ruxton et al. 1992; Cosner et al. 1999; Broom et al. 2010; Geritz and Gyllenberg 2012). This paper represents another effort to more rigorously derive a model of predator interference by borrowing concepts from chemical reaction kinetics (the

approach is similar to previous work: Ruxton et al. 1992). The main point of difference is that the model in the current manuscript allows for 'chasing pairs', where a predator and prey engage with one another to the exclusion of other interactions, a situation Ruxton et al. (1992) do not consider. While the resulting functional response is quite complex, the authors show that under certain conditions, one can get an analytical expression for the functional response of a predator as a function of predator and resource densities. They then go on to show that including intraspecific interference allows for the coexistence of multiple species on one or a few resources, and demonstrate that this result is robust to demographic stochasticity. This work provides additional support for the idea that predator interference allows multiple predators to persist with a shared resource.

Strengths:

I appreciate the effort to rigorously derive interaction rates from models of individual behaviors. As currently applied, functional responses (FRs) are estimated by fitting equations to feeding rate data across a range of prey or predator densities. In practice, such experiments are only possible for a limited set of species. This is problematic because whether a particular FR allows stability or coexistence depends on not just its functional form, but also its parameter values. The promise of the approach taken here is that one might be able to derive the functional response parameters of a particular predator species from species traits or more readily measurable behavioural data.

Weaknesses:

The main weakness of this paper is that while it is technically sound, it doesn't change the fundamental intuition gained from more phenomenological models of predator interference: as one species becomes more common, it limits its own growth (manifested by less time spent searching for/handling resources due to interference), such that it does not exclude the existence of a competitor species. However, given the authors use a different model formulation that has been used in past studies, it suggests that predator interference will likely tend to promote coexistence regardless of some of the technical details in how it is formulated in a model.

The formulation of chasing-pair engagements assumes that prey being chased by a predator are unavailable to other predators. While this may hold in some predator-prey, it does not hold for many others, perhaps limiting some results' generality.

Summary:

The manuscript by Kang et al investigates how the consideration of pairwise encounters (consumer-resource chasing, intraspecific consumer pair, and interspecific consumer pair) influences the community assembly results. To explore this, they presented a new model that considers pairwise encounters and intraspecific interference among consumer individuals, which is an extension of the classical Beddington-DeAngelis (B-D) phenomenological model, incorporating detailed considerations of pairwise encounters and intraspecific interference among consumer individuals. Later, they connected with several experimental datasets.

Strengths:

They found that the negative feedback loop created by the intraspecific interference allows a diverse range of consumer species to coexist with only one or a few types of resources. Additionally, they showed that some patterns of their model agree with experimental data, including time-series trajectories of two small in-lab community experiments and the rank-abundance curves from several natural communities. The presented results here are interesting and present another way to explain how the community overcomes the competitive exclusion principle.

Weaknesses:

The authors did a great job of satisfactorily addressing each of my concerns raised in the previous round. I did not detect additional weaknesses.

<https://doi.org/10.7554/eLife.93115.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

The manuscript considers a mechanistic extension of MacArthur's consumer-resource model to include chasing down food and potential encounters between the chasers (consumers) that lead to less efficient feeding in the form of negative feedback. After developing the model, a deterministic solution and two forms of stochastic solutions are presented, in agreement with each other. Finally, the model is applied to explain observed coexistence and rank-abundance data.

We thank the reviewer for the accurate summary of our manuscript.

Strengths:

The application of the theory to natural rank-abundance curves is impressive. The comparison with the experiments that reject the competitive exclusion principle is promising. It would be fascinating to see if in, e.g. insects, the specific interference dynamics could be observed and quantified and whether they would agree with the model.

The results are clearly presented; the methods adequately described; the supplement is rich with details.

There is much scope to build upon this expansion of the theory of consumer-resource models. This work can open up new avenues of research.

We appreciate the reviewer for the very positive comments. We have followed many of the suggestions raised by the reviewer, and the manuscript is much improved as a result.

Following the reviewer's suggestions, we have now used Shannon entropies to quantify the model comparison with experiments that reject the Competitive Exclusion Principle (CEP). Specifically, for each time point of each experimental or model-simulated community, we calculated the Shannon entropies using the formula:

$$H(t) = - \sum_{i=1}^{S_c} P_i(t) \log_2(P_i(t)), \text{ where } P_i(t) \text{ is the probability that a consumer individual}$$

belongs to species C_i at the time stamp of t . The comparison of Shannon entropies in the time series between those of the experimental data and SSA results shown in Fig. 2D-E is presented

in Appendix-fig. 7C-D. The time averages ($\bar{H}$) and standard deviations (δH) of the Shannon entropies for these experimental or SSA model-simulated communities are as follows:

$$\begin{aligned} \bar{H}_{\text{Exp(SSA)}}^{\text{Drosophila_AR_Grp1}} &= 0.95(0.97), \quad \delta H_{\text{Exp(SSA)}}^{\text{Drosophila_AR_Grp1}} = 0.06(0.04); \quad \bar{H}_{\text{Exp(SSA)}}^{\text{Drosophila_AR_Grp2}} = 0.94(0.92), \\ \delta H_{\text{Exp(SSA)}}^{\text{Drosophila_AR_Grp2}} &= 0.07(0.07), \quad \bar{H}_{\text{Exp(SSA)}}^{\text{Tribolium_24}^\circ\text{C}} = 0.96(0.92), \quad \delta H_{\text{Exp(SSA)}}^{\text{Tribolium_24}^\circ\text{C}} = 0.02(0.05). \end{aligned}$$

Meanwhile, we have calculated the time averages ($\bar{C}_i$) and standard deviations (δC_i) of the species' relative/absolute abundances for the experimental or SSA model-simulated communities shown in Fig. 2D-E, which are as follows:

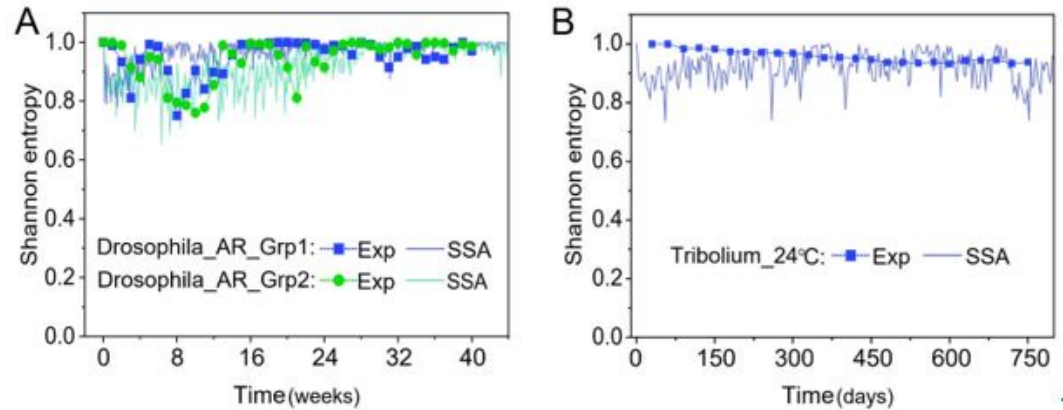
$$\begin{aligned} \overline{C_{D.serrata_AR_Grp1}^{\text{Exp(SSA)}}} &= 0.53(0.55), \quad \delta^{(R)} C_{D.serrata_AR_Grp1}^{\text{Exp(SSA)}} = 0.12(0.09); \\ \overline{C_{D.serrata_AR_Grp2}^{\text{Exp(SSA)}}} &= 0.59(0.61), \quad \delta^{(R)} C_{D.serrata_AR_Grp2}^{\text{Exp(SSA)}} = 0.10(0.12); \quad \overline{C_{T.confusum_24^\circ\text{C}}^{\text{Exp(SSA)}}} = 29.1(28.6), \\ \delta C_{T.confusum_24^\circ\text{C}}^{\text{Exp(SSA)}} &= 5.3(5.2), \quad \overline{C_{T.castaneuma_24^\circ\text{C}}^{\text{Exp(SSA)}}} = 45.9(54.5), \quad \delta C_{T.castaneuma_24^\circ\text{C}}^{\text{Exp(SSA)}} = 7.2(8.6), \text{ where the} \end{aligned}$$

superscript “(R)” represents relative abundances.

From the results of Shannon entropies shown in Author response image 1 (which are identical to those of Appendix-fig. 7C-D) and the quantitative comparison of the time average and standard deviation between the model and experiments presented above, it is evident that the model results in Fig. 2D-E exhibit good consistency with the experimental data. They share roughly identical time averages and standard deviations in both Shannon entropies and the species' relative/absolute abundances for most of the comparisons. All these analyses are included in the appendices and mentioned in the main text.

Author response image 1.

Shannon Entropies of the experimental data and SSA results in Fig. 2D-E, redrawn from Appendix-fig. 7C-D.



Weaknesses:

I am questioning the use of carrying capacity (Eq. 4) instead of using nutrient limitation directly through Monod consumption (e.g. Posfai et al. who the authors cite). I am curious to see how these results hold or are changed when Monod consumption is used.

We thank the reviewer for raising this question. To explain it more clearly, the equation combining the third equation in Eq. 1 and Eq. 4 of our manuscript is presented below as Eq. R1:

$$\dot{R}_l = \begin{cases} \eta_l R_l (1 - R_l / \kappa_l) - \sum_{i=1}^{S_C} k_{il} x_{il} & \text{(for biotic resources),} \\ \zeta_l (1 - R_l / \kappa_l) - \sum_{i=1}^{S_C} k_{il} x_{il} & \text{(for abiotic resources),} \end{cases} \quad (\text{R1})$$

where x_{il} represents the population abundance of the chasing pair $C_i^{(P)} \vee R_l^{(P)}$, κ_l stands for the steady-state population abundance of species R_l (the carrying capacity) in the absence of consumer species. In the case with no consumer species, then $x_{il} = 0$ since $C_i = 0$ ($i=1, \dots, S_C$), thus $R_l = \kappa_l$ when $R_l = 0$.

Eq. R1 for the case of abiotic resources is comparable to Eq. (1) in Posfai et al., which we present below as Eq. R2:

$$\frac{dc_i}{dt} = s_i - \left(\sum_{\sigma} n_{\sigma}(t) \alpha_{\sigma i} \right) r_i(c_i) - \mu_i c_i(t), \quad (\text{R2})$$

where c_i represents the concentration of nutrient i , and thus corresponds to our R_l ; $n_{\sigma}(t)$ is the population of species σ , which corresponds to our C_i ; s_i stands for the nutrient supply rate, which corresponds to our ζ_l ; μ_i denotes the nutrient loss rate, corresponding to our

ζ_l / κ_l ; $\alpha_{\sigma i}$ is the coefficient of the rate of species σ for consuming nutrient i , which

corresponds to our k_{il} ; $n_{\sigma}(t) \cdot r_i(c_i)$ in Posfai et al. is the consumption rate of nutrient i by the population of species σ , which corresponds to our x_{il} .

In Posfai et al., $r_i(c_i)$ is the Monod function: $r_i(c_i) = c_i / (K_i + c_i)$ and thus

$$n_{\sigma}(t) \cdot r_i(c_i) = n_{\sigma}(t) c_i / (K_i + c_i). \quad (\text{R3})$$

In our model, however, since predator interference is not involved in Posfai et al., we need to analyze the form of x_{il} presented in the functional form of $x_{il}(\{R_l\}, \{C_i\})$ in the case involving only chasing pairs. Specifically, for the case of abiotic resources, the population dynamics can be described by Eq. 1 combined with Eq. R1:

$$\begin{cases} \dot{x}_{il} = a_{il} C_i^{(F)} R_l^{(F)} - (k_{il} + d_{il}) x_{il}, \\ \dot{C}_i = \sum_{l=1}^{S_R} w_{il} k_{il} x_{il} - D_i C_i, \quad i = 1, \dots, S_C, \\ \dot{R}_l = \zeta_l (1 - R_l / \kappa_l) - \sum_{i=1}^{S_C} k_{il} x_{il}, \quad l = 1, \dots, S_R, \end{cases} \quad (\text{R4})$$

where $C_i = C_i^{(F)} + \sum_l x_{il}$ and $R_l = R_l^{(F)} + \sum_i x_{il}$. For convenience, we consider the case of $S_R = 1$

where the Monod form was derived (Monod, J. (1949). *Annu. Rev. Microbiol.*, 3, 371-394.).

From $dx_i/dt = 0$, we have

$$x_{il} = C_i^{(F)} R_l^{(F)} / K_{il}, \quad (\text{R5})$$

where $K_{il} = \frac{k_{il} + d_{il}}{a_{il}}$, and $l = 1$. If the population abundance of the resource species is much

larger than that of all consumer species (i.e., $R_l \gg \sum_{i=1}^{S_C} C_i$), then,

$R_l \gg \sum_{i=1}^{S_C} C_i > \sum_{i=1}^{S_C} x_{il}$ and $R_l^{(F)} \approx R_l$. Combined with R5, and noting that $C_i = C_i^{(F)} + x_{il}$ we can

solve for x_{il} :

$$x_{il} = C_i R_l / (K_{il} + R_l), \quad (\text{R6})$$

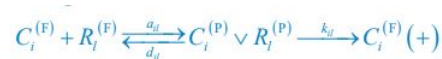
with $l=1$ since $S_R=1$. Comparing Eq. R6 with Eq. R3, and considering the symbol correspondence explained in the text above, it is now clear that our model can be reduced to the Monod consumption form in the case of $S_R=1$ where the Monod form was derived from.

Following on the previous comment, I am confused by the fact that the nutrient consumption term in Eq. 1 and how growth is modeled (Eq. 4) are not obviously compatible and would be hard to match directly to experimentally accessible quantities such as yield (nutrient to biomass conversion ratio). Ultimately, there is a conservation of mass ("flux balance"), and therefore the dynamics must obey it. I don't quite see how conservation of mass is imposed in this work.

We thank the reviewer for raising this question. Indeed, the population dynamics of our model must adhere to flux balance, with the most pertinent equation restated here as Eq. R7:

$$\begin{cases} \dot{C}_i = \sum_{l=1}^{S_R} w_{il} k_{il} x_{il} - D_i C_i, \quad i = 1, \dots, S_C, \\ \dot{R}_l = \zeta_l (1 - R_l / \kappa_l) - \sum_{i=1}^{S_C} k_{il} x_{il}, \quad l = 1, \dots, S_R. \end{cases} \quad (\text{R7})$$

Below is the explanation of how Eq. R7, and thus Eqs. 1 and 4 of our manuscript, adhere to the constraint of flux balance. The interactions and fluxes between consumer and resource species occur solely through chasing pairs. At the population level, the scenario of chasing pairs among consumer species C_i and resource species R_l is presented in the follow expression:



where the superscripts "(F)" and "(P)" represent the freely wandering individuals and those involved in chasing pairs, respectively, "(+)" stands for the gaining biomass of consumer C_i from resource R_l . In our manuscript, we use x_l to represent the population abundance (or equivalently, the concentration, for a well-mixed system with a given size) of the chasing pair $C_i^{(P)} \vee R_l^{(P)}$, and thus, the net flow from resource species R_l to consumer species C_i per unit time is $k_{il} x_{il}$. Noting that there is only one R_l individual within the chasing pair $C_i^{(P)} \vee R_l^{(P)}$,

then the net effect on the population dynamics of species R_l (i.e., dR_l/dt) is $-k_{il} x_{il}$.

However, since a consumer individual from species C_i could be much heavier than a species R_l individual, and energy dissipation would be involved from nutrient conversion into biomass, we introduce a mass conversion ratio w_l in our manuscript. For example, if a species C_i individual is ten times the weight of a species R_l individual, without energy dissipation, the mass conversion ratio w_l should be $1/10$ (i.e., $w_l = 0.1$), however, if half of the chemical energy is dissipated into heat from nutrient conversion into biomass, then $w_l = 0.1 \times 0.5 = 0.05$. Consequently, the net effect of the flux from resource species R_l to consumer

species C_i per unit time on the population dynamics dC_i/dt is $w_{il}k_{il}x_{il}$, and flux balance is clearly satisfied.

For the population dynamics of a consumer species C_i , we need to consider all the biomass influx from different resource species, and thus there is a summation over all species of

resources, which leads to the term of $\sum_{l=1}^{S_R} w_{il}k_{il}x_{il}$ in Eq. R7. Similarly, for the population

dynamics of a resource species R_l , we need to lump sum all the biomass outflow into different

consumer species, resulting in the term of $-\sum_{i=1}^{S_C} k_{il}x_{il}$ in Eq. R7.

Consequently, Eq. R7 and our model satisfy the constraint of flux balance.

These models could be better constrained by more data, in principle, thereby potential exists for a more compelling case of the relevance of this interference mechanism to natural systems.

We thank the reviewer for raising this question. Indeed, our model could benefit from the inclusion of more experimental data. In our manuscript, we primarily set the parameters by estimating their reasonable range. Following the reviewer's suggestions, we have now specified the data we used to set the parameters. For example, in Fig. 2D, we set $D_2=0.01$ with $\tau=0.4$ days, resulting in an expected lifespan of *Drosophila serrata* in our model setting of $\tau/D_2=40$ days, which roughly agrees with experimental data showing that the average lifespan of *D. serrata* is 34 days for males and 54 days for females (lines 321-325 in the appendices; reference: Narayan et al. J Evol Biol. 35: 657–663 (2022)). To explain biodiversity and quantitatively illustrate the rank-abundance curves across diverse communities, the competitive differences across consumer species, exemplified by the coefficient of variation of the mortality rates - a key parameter influencing the rank-abundance curve, were estimated from experimental data in the reference article (Patricia Menon et al., Water Research (2003) 37, 4151) using the two-sigma rule (lines 344-347 in the appendices).

Still, we admit that many factors other than intraspecific interference, such as temporal variation, spatial heterogeneity, etc., are involved in breaking the limits of CEP in natural systems, and it is still challenging to differentiate each contribution in wild systems. However, for the two classical experiments that break CEP (Francisco Ayala, 1969; Thomas Park, 1954), intraspecific interference could probably be the most relevant mechanism, since factors such as temporal variation, spatial heterogeneity, cross-feeding, and metabolic tradeoffs are not involved in those two experimental systems.

The underlying frameworks, B-D and MacArthur are not properly exposed in the introduction, and as a result, it is not obvious what is the specific contribution in this work as opposed to existing literature. One needs to dig into the literature a bit for that.

The specific contribution exists, but it might be more clearly separated and better explained. In the process, the introduction could be expanded a bit to make the paper more accessible, by reviewing key features from the literature that are used in this manuscript.

We thank the reviewer for these very insightful suggestions. Following these suggestions, we have now added a new paragraph and revised the introduction part of our manuscript (lines 51-67 in the main text) to address the relevant issues. Our paper is much improved as a result.

Reviewer #2 (Public Review):

Summary:

The manuscript by Kang et al investigates how the consideration of pairwise encounters (consumer-resource chasing, intraspecific consumer pair, and interspecific consumer pair) influences the community assembly results. To explore this, they presented a new model that considers pairwise encounters and intraspecific interference among consumer individuals, which is an extension of the classical Beddington-DeAngelis (BD) phenomenological model, incorporating detailed considerations of pairwise encounters and intraspecific interference among consumer individuals. Later, they connected with several experimental datasets.

Strengths:

They found that the negative feedback loop created by the intraspecific interference allows a diverse range of consumer species to coexist with only one or a few types of resources. Additionally, they showed that some patterns of their model agree with experimental data, including time-series trajectories of two small in-lab community experiments and the rank-abundance curves from several natural communities. The presented results here are interesting and present another way to explain how the community overcomes the competitive exclusion principle.

We appreciate the reviewer for the positive comments and the accurate summary of our manuscript.

Weaknesses:

The authors only explore the case with interspecific interference or intraspecific interference exists. I believe they need to systematically investigate the case when both interspecific and intraspecific interference exists. In addition, the text description, figures, and mathematical notations have to be improved to enhance the article's readability. I believe this manuscript can be improved by addressing my comments, which I describe in more detail below.

We thank the reviewer for these valuable suggestions. We have followed many of the suggestions raised by the reviewer, and the manuscript is much improved as a result.

(1) In nature, it is really hard for me to believe that only interspecific interference or intraspecific interference exists. I think a hybrid between interspecific interference and intraspecific interference is very likely. What would happen if both the interspecific and

intraspecific interference existed at the same time but with different encounter rates? Maybe the authors can systematically explore the hybrid between the two mechanisms by changing their encounter rates. I would appreciate it if the authors could explore this route.

We thank the reviewer for raising this question. Indeed, interspecific interference and intraspecific interference simultaneously exist in real cases. To differentiate the separate contributions of inter- and intra-specific interference on biodiversity, we considered different scenarios involving inter- or intra-specific interference. In fact, we have also considered the scenario involving both inter- and intra-specific interference in our old version for the case of $S_C = 2$ and $S_R = 1$, where two consumer species compete for one resource species (Appendix-fig. 5, and lines 147-148, 162-163 in the main text of the old version, or lines 160-161, 175-177 in the new version).

Following the reviewer's suggestions, we have now systematically investigated the cases of $S_C = 6$, $S_R = 1$, and $S_C = 20$, $S_R = 1$, where six or twenty consumer species compete for one resource species in scenarios involving chasing pairs and both inter- and intra-specific interference using both ordinary differential equations (ODEs) and stochastic simulation algorithm (SSA). These newly added ODE and SSA results are shown in Appendix-fig. 5 F-H, and we have added a new paragraph to describe these results in our manuscript (lines 212-215 in the main text). Consistent with our findings in the case of $S_C = 2$ and $S_R = 1$, the species coexistence behavior in the cases of both $S_C = 6$, $S_R = 1$, and $S_C = 20$, $S_R = 1$ is very similar to those without interspecific interference: all consumer species coexist with one type of resources at constant population densities in the ODE studies, and the SSA results fluctuate around the population dynamics of the ODEs.

As for the encounter rates of interspecific and intraspecific interference, in fact, in a well-mixed system, these encounter rates can be derived from the mobility rates of the consumer species using the mean field method. For a system with a size of L^2 , the interspecific

encounter rate between consumer species C_i and C_j ($i \neq j$) is $a'_{ij} = 2r^{(1)}L^{-2}\sqrt{v_{C_i}^2 + v_{C_j}^2}$ please

refer to lines 100-102, 293-317 in the main text, and see also Appendix-fig. 1), where $r^{(1)}$ is the upper distance for interference, while v_{C_i} and v_{C_j} represent the mobility rates of species C_i and C_j , respectively. Meanwhile, the intraspecific encounter rates within species C_i and

species C_j are $a'_{ii} = 2\sqrt{2}v_{C_i}r^{(1)}L^{-2}$ and $a'_{jj} = 2\sqrt{2}v_{C_j}r^{(1)}L^{-2}$, respectively.

Thus, once the intraspecific encounter rates a'_{ii} are a'_{jj} given, the interspecific encounter rate between species C_i and C_j is determined. Consequently, we could not tune the encounter rates of interspecific and intraspecific interference at will in our study, especially noting that for clarity reasons, we have used the mortality rate as the only parameter that varies among the consumer species throughout this study. Alternatively, we have made a systematic study on analyzing the influence of varying the separate rate and escape rate on species coexistence in the case of two consumers competing for a single type of resources (see Appendix-fig. 5A).

(2) In the first two paragraphs of the introduction, the authors describe the competitive exclusion principle (CEP) and past attempts to overcome the CEP. Moving on from the first two paragraphs to the third paragraph, I think there is a gap that needs to be filled to make the transition smoother and help readers understand the motivations. More specifically, I think the authors need to add one more paragraph dedicated to explaining why predator interference is important, how considering the mechanism of predator

interference may help overcome the CEP, and whether predator interference has been investigated or under-investigated in the past. Then building upon the more detailed introduction and movement of predator interference, the authors may briefly introduce the classical B-D phenomenological model and what are the conventional results derived from the classical B-D model as well as how they intend to extend the B-D model to consider the pairwise encounters.

We thank the reviewer for these very insightful suggestions. Following these suggestions, we have added a new paragraph and revised the introduction part of our paper (lines 51-67 in the main text). Our manuscript is significantly improved as a result.

(3) The notations for the species abundances are not very informative. I believe some improvements can be made to make them more meaningful. For example, I think using Greek letters for consumers and English letters for resources might improve readability. Some sub-scripts are not necessary. For instance, $R^{(I)}_0$ can be simplified to g_I to denote the intrinsic growth rate of resource I. Similarly, $K^{(I)}_0$ can be simplified to K_I . Another example is $R^{(I)}_a$, which can be simplified to s_I to denote the supply rate. In addition, right now, it is hard to find all definitions across the text. I would suggest adding a separate illustrative box with all mathematical equations and explanations of symbols.

We thank the reviewer for these very useful suggestions. We have now followed many of the suggestions to improve the readability of our manuscript. Given that we have used many English letters for consumers and there are already many symbols of English and Greek letters for different variables and parameters in the appendices, we have opted to use Greek letters for parameters specific to resource species and English letters for those specific to consumer species. Additionally, we have now added Appendix-tables 1-2 in the appendices (pages 16-17 in the appendices) to illustrate the symbols used throughout our manuscript.

(4) What is the $f_i(R^{(F)})$ on line 131? Does it refer to the growth rate of C_i ? I noticed that $f_i(R^{(F)})$ is defined in the supplementary information. But please ensure that readers can understand it even without reading the supplementary information. Otherwise, please directly refer to the supplementary information when $f_i(R^{(F)})$ occurs for the first time. Similarly, I don't think the readers can understand Ω_i and G_i on lines 135-136.

We thank the reviewer for raising these questions. We apologize for not illustrating those symbols and functions clearly enough in our previous version of the manuscript. $f_i(R^{(F)})$ is a

function of the variable $R^{(F)}$ with the index i , which is defined as $f_1(R^{(F)}) = \frac{R^{(F)}}{R^{(F)} + K_1}$ and

$f_2(R^{(F)}) = \frac{R^{(F)}}{R^{(F)} + K_2}$ for $i=2$. Following the reviewer's suggestions, we have now added

clear definitions for symbols and functions and resolved these issues. The definitions of Ω_i , Ω_i , G , and G are overly complex, and hence we directly refer to the Appendices when they occur for the first time in the main text.

Reviewer #3 (Public Review):

Summary:

A central question in ecology is: Why are there so many species? This question gained heightened interest after the development of influential models in theoretical ecology in the 1960s, demonstrating that under certain conditions, two consumer species cannot coexist on the same resource. Since then, several mechanisms have been shown to be capable of breaking the competitive exclusion principle (although, we still lack a general understanding of the relative importance of the various mechanisms in promoting biodiversity).

One mechanism that allows for breaking the competitive exclusion principle is predator interference. The Beddington-DeAngelis is a simple model that accounts for predator interference in the functional response of a predator. The B-D model is based on the idea that when two predators encounter one another, they waste some time engaging with one another which could otherwise be used to search for resources. While the model has been influential in theoretical ecology, it has also been criticized at times for several unusual assumptions, most critically, that predators interfere with each other regardless of whether they are already engaged in another interaction. However, there has been considerable work since then which has sought either to find sets of assumptions that lead to the B-D equation or to derive alternative equations from a more realistic set of assumptions (Ruxton et al. 1992; Cosner et al. 1999; Broom et al. 2010; Geritz and Gyllenberg 2012). This paper represents another attempt to more rigorously derive a model of predator interference by borrowing concepts from chemical reaction kinetics (the approach is similar to previous work: Ruxton et al. 1992). The main point of difference is that the model in the current manuscript allows for 'chasing pairs', where a predator and prey engage with one another to the exclusion of other interactions, a situation Ruxton et al. (1992) do not consider. While the resulting functional response is quite complex, the authors show that under certain conditions, one can get an analytical expression for the functional response of a predator as a function of predator and resource densities. They then go on to show that including intraspecific interference allows for the coexistence of multiple species on one or a few resources, and demonstrate that this result is robust to demographic stochasticity.

We thank the reviewer for carefully reading our manuscript and for the positive comments on the rigorously derived model of predator interference presented in our paper. We also appreciate the reviewer for providing a thorough introduction to the research background of our study, especially the studies related to the BeddingtonDeAngelis model. We apologize for our oversight in not fully appreciating the related study by Ruxton et al. (1992) at the time of our first submission. Indeed, as suggested by the reviewer, Ruxton et al. (1992) is relevant to our study in that we both borrowed concepts from chemical reaction kinetics. Now, we have reworked the introduction and discussion sections of our manuscript, cited, and acknowledged the contributions of related works, including Ruxton et al. (1992).

Strengths:

I appreciate the effort to rigorously derive interaction rates from models of individual behaviors. As currently applied, functional responses (FRs) are estimated by fitting equations to feeding rate data across a range of prey or predator densities. In practice, such experiments are only possible for a limited set of species. This is problematic because whether a particular FR allows stability or coexistence depends on not just its functional form, but also its parameter values. The promise of the approach taken here is that one might be able to derive the functional response parameters of a particular predator species from species traits or more readily measurable behavioral data.

We appreciate the reviewer's positive comments regarding the rigorous derivation of our model. Indeed, all parameters of our model can be derived from measurable behavioral data

for a specific set of predator species.

Weaknesses:

The main weakness of this paper is that it devotes the vast majority of its length to demonstrating results that are already widely known in ecology. We have known for some time that predator interference can relax the CEP (e.g., Cantrell, R. S., Cosner, C., & Ruan, S. 2004).

While the model presented in this paper differs from the functional form of the B-D in some cases, it would be difficult to formulate a model that includes intraspecific interference (that increases with predator density) that does not allow for coexistence under some parameter range. Thus, I find it strange that most of the main text of the paper deals with demonstrating that predator interference allows for coexistence, given that this result is already well known. A more useful contribution would focus on the extent to which the dynamics of this model differ from those of the B-D model.

We appreciate the reviewer for raising this question and apologize for not sufficiently clarifying the contribution of our manuscript in the context of existing knowledge upon our initial submission. We have now significantly revised the introduction part of our manuscript (lines 51-67 in the main text) to make this clearer. Indeed, with the application of the Beddington-DeAngelis (B-D) model, several studies (e.g., Cantrell, R. S., Cosner, C., & Ruan, S. 2004) have already shown that intraspecific interference promotes species coexistence, and it is certain that the mechanism of intraspecific interference could lead to species coexistence if modeled correctly. However, while we acknowledge that the B-D model is a brilliant phenomenological model of intraspecific interference, for the specific research topic of our manuscript on breaking the CEP and explaining the paradox of the plankton, it is highly questionable regarding the validity of applying the B-D model to obtain compelling results.

Specifically, the functional response in the B-D model of intraspecific interference can be formally derived from the scenario involving only chasing pairs without consideration of pairwise encounters between consumer individuals (Eq. S8 in Appendices; related references: Gert Huisman, Rob J De Boer, J. Theor. Biol. 185, 389 (1997) and Xin Wang and Yang-Yu Liu, iScience 23, 101009 (2020)). Since we have demonstrated that the scenario involving only chasing pairs is under the constraint of CEP (see lines 139-144 in the main text and Appendix-fig. 3A-C; related references: Xin Wang and Yang-Yu Liu, iScience 23, 101009 (2020)), and given the identical functional response mentioned above, it is thus highly questionable regarding the validity of the studies relying on the B-D model to break CEP or explain the paradox of the plankton.

Consequently, one of the major objectives of our manuscript is to resolve whether the mechanism of intraspecific interference can truly break CEP and explain the paradox of the plankton in a rigorous manner. By modeling intraspecific predator interference from a mechanistic perspective and applying rigorous mathematical analysis and numerical simulations, our work resolves these issues and demonstrates that intraspecific interference enables a wide range of consumer species to coexist with only one or a handful of resource species. This naturally breaks CEP, explains the paradox of plankton, and quantitatively illustrates a broad spectrum of experimental results.

For intuitive understanding, we introduced a functional response in our model (presented as Eq. 5 in the main text), which indeed involves approximations. However, to rigorously break the CEP or explain the paradox of plankton, all simulation results in our study were directly derived from equations 1 to 4 (main text), without relying on the approximate functional response presented in Eq. 5.

The formulation of chasing-pair engagements assumes that prey being chased by a predator are unavailable to other predators. For one, this seems inconsistent with the ecology of most predator-prey systems. In the system in which I work (coral reef fishes), prey under attack by one predator are much more likely to be attacked by other predators (whether it be a predator of the same species or otherwise). I find it challenging to think of a mechanism that would give rise to chased prey being unavailable to other predators. The authors also critique the B-D model: "However, the functional response of the B-D model involving intraspecific interference can be formally derived from the scenario involving only chasing pairs without predator interference (Wang and Liu, 2020; Huisman and De Boer, 1997) (see Eqs. S8 and S24). Therefore, the validity of applying the B-D model to break the CEP is questionable."

We appreciate the reviewer for raising this question. We fully agree with the reviewer that in many predator-prey systems (e.g., coral reef fishes as mentioned by the reviewer, wolves, and even microbial species such as *Myxococcus xanthus*; related references: Berleman et al., FEMS Microbiol. Rev. 33, 942-957 (2009)), prey under attack by one predator can be targeted by another predator (which we term as a chasing triplet) or even by additional predator individuals (which we define as higher-order terms). However, since we have already demonstrated in a previous study (Xin Wang, Yang-Yu Liu, iScience 23, 101009 (2020)) from a mechanistic perspective that a scenario involving chasing triplets or higher-order terms can naturally break the CEP, while our manuscript focuses on whether pairwise encounters between individuals can break the CEP and explain the paradox of plankton, we deliberately excluded confounding factors that are already known to promote biodiversity, just as we excluded prevalent factors such as cross-feeding and temporal variations in our model.

However, the way "chasing pairs" are formulated does result in predator interference because a predator attacking prey interferes with the ability of other predators to encounter the prey. I don't follow the author's logic that B-D isn't a valid explanation for coexistence because a model incorporating chasing pairs engagements results in the same functional form as B-D.

We thank the reviewer for raising this question, and we apologize for not making this point clear enough at the time of our initial submission. We have now revised the related part of our manuscript (lines 56-62 in the main text) to make this clearer.

In our definition, predator interference means the pairwise encounter between consumer individuals, while a chasing pair is formed by a pairwise encounter between a consumer individual and a resource individual. Thus, in these definitions, a scenario involving only chasing pairs does not involve pairwise encounters between consumer individuals (which is our definition of predator interference).

We acknowledge that there can be different definitions of predator interference, and the reviewer's interpretation is based on a definition of predator interference that incorporates indirect interference without pairwise encounters between consumer individuals. We do not wish to argue about the appropriateness of definitions. However, since we have proven that scenarios involving only chasing pairs are under the constraint of CEP (see lines 139-144 in the main text and Appendix-fig. 3A-C; related references: Xin Wang and Yang-Yu Liu, iScience 23, 101009 (2020)), while the functional response of the B-D model can be derived from the scenario involving only chasing pairs without consideration of pairwise encounters between consumer individuals (Eq. S8 in Appendices; related references: Gert Huisman, Rob J De Boer, J. Theor. Biol. 185, 389 (1997) and Xin Wang and Yang-Yu Liu, iScience 23, 101009 (2020)), it is thus highly questionable regarding the validity of applying the B-D model to break CEP.

More broadly, the specific functional form used to model predator interference is of secondary importance to the general insight that intraspecific interference (however it is modeled) can allow for coexistence. Mechanisms of predator interference are complex and vary substantially across species. Thus it is unlikely that any one specific functional form is generally applicable.

We thank the reviewer for raising this issue. We agree that the general insight that intraspecific predator interference can facilitate species coexistence is of great importance. We also acknowledge that any functional form of a functional response is unlikely to be universally applicable, as explicit functional responses inevitably involve approximations. However, we must reemphasize the importance of verifying whether intraspecific predator interference can truly break CEP and explain the paradox of plankton, which is one of the primary objectives of our study. As mentioned above, since the B-D model can be derived from the scenario involving only chasing pairs (Eq. S8 in Appendices; related references: Gert Huisman, Rob J De Boer, J. Theor. Biol. 185, 389 (1997) and Xin Wang and Yang-Yu Liu, iScience 23, 101009 (2020)), while we have demonstrated that scenarios involving only chasing pairs are subject to the constraint of CEP (see lines 139-144 in the main text and Appendix-fig. 3A-C; related references: Xin Wang and Yang-Yu Liu, iScience 23, 101009 (2020)), it is highly questionable regarding the validity of applying the B-D model to break CEP.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

I do not see any code or data sharing. They should exist in a prominent place. The authors should make their simulations and the analysis scripts freely available to download, e.g. by GitHub. This is always true but especially so in a journal like eLife.

We appreciate the reviewer for these recommendations. We apologize for our oversight regarding the unsuccessful upload of the data in our initial submission, as the data size was considerable and we neglected to double-check for this issue. Following the reviewer's recommendation, we have now uploaded the code and dataset to GitHub (accessible at <https://github.com/SchordK/Intraspecific-predator-interference-promotesbiodiversity-in-ecosystems>), where they are freely available for download.

The introduction section should include more background, including about BD but also about consumer-resource models. Part of the results section could be moved/edited to the introduction. You should try that the results section should contain only "new" stuff whereas the "old" stuff should go in the introduction.

We thank the reviewer for these recommendations. Following these suggestions, we have now reorganized our manuscript by adding a new paragraph to the introduction section (lines 51-62 in the main text) and revising related content in both the introduction and results sections (lines 63-67, 81-83 in the main text).

I found myself getting a little bogged down in the general/formal description of the model before you go to specific cases. I found the most interesting part of the paper to be its second half. This is a dangerous strategy, a casual reader may miss out on the most interesting part of the paper. It's your paper and do what you think is best, but my opinion is that you could improve the presentation of the model and background to get to the specific contribution and specific use case quickly and easily, then immediately to the data. You can leave the more general formulation and the details to later in the paper or even the appendix. Ultimately, you have a simple idea and a beautiful

application on interesting data-that is your strength I think, and so, I would focus on that.

We appreciate the reviewer for the positive comments and valuable suggestions. Following these recommendations, we have revised the presentation of the background information to clarify the contribution of our manuscript, and we have refined our model presentation to enhance clarity. Meanwhile, as we need to address the concerns raised by other reviewers, we continue to maintain systematic investigations for scenarios involving different forms of pairwise encounters in the case of $S_C = 2$ and $S_R = 1$ before applying our model to the experimental data.

Reviewer #2 (Recommendations For The Authors):

(1) I believe the surfaces in Figs. 1F-H corresponds to the zero-growth isoclines. The authors should directly point it out in the figure captions and text descriptions.

We thank the reviewer for this suggestion, and we have followed it to address the issue.

(2) After showing equations 1 or 2, I believe it will help readers understand the mechanism of equations by adding text such as "(see Fig. 1B)" to the sentences following the equations.

We appreciate the reviewer's suggestion, and we have implemented it to address the issue.

(3) Lines 12, 129 143 & 188: "at steady state" -> "at a steady state"

(4) Line 138: "is doom to extinct" -> "is doomed to extinct"

(5) Line 170: "intraspecific interference promotes species coexistence along with stochasticity" -> "intraspecific interference still robustly promotes species coexistence when stochasticity is considered"

(6) Line 190: "The long-term coexistence behavior are exemplified" -> "The long-term coexistence behavior is exemplified"

(7) Line 227: "the coefficient of variation was taken round 0.3" -> "the coefficient of variation was taken around 0.3"?

(8) Line 235: "tend to extinct" -> "tend to be extinct"

We thank the reviewer for all these suggestions, and we have implemented each of them to revise our manuscript.

Reviewer #3 (Recommendations For The Authors):

I think this would be a much more useful paper if the authors focused on how the behavior of this model differs from existing models rather than showing that the new formation also generates the same dynamics as the existing theory.

We thank the reviewers for this suggestion, and we apologize for not explaining the limitations of the B-D model and the related studies on the topic of CEP clearly enough at the time of our initial submission. As we have explained in the responses above, we have now revised the introduction part of our manuscript (lines 5167 in the main text) to make it clear that since the functional response in the B-D model can be derived from the scenario involving only chasing pairs without consideration of pairwise encounters between consumer individuals, while we have demonstrated that a scenario involving only chasing

pairs is under the constraint of CEP, it is thus highly questionable regarding the validity of the studies relying on the B-D model to break CEP or explain the paradox of the plankton. Consequently, one of the major objectives of our manuscript is to resolve whether the mechanism of intraspecific interference can truly break CEP and explain the paradox of the plankton in a rigorous manner. By modeling from a mechanistic perspective, we resolve the above issues and quantitatively illustrate a broad spectrum of experimental results, including two classical experiments that violate CEP and the rank-abundance curves across diverse ecological communities.

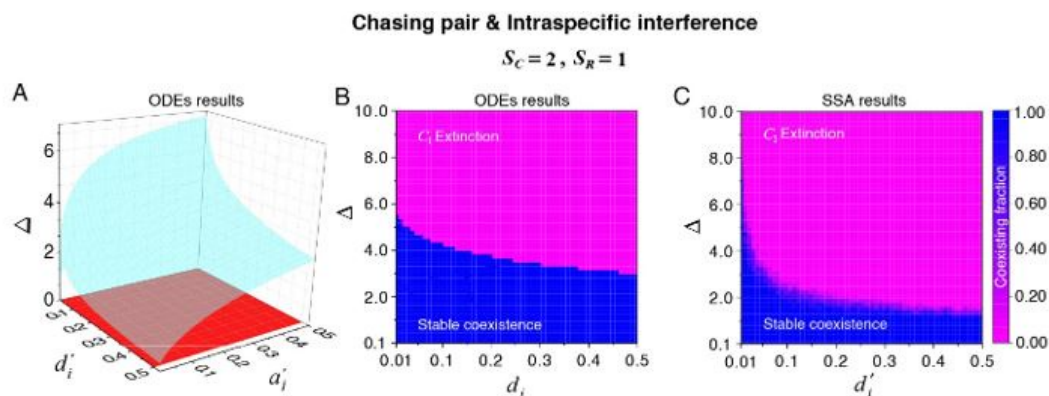
Things that would be of interest:

What are the conditions for coexistence in this model? Presumably, it depends heavily on the equilibrium abundances of the consumers and resources as well as the engagement times/rates.

We thank the reviewer for raising this question. We have shown that there is a wide range of parameter space for species coexistence in our model. Specifically, for the case involving two consumer species and one resource species ($S_C = 2$ and $S_R = 1$), we have conducted a systematic study on the parameter region for promoting species coexistence. For clarity, we set the mortality rate D_i ($i = 1, 2$) as the only parameter that varies with the consumer species, and the order of magnitude of all model parameters was estimated from behavioral data. The results for scenarios involving intraspecific predator interference are shown in Appendix-figs. 4B-D, 5A, 6C-D and we redraw some of them here as Fig. R2, including both ODEs and SSA results, wherein $\Delta = (D_1 - D_2) / D_2$ represents the competitive difference between the two consumer species. For example, $\Delta = 1$ means that species C2 is twice the competitiveness of species C1. In Fig. R2 (see also Appendix-figs. 4B-D, 5A, 6C-D), we see that the two consumer species can coexist with a large competitive difference in either ODEs and SSA simulation studies.

Author response image 2.

The parameter region for two consumer species coexisting with one type of abiotic resource species ($S_C = 2$ and $S_R = 1$). (A) The region below the blue surface and above the red surface represents stable coexistence of the three species at constant population densities. (B) The blue region represents stable coexistence at a steady state for the three species. (C) The color indicates (refer to the color bar) the coexisting fraction for long-term coexistence of the three species. Figure redrawn from Appendixfigs. 4B, 6C-D.



For systems shown in Fig. 3A-D, where the number of consumer species is much larger than that of the resource species, we set each consumer species with unique competitiveness through a distinctive D_i ($i = 1, \dots, S_C$). In Fig. 3A-D (see also Appendix fig. 10), we see that

hundreds of consumer species may coexist with one or three types of resources when the coefficient of variation (CV) of the consumer species' competitiveness was taken around 0.3, which indicates a large parameter region for promoting species coexistence.

Is there existing data to estimate the parameters in the model directly from behavioral data? Do these parameter ranges support the hypothesis that predator interference is significant enough to allow for the coexistence of natural predator populations?

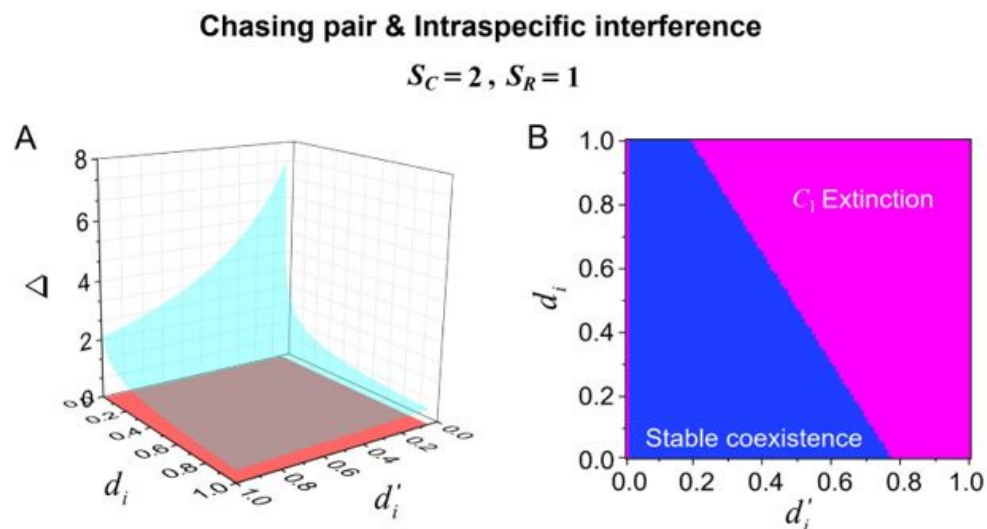
We appreciate the reviewer for raising this question. Indeed, the parameters in our model were primarily determined by estimating their reasonable range from behavioral data. Following the reviewer's suggestions, we have now specified the data we used to set the parameters. For instance, in Fig. 2D, we set $D_2=0.01$ with $\tau=0.4$ Day, resulting in an expected lifespan of *Drosophila serrata* in our model setting of $\tau/D_2=40$ days, which roughly agrees with experimental behavioral data showing that the average lifespan of *D. serrata* is 34 days for males and 54 days for females (lines 321325 in the appendices; reference: Narayan et al. J Evol Biol. 35: 657–663 (2022)). To account for competitive differences, we set the mortality rate as the only parameter that varies among the consumer species. As specified in the Appendices, the CV of the mortality rate is the only parameter that was used to fit the experiments within the range of 0.15-0.43. This parameter range (i.e., 0.15-0.43) was directly estimated from experimental data in the reference article (Patricia Menon et al., Water Research 37, 4151(2003)) using the two-sigma rule (lines 344-347 in the appendices).

Given the high consistency between the model results and experiments shown in Figs. 2D-E and 3C-D, where all the key model parameters were estimated from experimental data in references, and considering that the rank-abundance curves shown in Fig. 3C-D include a wide range of ecological communities, there is no doubt that predator interference is significant enough to allow for the coexistence of natural predator populations within the parameter ranges estimated from experimental references.

Bifurcation analyses for the novel parameters of this model. Does the fact that prey can escape lead to qualitatively different model behaviors?

Author response image 3.

Bifurcation analyses for the separate rate d'_i and escape rate d_i ($i=1, 2$) of our model in the case of two consumer species competing for one abiotic resource species ($S_C=2$ and $S_R=1$). (A) A 3D representation: the region above the blue surface signifies competitive exclusion where C_1 species extinct, while the region below the blue surface and above the red surface represents stable coexistence of the three species at constant population densities. (B) a 2D representation: the blue region represents stable coexistence at a steady state for the three species. Figure redrawn from Appendix-fig. 4C-D.



We appreciate the reviewer for this suggestion. Following this suggestion, we have conducted bifurcation analyses for the separate rate d'_i and escape rate d_i of our model in the case where two consumer species compete for one resource species ($S_C=2$ and $S_R=1$). Both 2D and 3D representations of these results have been included in Appendix-fig. 4, and we redraw them here as Fig. R3. In Fig. R3, we set the mortality rate D_i ($i=1, 2$) as the only parameter that varies between the consumer species, and thus $\Delta = -(D_1-D_2)/D_2$ represents the competitive difference between the two species.

As shown in Fig. R3A-B, the smaller the escape rate d_i , the larger the competitive difference Δ tolerated for species coexistence at steady state. A similar trend is observed for the separate rate d'_i . However, there is an abrupt change for both 2D and 3D representations at the area where $d'_i=0$, since if $d'_i=0$, all consumer individuals would be trapped in interference pairs, and then no consumer species could exist. On the contrary, there is no abrupt change for both 2D and 3D representations at the area where $d_i=0$, since even if $d_i=0$, the consumer individuals could still leave the chasing pair through the capture process.

Figures: I found the 3D plots especially Appendix Figure 2 very difficult to interpret. I think 2D plots with multiple lines to represent predator densities would be more clear.

We thank the reviewer for this suggestion. Following this suggestion, we have added a 2D diagram to Appendix-fig. 2.

<https://doi.org/10.7554/eLife.93115.2.sa0>